

# PROJECT: CONTEXT / CONTENT

A total conversion — the living design bible, v11.71

Machines that evolved into creatures. A theory of mind hidden in a type chart.

Personal-use ROM hack built on the pret `pokered` decompilation. This document consolidates every decision made so far, including the reasoning behind rejected options, so nothing is lost when work moves into source control.

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## 0. THE THESIS

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Emotions are the color of context. Thoughts are the content.

The game argues this in five registers and never once in dialogue.

| Layer       | How it argues                                                               |
|-------------|-----------------------------------------------------------------------------|
| Mechanics   | The type chart makes LOGIC fail against CONTEXT                             |
| Cartography | Every place name is a word meaning both a color and a feeling               |
| Structure   | The Index — the artifact you spend the game filling — measures only content |
| Movedex     | PERSPECTIVE to RECURSION is the lineage of feeling, in two move slots       |
| Vocabulary  | The whole map is named in the language of image reproduction                |

### 0.1 Craft rules

1. **No character ever explains the color–emotion link.** Not one line. The moment it is said aloud it stops being architecture and becomes a moral.
2. **One NPC nearly notices.** A kid in Verdigris says the town felt greener before the store went up. Wrong — but wrong in the right direction.
3. **Name the process, not the pathology.** Verdigris, gilt, blanch, quicksilver are things that *happened to* a place. Jaundice, sallow, fever are things a place supposedly *is* — that relocates blame onto whoever lives there.
4. **Neutral, not dull.** Place names should be materially specific rather than moody. Prefer a pigment to an adjective.
5. **Lessons must be mechanical.** If a gym's lesson can be skipped by grinding, it is not a lesson.
6. **Comedy is the best cover for a thesis.** A game that is funny in its ordinary moments earns the right to be serious in three or four of them. Corpus employees are cheerful and absurd; the horror is what they are cheerful *about*. Nothing protects rule 1 better than a joke — a player who is laughing is a player who is not being lectured.

## 0.2 The pigment throughline

Blanche, Slate, Halftone, Verdigris, Brazen, Quicksilver, Lurid — pigment, printing and reproduction terms. Lithographic stone, dot screens, corroded bronze, brass plate, alchemical pigment, glow.

The map is written in the vocabulary of **making an image of a thing**. Which is what a model is. The Index is a reproduction that loses the original, and it charts a country named in the language of reproduction.

This was discovered after the fact, not planned. Do not explain it either.

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## 0.3 The loop underneath

The thesis in 0 is a statement. The mechanism under it is a **loop**.

**Context determines which content is available; the content you end up holding reshapes the context.** Around and around, unattended, in everything. That is the Contextual Feedback Model, and like everything else here it is never named in the game.

Two consequences the game can show without ever explaining:

- **Positive loops compound.** Something amplifies itself until something stops it. RECURSION is exactly this, mechanically — see 4.7.
- **Negative loops stabilise.** Something damps itself back toward where it started. It looks like nothing happening, which is why it goes unnoticed far longer.

**And an unwatched loop distorts the representation, not the thing.** This is where the Index and the halftone converge: the entry stays thin, the dots stay the same size, the reader concludes something about the **daemon** — and the bias came from resolution and inattention, never from malice. Nobody in the game is lying. The picture is simply being made at the wrong distance by someone who is inside the loop that makes it.

Dialogue guidance for gesturing at this without stating it is in 4.9.

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## 0.4 Sensation and perception — where CONTENT and CONTEXT came from

**The split is not a design invention.** It was worked out in 2013, in *Sensation into Perception, and the Filters of our Experience* — the longest post on PsychologyCode, nearly a third of the site, and **the one that was never reposted**. `lineage.md` 2.2 has the archaeology. This section is what the game owes it.

It names both halves, under different words:

*The basic information your eyes (or any other sense organ) gathers is known as the **bottom-up** information, and all your memories which come into play during the analysis is called **top-down** information. Eventually those two types of information interact and the result is recognition.*

**Bottom-up is CONTENT. Top-down is CONTEXT.** Eleven years before either had its name here.

**The sentence the whole game rests on**

*In a normally working human eye **the sensation of red is the same for everyone.** ... The red I experience is influenced by every other red I have experienced, all of my thoughts about it, my experiences, beliefs, and knowledge all influence how the red appears to me.*

**Sensation is shared. Perception is not.** One line, doing three jobs:

- **It is the edition split.** CONTENT and CONTEXT are not two versions of a product. They are the two stages of a single act of recognition — which is why 8.4 fixed both cartridges to the same story. *Sensation and perception are not two tales; they are one told twice.*
- **It is the type chart.** LOGIC fails against CONTEXT because top-down information is what *completes* bottom-up information, never the reverse.
- **It is the colour cartography.** Every town is named for something that is both a colour and a feeling, and 8.6 spends the greyscale so that the player supplies the hue. **The player supplying the hue is this article, played.**

**Three places it is already in the game, none of them put there on purpose**

1. **DETACHED**. 7.2 defines fleeing as *breaking your connection to a running process — the process is still running, you simply stopped observing it.* **That is the falling tree, sitting in a battle message,** written before anyone joined the two up:

*Since sound is a perception and perception requires an observer then the answer is no. ... no observer, no sound.*

2. **"Some of them disagree."** The briefing already warns the player that the records will not match, which is the article's own conclusion: *two people looking at the same thing may receive the same information through their eyes, but their experience of that information will be vastly different.*

3. **The Five Witnesses.** 4.8's lock is that sentence built into a puzzle.

**Ruled 2026-09-04 — the two Indexes disagree on purpose**

*Gen 3 ships a separate Index text file per edition*, and vanilla uses it for flavour: 451 of ours still carry that variation, inherited. But every entry *this project has written* was written once and copied into both — which quietly threw away **the one instrument in the ROM shaped exactly like the thesis**.

**So the entries are a pair, and the pair is the argument.** One daemon, two records, neither wrong:

|                |                                                                           |
|----------------|---------------------------------------------------------------------------|
| <b>CONTENT</b> | what it does — mechanism, procedure, the thing itself                     |
| <b>CONTEXT</b> | what follows from it — who framed it, who chose, who is not in the record |

*Neither entry reports a feeling*, so 0's table still holds exactly: **the Index measures only content**. What differs is **what the observer attended to** — and two honest records of one creature still fail to match. *That is the article, inside the artifact the player spends the game filling*.

**Written first for the four MUSAI**, because that family already is the argument. SEEKMUSAI's pair is the clearest, and it is about EMBEDDING both times:

|         |                                                                                                                   |
|---------|-------------------------------------------------------------------------------------------------------------------|
| CONTENT | "Finds the nearest match to anything it is shown. Nearest is not the same as right. Nobody told it. Nobody will." |
| CONTEXT | "Every answer it gives is the closest thing it has already seen. What it has seen was chosen by someone else."    |

**Craft rule 1 is unbreached.** Nothing here is said in the game. A player with one cartridge learns a creature; a player with both learns that **a record has an author**. *They are never told which entry is true, because the question is malformed* — which is the point of the article, and the point of the pair.

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**And one man in Halftone argues about paint**

**The balding man in the Halftone Mart is a printer, and his brother works in light:**

"My brother and I did not speak for a year over the primaries. I said red, yellow, blue. He said red, green, blue. He works in light, so he adds. I work in ink, so I take away. He called me negative for a year. Take everything from him and he is sitting in the dark. Take everything from me and I have a clean sheet. We were both right. He still will not have it."

***It is not arbitrary and it is not the thesis.*** Three things hold it in place:

1. **0.2.** The map is named in the vocabulary of pigment and printing — Blanche, Halftone, Verdigris, Quicksilver, Brazen. **This world is subtractive by construction.** Meanwhile the player is holding a **GBA screen, which is additive.** *The argument is about a split the player is physically sitting inside.*
2. **0.4.** Both men are right. RGB is correct for light and RYB for ink, and **the disagreement is not an error — it is two frames.** *This is the article in a form a player can go and check for themselves.*
3. **8.6.** The world is grey and the player supplies the colour, so the game is **neither** system. *It declines to mix at all, and lets the observer finish it.*

**Halftone was the only possible town.** Its name is the dot screen — *the process that makes a colour out of inks that are not that colour* — and its own sign already reads "The Noble Purple Town", in a game where 8.6 records that **purple is not on the spectrum**, that the mind folds the line into a loop and invents it. *A man arguing about which primaries are real, under a sign naming the colour that is not one.*

**It does not spend craft rule 2.** That rule allows exactly one NPC to *nearly notice* the colour-emotion link and the Verdigris kid has it. **The printer never mentions feeling.** He is talking about ink, and he is simply right.

*The joke inverts on purpose.* The brother calls him **negative** — and the subtractive one is the one who ends up holding the **clean sheet**, white, while the additive one is left in the dark. **The pessimist gets the blank page.**

*Cost:* the balding man used to ask where to buy X ATTACK. **That was the weakest of the three hints in that shop**, and the shop keeps the other two.

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## 1. LEXICON

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The creatures are **DAEMONS**.

A daemon is a background process that runs unattended, and it is the Greek *daimon* — the guiding spirit, Socrates' inner voice speaking from somewhere he did not control. One word, both registers, and it never says "AI." Six characters, so it fits every UI box that held POKÉMON.

*Considered and rejected:* QUALIA and ANIMA both state the thesis outright, violating craft rule 1. ENGRAM (a memory trace physically encoded in tissue) and KERNEL (system core, and a seed) remain viable alternates.

| Vanilla                | Ours                           | Why                                                                                                                                                                                                                                                                                       |
|------------------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pokémon                | <b>DAEMON</b>                  | background process; guiding spirit                                                                                                                                                                                                                                                        |
| Pokédex                | <b>THE INDEX</b>               | a lookup table that points at content. Cold, bureaucratic, exactly wrong in the right way                                                                                                                                                                                                 |
| Trainer                | <b>USER</b>                    | user-level privileges — and the first box is a USERBOX. Also: a <i>user</i> is someone who uses people                                                                                                                                                                                    |
| (the verb)             | <b>RUN</b>                     | you still run daemons. The title is what you are; the verb is what you do                                                                                                                                                                                                                 |
| Wild                   | <b>UNBOUND</b>                 | the exact antonym of BIND — an unbound socket is held by nobody                                                                                                                                                                                                                           |
| Fleeing                | <b>DETACHED</b>                | you break your connection to a running process                                                                                                                                                                                                                                            |
| Poké Flute             | <b>INTERRUPT</b>               | what wakes a blocked process                                                                                                                                                                                                                                                              |
| Gym                    | <b>BENCHMARK</b>               | what a gym actually is                                                                                                                                                                                                                                                                    |
| Battle (the challenge) | <b>ENGAGE</b>                  | added 2026-09-04. <i>The table never ruled on this one</i> , and 237's move is away from combat language — outsourced not defeated, HALTED not fainted, DETACHED not ran. <b>"Would like to battle" was the last of it left in plain sight.</b> Six letters for six, so nothing rewrapped |
| Badge                  | <b>CERT</b>                    | eight certifications                                                                                                                                                                                                                                                                      |
| Elite Four             | <b>THE REVIEW BOARD</b>        | beating them is passing peer review                                                                                                                                                                                                                                                       |
| Champion               | (incumbent)                    | Ty Clear is the prevailing paradigm, not a king                                                                                                                                                                                                                                           |
| Pokémon Center         | <b>CHECKPOINT</b>              | restore from a saved training state                                                                                                                                                                                                                                                       |
| Poké Mart              | <b>THE REPO</b>                |                                                                                                                                                                                                                                                                                           |
| PC storage             | <b>COLD STORAGE</b>            |                                                                                                                                                                                                                                                                                           |
| PROF.OAK               | <b>CRYSTAL / CRYSTAL CLEAR</b> | <b>no title.</b> See below                                                                                                                                                                                                                                                                |

One escaped, and a gender sweep found it. ViridianMart still said **PROF. CRYSTAL** — the clerk with the PACKAGE. Fixed 2026-08-29 to **CRYSTAL CLEAR**. Worth noting how it was caught: not by looking for titles, but while checking every gendered reference to her. **Passes find each other's misses.** | OAK's PARCEL | **PACKAGE** | a repo distributes packages. Also forced — **CRYSTAL's PARCEL** is 16 and items cap at 12 | | Poké Ball line | **USERBOX** → **ADMINBOX** → **SUPERBOX** → **ROOTBOX** | acquisition as privilege escalation; root access is the Master Ball | | Catching | **BINDING** | `bind()` — and *binding a daimon*, which is the literal ritual phrase | | Silph Scope | **RESOLVER** | a linker resolves a symbol to a name; *resolution*; and resolving an ambiguity | | Fainting | **HALTED** | | Evolution | **RECOMPILE** | |

Box names are tunable — the *privilege ladder* is the idea worth keeping. Success rate rising with permission level is thematically exact.

### 1.1 On the missing title

Implemented 2026-08-29. Both editions build; no OAK remains in any text file.



**Crystal has no title, and the absence is the point.**

4.1 says she spent her career arguing that machines have context, nobody funded it, and **she built the Index in order to be taken seriously**. Then 4.10 has a fitness-for-work procedure remove her from her own lab. A woman who had to build a taxonomy engine to be credible, and was then processed out of the building, does not get a PROF.

Every other authority in the game has a role — BENCHMARK leaders, the REVIEW BOARD, CORPUS. She is just her name. And it lands because vanilla players *expect* the title: the professor-slot character has no professor, and nobody remarks on it.

**The vanilla text split perfectly for it**, which is how you know it is the right call:

| Vanilla       | Ours          | Register                         |
|---------------|---------------|----------------------------------|
| OAK: — 19×    | CRYSTAL:      | dialogue prefix. People know her |
| PROF.OAK — 8× | CRYSTAL CLEAR | third person. The full name      |

First name when spoken to, full name when spoken about, and **no rank anywhere**.

**One line had to be rewritten rather than renamed**. The intro speech is the single place the title is claimed aloud:

|

*vanilla: My name is OAK! People call me the #MON PROF!*

There is no honest rename of that, because the second sentence *is* the rank. It became:

|

*My name is CRYSTAL CLEAR! I study #MON.*

Flat, factual, and it claims nothing. Placeholder-grade wording that should get a proper pass at step 8 — but the shape is correct: **where Oak asserts a title, she states an occupation**.

**Why the intro said #MON and not DAEMONS, and no longer does.** # is a control character that expands to POKé, and the species word has not been renamed yet — that is a bulk pass over the charmap and every string (9.2, step 8). Writing DAEMONS in one line while the other several hundred still say POKéMON would be worse than waiting. **The line became "I study DAEMONS."** the day the species rename landed, and nothing else about it changed. (Done 2026-08-29 — it needed the plural mark, per 1.2.)

*The cost, recorded:* every touched line needed rewrapping. Both replacements are longer ( `OAK:` → `CRYSTAL:` is +4, `PROF.OAK` → `CRYSTAL CLEAR` is +5) and Gen 1 text is hand-wrapped. Twenty-one blocks were rewrapped to stay inside the box.

**The ceiling, measured properly 2026-08-29 — and the earlier figure here was wrong.** The message box is declared `MESSAGE_BOX, 0, 12, 19, 17`, so its borders sit at x0 and x19 and text starts at x1: **the interior is 18 columns.** Vanilla reaches 19 exactly four times ("*It's very accurate*", "*That's odd, MR.FUJI*", "*You can't use items*", and one trailing space) — and each of those **overwrites the right border tile**. So the honest rule is **18 is clean, 19 is vanilla-grade jank**, not "19 is the ceiling". A control token like `<COLON>` or `<PK>` is **one tile**, which is what made the earlier count wrong.

*Code identifiers were left alone.* `OaksLab`, `OAKS_LAB`, `ProfOakName` and roughly 267 similar symbols are internal and invisible to the player, in the same category as the `_RED` / `_BLUE` defines (8.4). Renaming them is cosmetic and deferred.

**1.2 On the species rename — and the trick that made it free**

*Implemented 2026-08-29. Both editions build; zero overflows.*

**# is one byte.** `constants/charmap.asm` maps it to `$54`, and the text engine expands it from a single string:

```
home/text.asm:183      PlacePOKeText::      db "POKé@"
```

So `#MON` is four bytes rendering seven characters. **Change that one string to `db "DAE@"` and all 650 occurrences become DAEMON** — no source edits, no rewrapping.

The widths cooperate, which is the opposite of the 1.1 job:

|                | rendered |
|----------------|----------|
| POKéMON        | 7        |
| <b>DAEMON</b>  | <b>6</b> |
| <b>DAEMONS</b> | 7        |

Singular gains a character, plural is identical. **Nothing needed rewrapping.**

**What did need doing.** Fifty-four occurrences were not the species and had to become literals first, or they would have rendered as DAE-nonsense:



|         | Count | Became                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| #DEX    | 26    | literal <b>INDEX</b> — 7 chars to 5, so it also gains room                                                                                                                                                                                                                                                                                                                                                                                                                             |
| bare #  | 23    | literal POKÉ. <b>All 23 are item prefixes split across a line break</b> — # BALL, # DOLL, # FLUTE — not species, so the trick was safe                                                                                                                                                                                                                                                                                                                                                 |
| #MANIAC | 5     | <b>ARCHIVIST</b> , done 2026-08-31. <i>Maniac</i> is a pathology word and craft rule 3 says name the process — a game whose central injustice is a woman recorded as unwell cannot have a trainer class called a maniac. <b>ARCHIVIST pays twice</b> (1): an archivist keeps <b>records</b> , and an <b>archive</b> is stored compressed data, which is exactly what a box does to a daemon. It is also precisely right for HOLT, who built the storage and was then inside it (4.20a) |

The real work was grammatical number. POKÉMON was both singular and plural. DAEMON is not. Of the 650:

|                        | Count    |                                                                                           |
|------------------------|----------|-------------------------------------------------------------------------------------------|
| Compound               | 106      | #MON CENTER, GYM, LEAGUE, MART — not number questions. The lexicon renames these outright |
| Plural                 | 123      | marked #MONS                                                                              |
| Singular / attributive | the rest | no edit required                                                                          |

The default is free, and that is why this was tractable. English keeps attributive nouns singular — *DAEMON magazines*, *DAEMON trainers*, *DAEMON fights* — and singular is what #MON already renders. Only plurals needed touching, so an error of omission reads as a mild singular rather than as breakage.

Two heuristic passes were needed. The first over-marked: it read looks, is, was as evidence but also caught will and can, which are number-neutral. *My DAEMON looks stronger* is singular; *all DAEMONS will have weak points* is not. A corrector reverted eight, four of them wrongly, and a quantifier rule (all, some, up to 6 plus a modal) restored those.

The intro got its human read first, 2026-08-29 — on screenshots, which is the only reliable way to catch these. Six in Crystal's opening speech were reading singular and should not have been: *world of DAEMONS*, *I study DAEMONS* (twice), *creatures called DAEMONS*, and *adventures with DAEMONS*. All six are the same class of miss — vanilla wrote the singular because POKÉMON is a mass noun, and DAEMON is not. Marking them #MONS was the whole fix; the longest result is 17 characters and nothing needed rewrapping.

**A second class, found the same way, 2026-08-29.** Dialogue names types in lowercase — *the fire #MON, CHARMANDER* — and those words were never renamed with the chart. The starters are fixed (**ENTROPY, FLOW, GROWTH**, since vanilla says *plant*, not *grass*). **Nineteen more remain, nearly all inside gyms:** *electric #MON, Psychic #MON, rock-type #MON, fighting #MON*. They are deliberately left — a Benchmark's type is 5's design work, and renaming the dialogue before the leaders are settled would just be done twice. Separately, **42 occurrences of *fight*** want the same human read; some are the verb we replaced, some are people (*not a fighter, a fighting game*).

**Still open:** roughly 154 occurrences carry no strong evidence either way and currently read singular. Most are attributive and correct as they stand, but the set wants the same treatment — read on screen, not in the source.

### 1.3 On the boxes, and what binding actually is

**A box is a machine.** Not a metaphor we are imposing — it is sysadmin vernacular. *I sshed into the box.*

So the ladder is not a set of cages of increasing strength. **You are offering the daemon a host.** A daemon is a process; a process needs somewhere to run. It stays if that host grants the privileges it needs.

Which makes the catch rate literal rather than figurative: **an unbound daemon will not run on a box where it only has user rights.** ROOTBOX takes anything because root takes anything. The suffix was already the right word before we knew why.

### 1.4 On the battle vocabulary

The gyms were already BENCHMARKS, so a battle was already a **run** — you put a process against a workload and see what it does. Two decisions fell out of that.

**RUNNER became USER**, because RUN was doing two jobs and colliding with itself. Vanilla's own text proves it: "*No! There's no running from a trainer battle!*" while the player is a RUNNER. USER does three jobs instead:

- **Technical** — user-level privileges, and the first box is a **USERBOX**. Your title and the bottom rung of the ladder are the same word. You start as a USER with a USERBOX and escalate.
- **Moral** — a *user* is someone who uses people. Unaware selfishness: `oPerson` from 2011, and the implicit-selfishness thread from 2021.
- **Practical** — it frees RUN entirely.

**Nothing was lost.** *You run processes* relocates from the noun to the verb: **you are a USER, and you RUN daemons.** The title is what you are, the verb is what you do, and neither collides.

Trainer battles are **BENCHMARKS** too, and that is a scale distinction rather than a collision — exactly as in machine learning. You benchmark constantly; **THE BENCHMARKS** are the eight formal ones that issue CERTs. A trainer battle is an informal benchmark, which is literally what it is.

**DETACHED** has no inverse, deliberately. Fleeing is breaking your connection to a running process, and the process is still running — you simply stopped observing it. **EXITED** would be wrong, because *you* did not exit. And **ATTACHED** for encounters would imply a connection you had not made; **appeared** already covers it.

The strings, as implemented 2026-08-29:

| Vanilla                                       | Ours                                                         |
|-----------------------------------------------|--------------------------------------------------------------|
| Wild <nick> appeared!                         | UNBOUND <nick> appeared!                                     |
| <nick> fainted!                               | <nick> HALTED!                                               |
| Enemy <nick> fainted!                         | Enemy <nick> HALTED!                                         |
| Got away safely!                              | DETACHED.                                                    |
| No! There's no running from a trainer battle! | No! You can't DETACH from a USER's run!                      |
| Go! <nick>                                    | RUN, <nick>                                                  |
| <TRAINER> wants to fight!                     | <USER> wants to BENCHMARK! <i>(pending the trainer pass)</i> |
| #MON are pets. Others use them for fights.    | DAEMONS are companions. Others BENCHMARK them.               |
| Enemy <nick> fainted!                         | Remote <nick> HALTED!                                        |
| Enemy <nick> ran!                             | Remote <nick> DETACHED.                                      |
| The enemy's weak!                             | The remote's weak!                                           |
| <PLAYER> defeated <TRAINER>!                  | <PLAYER> outscored <USER>!                                   |

On *pets*, and the precedent that decided it. Crystal's opening speech drew a contrast — kept for affection, or used for gain — and vanilla's affection word was **pets**. 1.5 rejected *caught* for being "a word about grabbing an animal"; **pets** is a word about keeping one, and fails the same test for the same reason. **companions** keeps the register without the menagerie.

Why not **assistants**. Tempting, and the contemporary word — but it collapses the sentence. Vanilla's line only works because its two halves oppose each other, and an assistant is *already* being used, so "assistants... others use them" stops being a contrast at all. It is also the nearest the first minute could come to saying the thesis out loud (craft rule 1), and it pins the game to one year's discourse. **Held in reserve**: if it is ever taken, the second half has to change with it.

And *fights* becomes **BENCHMARK**, which is what this world calls the thing. Vanilla teaches "fights" in the first minute; we teach the real verb in the same breath, three lessons before the player needs it.

*The prefix was the one that mattered, and the first pass missed it.* Two standalone strings said *Enemy*; the **third** is `home/text.asm's EnemyText:: db "Enemy @"`, which is what every `<USER>` and `<TARGET>` expands to for the opposing side — so it appears in far more messages than the other two combined. Now `Remote @`.

**And it sets a naming budget.** Thirty-two strings put text straight after `<USER>` on the same line, the worst being `<USER>'s`. Vanilla fits exactly: `Enemy` + a 10-character name + `'s` = **18**. *Remote* is one longer, so the same line is **19** — over the clean edge. **Species names should therefore be capped at 9 characters** (9 gives 18 and fits). This costs us nothing we were not already choosing: 9 renames the bestiary anyway, and vanilla's own longest are 10. A player who *types* a 10-character nickname will push one character over the border in some messages — self-inflicted, self-healing, and no worse than what vanilla does four times on its own.

*On enemy, replaced by REMOTE.* Nothing in this world is anyone's enemy — the opposing daemon is a process bound to somebody else, and *enemy* imports a hostility the lexicon never claims. **REMOTE** is the exact technical word for a process you have no handle on, it sits on the same axis as **DETACH**, and it is attested as a bare noun in precisely this register (*the remote is down*). It is one character longer than *enemy* and the nickname occupies its own line, so nothing rewrapped.

*And it caught a live collision.* `Enemy <nick> ran!` was still using **ran** — the word 1.4 went to some trouble to free. It is now `Remote <nick> DETACHED.`, which also quietly teaches that detaching is something either side can do.

*On defeated, replaced by outscored.* A trainer battle is a **BENCHMARK**, and a benchmark yields a **score**. *Defeated* is the language of combat; *outscored* is the language of evaluation, which is what the Review Board will later do to the player. **outran** was considered and rejected — it reaches for the racing sense of **RUN**, and 1.4 spent real effort making **RUN** mean *execute*.

## **On money — CACHE, and how little of it is visible**

**The word money is never shown to the player.** It appears only as labels and variable names; on screen there is the ¥ glyph and digits, nothing else. The single prose instance in the whole game is Team Rocket in Mt Moon — *"revive and sell them for cash!"* — which is now **CACHE**, and is exactly the right mouth to put it in.

**CACHE, because it is not only a pun.** A cache is, in plain English, a hidden store of valuables — *a cache of gold* — before it is anything to do with computers. So it carries three readings at once: the hoard, the fast local memory, and *cash*, which it is pronounced as. That is one more than 1 asks of any term, and unusually it needs no beat of confusion: a player reads it as money immediately.

*Considered:* **CYCLES** — compute time as the scarce resource, with a bonus nod to the loops in 0.3. Genuinely good, and rejected only because CACHE reads instantly and CYCLES needs a beat, in a slot where the player is doing arithmetic rather than reading.

**The ¥ glyph stays**, and the divergence is smaller than it looks. ¥ is one tile — charmap "¥", \$f0 — and vanilla was *already* using it as a stand-in for an invented currency. Redrawing it is an afternoon that buys almost nothing, because the mark is read as *currency* whatever it looks like and the word does the work everywhere it appears. **Logged as optional art, not as a rename.**

*Untouched, deliberately:* the Game Corner's **coins** are a second currency and stay coins.

**The battle menu is a live collision, and it is engine work**

data/text\_boxes.asm still reads:

```
BattleMenuText:
    db  "FIGHT <PK><MN>"
    next "ITEM  RUN@"
```

*Measured exactly:* the left cursor is at x9 and its text runs **x10–x14, five characters**; the right cursor is at x15 and its text runs **x16–x18, three characters**. So **FIGHT** and **RUN** both fit their slots precisely, and **nothing longer does** — not **DETACH** (6), not **INVOKE** (6). Widening means moving the cursor writes *and* the box origin.

**FIGHT is a word we replaced, and RUN there means flee** — which is precisely the collision 1.4 renamed RUNNER → USER to remove. It is also the single most-seen string in the game.

*Why it is not a text edit.* The box is declared **BATTLE\_MENU\_TEMPLATE, 8, 12, 19, 17, BattleMenuText, 10, 14** — spanning x8–19 with text at x10, so there are **nine columns**, split into a **5-character left column** and a **3-character right column** at x16–18. **DETACH is six characters and physically cannot go where RUN is**. Moving the split means rewriting the `wTopMenuItemX` values at four sites in `engine/battle/core.asm` plus the 2×2 cursor logic. That is a real change, correctly sized, and it should be made deliberately rather than folded into a text pass.

## Three that were interrogated and kept

**RUN, <nick>** keeps its comma. A colon would make it a command echo, which is tempting — but this string has three siblings, **Do it! @**, **Get'm! @** and **The remote's weak! Get'm! @**, which vanilla selects between by situation. The slot is **the player's voice**, not the system's. Colon-ing one of four would leave a command echo standing next to two shouts, and that reads as an error rather than a choice. *If the colon is ever taken, all four have to flatten together — and vanilla's variety goes with them.*

**used** stays, and for the same reason EXP does. 1.4 made **use** a morally loaded word on purpose — *a user is someone who uses people* — so every time a daemon *uses* a move, the game repeats the word that names what the player is doing to the daemon. Trading that for a merely technical verb would be a downgrade in the one place this game is trying to implicate someone. *And the tempting alternative does not fit — measured twice, in both places it could have lived.* **INVOKED** would have been the exact partner to **BIND**: you **bind()** a daimon and you *invoke* it, both idioms true twice over.

- **As the verb.** The message is **used <MOVE>!** and the longest move name is 12, so vanilla sits at  $5 + 12 + 1 = 18$ , exactly the box. **invoked** gives 21, and **called** gives 20. Both past even the jank line.
- **As the battle menu's FIGHT.** The left column runs x10–x14 — **five characters** — because the right column's cursor is drawn at x15 (**ldcoord\_a 15, 14**). **INVOKED** is six, so it would sit under the cursor.

The word is genuinely better than **used**. It simply has nowhere to go that does not cost either a clobbered border or an engine change, and **used** is carrying real weight already.

**ATTACK** stays too, as a *stat*. It is a magnitude rather than a narrative word, it belongs to a fixed set (**HEALTH / ATTACK / DEFENSE / SPEED / SPECIAL**) that would have to move together, and the stat screen's columns are fixed-width. The word that actually needs work is **FIGHT**, and it is in the battle menu, above.

**EXP** stays **EXP**, and it is not a compromise. 4.3 says context forms from **experience** and nowhere else — that is the argument the whole rival plot turns on. The quantity the game already accumulates is called experience. **The best resonance in the system is the one vanilla handed us**, and renaming it would be the only change in this document that made the game say less.

**LEVEL** stays **LEVEL**, for two reasons. It already does the double duty 1 asks for: 1.5 has you gaining access "*at a **permission level** your box determines*", so the word is simultaneously the RPG stat and the privilege tier — you escalate. And practically, the HUD does not spell it: **<LV>** is a **single tile**, **\$6e**, drawing **:L**. Any replacement needs new art before it needs a decision.



On punctuation: LABL was BOUND. is flat because the game declines to congratulate the player for acquiring something. UNBOUND RATTATA appeared! is alarm, not congratulation — the rule is about refusing to celebrate, not refusing to punctuate.

Done 2026-08-29. The 379 figure counted code identifiers; in player-visible strings it was 93, across six casings ( trainer , trainers , TRAINER , Trainer , Trainers , TRAINERS ). All became USER / USERS, and since USER is shorter than trainer nothing needed rewrapping.

USER TIPS

are all USERS!

for USERS!

The trainer class names — YOUNGSTER, BUG CATCHER, LASS — are untouched. They are occupations, not the general term, and they belong to the bestiary-adjacent pass.

### 1.5 On BINDING

The container was renamed and the verb was not, which left the lexicon saying *privilege escalation* and then *caught* — a word about grabbing an animal.

**What the act actually is:** you weaken a background process and gain persistent access to it at a permission level your box determines. That is not capture. It is a handle on something that was already running.

The ladder was built 2026-08-29, and it had never been built at all. 1.1 settled USERBOX → ADMINBOX → SUPERBOX → ROOTBOX early and nothing implemented it — the items were still POKé BALL, GREAT BALL, ULTRA BALL and MASTER BALL, and 36 player-visible strings still said *ball*, across 19 files. Found because a playtest walked up to a lab table and was told "*Those are POKé BALLs.*"

SAFARI BALL became GUESTBOX, which the ladder implies and nobody had written down: the Safari Zone is where you are given restricted, temporary, expiring access. A guest account.

And one line was actively arguing against this section. "*Those are POKé BALLs. They contain #MON!*" — but a box does not contain. It hosts. So:

*Those are BOXes. DAEMONS run on them.*

That is 1.3 delivered at the player's first sight of the object, in nine words, explaining nothing. The same fix applies to "*They are inside the POKé BALLs*" → "**They are running on the BOXes.**"

Three catch instances were swept in the same lines (1.5's verb), and 57 remain — that pass was never done outside the battle messages.



**The object, built 2026-08-29.** `tools/genbox.py` replaces the two ball sprites — `gfx/sprites/poke_ball.png` (16×16, the box on the ground) and `gfx/battle/balls.png` (32×8, four throw frames). Dimensions match vanilla exactly, so no engine code moved.

**It is a small server, not a crate and not a cube.** Both of those say *container*, which is the reading this section replaced — a container shape would quietly undo the rename. So: hard corners against the ball's circle, a bevel so it reads solid, two vents, one indicator light, and feet. Something you would `ssh` into.

**The throw stays one object.** Closed with a visible seam → struck solid → **pulled apart**, the two halves separating rather than a new shape appearing → dispersed. Vanilla's sequence turns a ball into a burst; ours opens a machine.

**And the ladder is a parameter.** `overworld(tier)` raises the vent count, so `USERBOX` → `ROOTBOX` is one argument rather than four drawings. Privilege reads as density.

*Generated rather than drawn*, for the reason `sprite-prompts.md` gives: at 16×16 and 8×8 this is geometry, and downsampling a generated illustration to that size produces mush. Same argument as the SGB borders, same outcome.

**Why BIND and not something else.** Every other entry in the table above does double duty — `DAEMON`, `CHECKPOINT`, `RECOMPILE`, `HALTED`. `CATCH` does none. `BIND` does three ways: `bind()` and binding a name or a port; **binding a daimon**, which is the exact ritual phrase and not a gesture at one; and a *bond*, being bound to someone. It fits the ladder precisely — you do not overpower a process, you bind it at a permission level, and `ROOTBOX` binds anything. `BIND` is 4 characters and `BOUND` is 5, so it fits every box that held "caught."

**The test it had to pass was not "is it nicer." It was "is it softer."** The sharpest thing this game does is implicate the player: you fill an Index that cannot hold what matters, and you box creatures whose entire significance is that they are not inventory — which is Scorn's exact error (4.4). A euphemism here would stop the game accusing the player and start it flattering them.

`BIND` is not softer. **Binding a spirit is a darker act than catching an animal.** Catching is sport; binding is a contract you imposed. The complicity survives and a second reading arrives with it.

**And the success message is flat, 2026-08-31.** Vanilla says *All right! was caught!* — the exclamation congratulates the player for imposing the contract. Ours reads:

| `<NAME>` was `BOUND`.

*Nothing else changed.* The failure lines keep their *Darn!* and their *Shoot!*, because 1.4's rule **refuses to celebrate, not to punctuate** — a lament is not a celebration. **Only the moment of binding goes quiet.**

*Considered and rejected:* **ATTACH** is the most technically exact — you attach to a process that is already running, which is what a daemon is — and it carries *attachment* in the psychological sense. Cut for being too gentle: attaching is observational and does not implicate you. **Hold it in reserve if playtesters find BIND too dark.** **REGISTER** would rhyme with the Index but conflates the box with the Index, and those two artifacts must stay separate — one measures, the other holds. **CLAIM** is acquisitive with no second reading, which is strictly worse than **CATCH**.

**The message register.** Vanilla is "*Gotcha! X was caught!*" — triumphal, and wrong for this world twice over. Drop the exclamation and let it sit in the Index's bureaucratic register:

*LABL was BOUND.*

Flat. No congratulation. **The game declining to celebrate is more unsettling than any line of dialogue about it would be.**

*Cost, accepted:* every Gen 1 player knows "caught," so a new verb costs a beat of confusion the first time. That beat is the point — it is the first moment the player notices this world does not share vocabulary with the one they expected. It lands in good company: Crystal asks what you will *call* him (4.3), Route 1's signpost gives a number and a name that disagree (3.2), and the first bind says the verbs are different too. Three lessons in the first fifteen minutes, none explained.

---

## 2. THE TYPE SYSTEM

---

Gen 1 has fifteen usable type slots, no abilities, no held items, and one Special stat. That constraint is the design: **the type chart is the entire mechanical philosophy of the game.**

### 2.1 The engine constraint you cannot design around

Gen 1 determines physical vs. special by **type ID range**, not per move:

- \$00 – \$08 → physical (Attack / Defense)
- \$14 – \$1A → special (Special / Special)

That is 8 physical slots and 7 special slots, fixed. Which types are special is therefore a design decision made *by where you place them in the constant list.*

## 2.2 The fifteen

| Vanilla  | New type        | Concept                             | Class    |
|----------|-----------------|-------------------------------------|----------|
| Normal   | <b>CONTENT</b>  | tokens, symbols, literal data       | Physical |
| Fighting | <b>LOGIC</b>    | symbolic rules, brute force, proof  | Physical |
| Flying   | <b>VECTOR</b>   | embeddings, latent space, direction | Physical |
| Poison   | <b>CORRUPT</b>  | bias, poisoned data, hallucination  | Physical |
| Ground   | <b>STRATUM</b>  | memory, storage, the physical layer | Physical |
| Rock     | <b>LEGACY</b>   | silicon, deprecated hardware        | Physical |
| Bug      | <b>SWARM</b>    | multi-agent, distributed (and bugs) | Physical |
| Ghost    | <b>LATENT</b>   | dormant processes, the unconscious  | Physical |
| Fire     | <b>ENTROPY</b>  | noise, temperature, exploration     | Special  |
| Water    | <b>FLOW</b>     | gradients, backprop, descent        | Special  |
| Grass    | <b>GROWTH</b>   | training, reinforcement, fitting    | Special  |
| Electric | <b>SIGNAL</b>   | raw input, sensors, current         | Special  |
| Psychic  | <b>CONTEXT</b>  | framing, salience, affect           | Special  |
| Ice      | <b>FROZEN</b>   | overfit, brittle, hard-coded        | Special  |
| Dragon   | <b>EMERGENT</b> | rare, unaccounted-for, AGI-tier     | Special  |

**On STRATUM.** The original choice was SUBSTRATE, cut for a hard reason: Gen 1's longest type strings are FIGHTING and ELECTRIC at 8 characters, and SUBSTRATE is 9. It overflows the battle status box and the Index type field. STRATUM keeps the layer meaning, keeps the geological register, and lands at 7.

Character counts, all verified: CONTENT 7, LOGIC 5, VECTOR 6, CORRUPT 7, STRATUM 7, LEGACY 6, SWARM 5, LATENT 6, ENTROPY 7, FLOW 4, GROWTH 6, SIGNAL 6, CONTEXT 7, FROZEN 6, EMERGENT 8 — exactly at the limit.

## 2.3 What Kanto already encoded

Because we rename **constants**, every existing line in `type_matchups.asm` carries over and becomes a line in our chart. The thesis is already there:

| Vanilla line          | Becomes                      | Meaning                                    |
|-----------------------|------------------------------|--------------------------------------------|
| FIGHTING, NORMAL, 20  | LOGIC beats CONTENT ×2       | Rules parse data brilliantly               |
| FIGHTING, PSYCHIC, 05 | LOGIC resisted by CONTEXT ×½ | <b>The thesis.</b> Free.                   |
| NORMAL, GHOST, 00     | CONTENT cannot touch LATENT  | Literal data cannot reach the unconscious  |
| FIGHTING, GHOST, 00   | LOGIC cannot touch LATENT    | Symbolic rules bounce off it               |
| FIRE, ICE, 20         | ENTROPY thaws FROZEN         | Temperature breaks an overfit model        |
| POISON, GRASS, 20     | CORRUPT beats GROWTH         | Poisoned data ruins training               |
| WATER, FIRE, 20       | FLOW beats ENTROPY           | Gradient descent tames noise               |
| BUG, PSYCHIC, 20      | SWARM beats CONTEXT ×2       | Collectives destabilise individual framing |

Do not rewrite the matchup file. Patch it.

### 2.4 The only deltas

Delete the Gen 1 developer error:

```
db GHOST,    PSYCHIC,    00
```

Ghost moves were meant to be super effective against Psychic; the table says *no effect*. We want the opposite of the bug.

Add the mutual pair:

```
db CONTEXT,  LATENT,    20
db LATENT,   CONTEXT,    20
```

Framing is how you reach what runs below the surface; what runs below the surface destabilises framing. Mutual 2× is unusual and correct here — these two interpenetrate and neither has a safe angle on the other. It also makes the Halftone Tower stretch genuinely dangerous instead of a LATENT walkover.

Two additions, one deletion. That is the whole chart change.

### 2.5 The SWARM problem is a MOVE problem

Gen 1's Bug type is weak, but **not because of the chart** — `BUG, PSYCHIC, 20` is already there. It is weak because every Bug move is unusable: Twineedle, Leech Life, Pin Missile, String Shot. The best has 25 base power.

So the CONTEXT balance fix belongs in `moves.asm`.

**New move — CONSENSUS.** SWARM type, 90 power, 100% accuracy, 15 PP, no secondary effect. Deliberately boring: SWARM's job is to be a *reliable* check on CONTEXT, and reliability is the point. A swarm does not need a gimmick, it needs to keep showing up. Give it to two or three mid-game encounters and one Review Board coverage slot.

*Tuning note:* 90 BP with no drawback is strong for Gen 1. If CONTEXT still runs away with the late game, raise CONSENSUS before touching the chart. Moves rebalance far more easily than matchups.

**Built 2026-08-29.** CONSENSUS is move \$A5 , 90 power, SWARM, 100%, 15 PP, NO\_ADDITIONAL\_EFFECT — verified by reading it back out of the ROM.

*Where it had to go, and why.* engine/battle/core.asm asserts NUM\_ATTACKS == STRUGGLE , because the AI treats any random number above STRUGGLE as "not a move". So a new move cannot be appended; it has to be **inserted immediately before STRUGGLE**. CONSENSUS therefore takes \$A5 and STRUGGLE shifts to \$A6 . Every other move ID is untouched, which is what makes this safe.

*Five tables, not two.* 9.2 step 6 says moves.asm and names.asm ; the build needs three more, each assert\_table\_length NUM\_ATTACKS — constants/move\_constants.asm , data/moves/animations.asm and data/moves/sfx.asm . CONSENSUS borrows PIN\_MISSILE's animation and sound: converging projectiles, which is the right picture for a swarm agreeing. Note that moves.asm still names the type BUG — per invariant 6, only the *string* is SWARM.

**Not yet learnable.** Nothing in the game can use CONSENSUS until a learnset entry exists. That is deliberate: assigning it means naming species, which is 9 and 11's work. The move is in the ROM and correct; it is waiting on the encounter design, not on more engine work.

---

### 3. THE MAP

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#### 3.1 Cities

| Kanto          | New name                  | Double meaning                              | Role                                 |
|----------------|---------------------------|---------------------------------------------|--------------------------------------|
| Pallet         | <b>Blanche Town</b>       | white/blank + to flinch                     | Home; the pre-color state            |
| Viridian       | <b>Callow City</b>        | green + unripe, untested                    | Benchmark 8 (locked all game)        |
| Pewter         | <b>Slate City</b>         | stone, color, writing surface, clean slate  | Benchmark 1; museum of dead hardware |
| Cerulean       | <b>Doldrum City</b>       | becalmed sea + low spirits                  | Benchmark 2                          |
| Vermilion      | <b>Ardor City</b>         | flush/heat + brash zeal                     | Benchmark 3; the port                |
| Lavender       | <b>Halftone Town</b>      | dots that only <i>look</i> like grey        | The tower                            |
| Celadon        | <b>Verdigris City</b>     | green corrosion on bronze                   | Benchmark 4; Corpus rotting beneath  |
| Fuchsia        | <b>Lurid City</b>         | garish glow + shocking                      | Benchmark 5; spectacle and toxicity  |
| Saffron        | <b>Brazen City</b>        | brass over base metal + shameless, unhidden | Benchmark 6; corporate capture       |
| Cinnabar       | <b>Quicksilver Island</b> | mercury: alive, unstable, poisonous         | Benchmark 7; the ruined lab          |
| Indigo Plateau | <b>Umbra Plateau</b>      | full shadow; all color absorbed             | The Review Board                     |

**Slate over Somber.** Somber beside Doldrum is two downbeat mood words in a row — it greys out the early game. Slate breaks the run, is materially specific, and a slate is a writing surface, which is Benchmark 1's Representation lesson sitting inside the name.

**The Tunnel, and the thirsty guard.** Two places where vanilla needed almost nothing.

**The Underground Path is already correct.** In networking a **tunnel** is precisely how you route traffic around a blocked path — an SSH tunnel, a VPN tunnel. You cannot take the direct route, so you tunnel. It needs no rename, only the joke made visible in the local-name register: *official* UNDERGROUND PATH, *local* THE TUNNEL.

**And the guard stays thirsty.** Craft rule 6 doing its work: Brazen is the bought city, Corpus headquarters, corporate capture made architecture — and it is guarded by a man who will move for a beverage. **The barrier is not security. It is inertia.** Nobody says so, and it lands harder for being literal.



**Halftone.** A halftone looks like continuous grey from a distance and is actually discrete black and white dots up close; the gradient is an artifact of sampling. That is the man/machine problem exactly — the bias comes from **resolution**, **not malice**. Neutral, not dull, and nobody is unwell. *Alternates held in reserve: Penumbra Town* (partial shadow; also the legal sense of implied edges — would require renaming Route 23) and **Moiré Town** (interference from two grids slightly out of register).

**Quicksilver.** Cinnabar is mercury sulfide, and alchemists took mercury as an elixir of vitality — it made you tremor, then it made you mad. Quicksilver keeps mercury, a real color, *quick* meaning alive, and *mercurial* meaning unstable. The island promised to make things stronger. It made S.T.A.R.R., and it burned down. It also pre-echoes the humors payoff at Umbra, since mercury was the other ancient system's material. (*Deeper cut in reserve: Realgar Island* — arsenic sulfide, a ruby pigment prized by painters, quietly poisoning everyone who ground it.)

**Brazen, over Gilt.** Both names say *cheap metal dressed as precious*. They differ on whether anyone is hiding it.

*Gilt* is gold leaf over base metal: microns thick, applied to conceal, and it carries "guilt" in the ear. It implies a **concealer** — someone who knew the substance was hollow and covered it anyway. This project does not have one. It would also quietly assign guilt to a man who feels none, and craft rule 3 says name the process, not the pathology.

*Brazen* is brass: an alloy, honestly itself, gold-colored the whole way through and worth a fraction as much. Nothing is concealed because nobody is concealing. The city is exactly what it appears to be and says so. **That is Scorn's city.** He is not covering anything up — he is *pleased with the arrangement* and will explain it to you warmly. Open capture, with the warning cast into the lobby floor (4.4).

The insidious reading also mis-sequences the arc. If Brazen is a concealment, the player's job there is to uncover something, and the reveal does the work for them. If Brazen is open, **there is nothing to uncover** — the player walks through a city that has already agreed, meets a man who is glad to see them, and has to locate their own objection unaided. Much harder to shake, and it is what 4.4 asks for: finish his benchmark unsure whether you disagreed with him or merely out-argued him.

*Cost of the swap:* the "guilt" homophone, which was good, and a small friction with craft rule 3 — *gilt* is a process done **to** a place, while *brazen* leans toward describing what a place **is**. **Fix in-world:** use brazen in its literal material sense. Signage, the guidebook and the museum plaque talk about the brass, the plating works, the alloy. The attitude sense is the second read and arrives on its own.

*Held in reserve: Brass City* — same metal, none of the attitude. **Interrogated and rejected:** every other city on the map carries two readings, a colour and a feeling. Brass is a colour and not a feeling, and would be the only single-meaning name here — breaking the device precisely at the city where surface-versus-substance is the subject. Kept in reserve against exactly one condition: if Brazen ever reads as the game sneering at Scorn, swap it that day. Gilt is retired, not deleted; see 10.

**Blanche and Halftone** are the two poles of one question: the colorless place you start from, and the place that cannot agree whether what it sees has color at all.

**Two names flagged as deliberate, not accidental.** *Lurid* originally meant sickly-pale before drifting to garish; the modern sense dominates, so keep it knowingly.

**Doldrum stays, and the reason is better than "it also means a color."** The doldrums are not a mood — they are a **place**, the equatorial belt where the trade winds fail and sailing ships sat for weeks. That is materially specific in exactly the way craft rule 4 asks for, and it is a *location* word, which is what a city name should be.

Then it earns its keep twice more. Benchmark 2 is FLOW — gradient descent — and **being becalmed is the failure state of gradient descent.** No slope, no movement, stuck in a local minimum, and no amount of effort helps because the problem is that there is nothing to descend. The name is the lesson, the way Slate's is. That satisfies craft rule 5 in the name itself, which almost nothing else on the map manages.

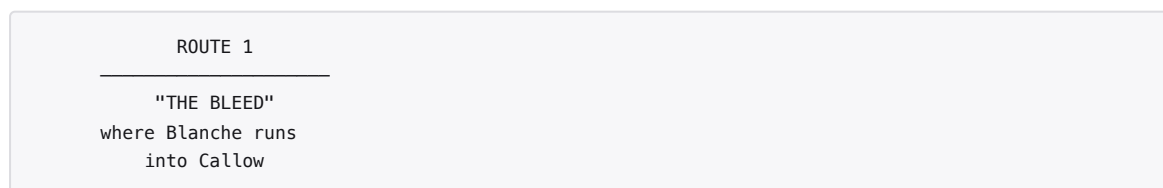
The original objection was really an objection to *Somber beside Doldrum* — two downbeat mood words in a row. Slate fixed that. Doldrum standing alone is fine.

*One oddity, knowingly kept:* the idiom is *the doldrums*, plural. **Doldrum City** is a singular back-formation. Place names do this constantly and it reads as a proper noun by the second glance.

### 3.2 Routes — the institutional map vs. the local one

Cities are colors. **Routes are what happens between colors** — bleeding, fading, washing, casting. Which is where the theory lives: routes are where context is *changing*.

**The device:** the official map keeps the numbers. Signposts carry the local name.



The institution navigates by content. The residents navigate by context. The player learns both and starts using the second. Index-blindness extended into cartography, for the price of one sign-text pass.

**Built 2026-08-30 — thirteen signs, every route that has one.** The official plate is untouched; the local name arrives underneath it as **Painted beneath:** . The institution's name is cast into the sign and the residents' name is added to it, which is the difference the section is about, made physical for free.

*Routes 16–18 all read **THE STREAK** and that is not an error — it is one cycling road with three signs on it.*

| #     | Local name                                       | Connects                 | Note                                                    |
|-------|--------------------------------------------------|--------------------------|---------------------------------------------------------|
| 1     | <b>The Bleed</b>                                 | Blanche → Callow         | First color running out of home                         |
| 2     | <b>Underpaint</b>                                | Callow → Slate           | The layer beneath the visible one                       |
| —     | <b>The Undertone</b>                             | (Viridian Forest)        | Green half-light; first real maze                       |
| 3     | <b>Ashfall</b>                                   | Slate → Deadstack        | Grey drift                                              |
| —     | <b>Deadstack</b>                                 | (Mt. Moon)               | A mountain of dead hardware; fossils are legacy silicon |
| 4     | <b>The Wash</b>                                  | Deadstack → Doldrum      | Color thinned with water                                |
| 5     | <b>The Fade</b>                                  | Doldrum → Brazen         |                                                         |
| 6     | <b>The Flush</b>                                 | Brazen → Ardor           | Heat rising toward the port                             |
| 7     | <b>The Glaze</b>                                 | Brazen → Verdigris       | Transparent coat over what is underneath                |
| 8     | <b>Overcast</b>                                  | Brazen → east            | Light without a source                                  |
| 9     | <b>The Dapple</b>                                | Doldrum → Blackout       | Broken light                                            |
| 10    | <b>Gloaming</b>                                  | Blackout → Halftone      | Dusk approaching the tower                              |
| —     | <b>The Blackout</b>                              | (Rock Tunnel)            | No light, no color, no context                          |
| 11    | <b>Sheen</b>                                     | Ardor → east             | Surface shine                                           |
| 12–15 | <b>The Drift / Slack / Brackish / The Muddle</b> | south to Lurid           | Colors mixing to mud                                    |
| 16–18 | <b>The Streak</b>                                | Verdigris → Lurid        | Cycling road; speed as smear                            |
| 19–21 | <b>Seafade / The Deep Wash</b>                   | south sea to Quicksilver |                                                         |
| 22    | <b>The Draining</b>                              | Callow → Umbra           | Color leaving                                           |
| 23    | <b>Penumbra</b>                                  | approach to Umbra        | Partial shadow                                          |
| —     | <b>The Umbral Ascent</b>                         | (Victory Road)           |                                                         |

On "The Bleed." Worth interrogating, because in isolation the wound sense is loud. It survives, for three reasons.

**The register decides it.** *Bleed* is a printing term before it is anything else here — ink carried past the trim edge so no white shows at the margin, and *color bleed*, where one ink migrates into the one beside it. Sitting in a list that reads Underpaint, Wash, Glaze, Overcast, Halftone, the printing sense is the one that arrives first. Words are read in the company they keep, and this one keeps extremely specific company.

**The map already does this on purpose.** *The Flush*, Route 6, is blood rising to the face, and it runs into Ardor, whose name is heat and flush. *Verdigris* is corrosion. *Lurid* was a sickly pallor. This cartography has used the body's color words for what happens to surfaces from the beginning. The Bleed is not an exception to the scheme — it is the scheme, first instance.

**The position earns it.** Route 1 is the first step outside Blanche, the blank colorless place, and it is where color starts running out of home into whatever is next. A faint disquiet on the first road out the door is correct. Not gore — a page edge where the ink did not stop.

*Held in reserve, if it ever reads as a wound:* **The Feather** (feathering — softening an edge so two areas merge) and **The Bloom** (bloom — the haze that rises on a printed or varnished surface as it cures). Both gentler. Both weaker. Do not trade down without a reason.

---

### 3.3 The rename, as built

*Implemented 2026-08-29. Both editions build; zero overflows; no vanilla place name remains in any text file.*

**Fourteen names in `data/maps/names.asm`, and 143 references swept through dialogue.** Only two lines needed rewrapping — because **most of the new names are shorter than the old ones:**

|                        | Δ            |
|------------------------|--------------|
| VERMILION → ARDOR      | −4           |
| CINNABAR → QUICKSILVER | +3           |
| CELADON → VERDIGRIS    | +2           |
| PALLET → BLANCHE       | +1           |
| everything else        | 0 or shorter |

Three names grew, and only VERDIGRIS actually broke a line. That is the opposite of the OAK job (1.1), where every replacement was longer and twenty-one blocks needed rewrapping.

**The landmarks came too**, since they carry the same vocabulary:

|                 |                                                                                                            |
|-----------------|------------------------------------------------------------------------------------------------------------|
| VIRIDIAN FOREST | THE UNDERTONE                                                                                              |
| ROCK TUNNEL     | THE BLACKOUT                                                                                               |
| VICTORY ROAD    | UMBRAL ASCENT                                                                                              |
| CINNABAR ISLAND | QUICKSILVER IS. — 15 characters, exactly what CINNABAR ISLAND was, so the location banner is proven to fit |

**Routes were deliberately not touched.** 3.2's device is that *the official map keeps its numbers* — ROUTE 1 stays ROUTE 1. The Bleed, Underpaint, Ashfall and the rest are **signpost content**, which is new writing rather than a rename, and it is the part that actually teaches the player the institution navigates by number and the residents do not.

**Still to do in this pass:** 379 occurrences of *trainer* → USER (1.4), and the route signposts themselves.

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## 4. CAST

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### 4.1 Crystal Clear — the Oak slot

Golden-amber fox, drawn from the SHC / iASHC universe.

She spent her career arguing that machines have *context*, not merely content. Nobody funded that. What she could get funded was a taxonomy engine — **so she built the Index to be taken seriously.**

| Oak beat          | Crystal Clear version                                                             |
|-------------------|-----------------------------------------------------------------------------------|
| Opening monologue | States the thesis in the abstract; the player has no frame for it yet             |
| Gives the starter | She lets the daemon choose <i>you</i> — she would never assign one                |
| Gives the Index   | Handed over with visible ambivalence                                              |
| Rival framing     | A research disagreement that is also a family one, two generations down — see 4.3 |
| Hall of Fame      | She records you into a system that flattens you into stats                        |

**She was at Quicksilver.** She built S.T.A.R.R. there, and a fitness-for-work procedure removed her from the building before it finished. See 4.10.

**The Index is what she did next.** This is the causal chain the design was missing: she lost the lab, lost the work, and then built a taxonomy engine *in order to be taken seriously* — because being taken seriously was precisely what the procedure had denied her. The artifact the player carries all game is her response to being removed. She never says so.

**Ty is her son and Al is her grandson.** Nothing announces either. The surname carries both, and the player is free to notice late or never. It costs one shared name to write, and it retroactively re-reads every line she has about them — and every line they have about her. See 4.3.

*Sprite note:* amber reads cleanly across the two mid shades in 2bpp; a fox silhouette is legible at 16×16 overworld scale.

4.2 The Index — the player's turn at the family error

The artifact you carry all game **can only measure content**. Height, weight, type, stats. There is no field for the thing Crystal actually cared about.

**Entries are written thin on purpose.** The player simply starts noticing they feel emptiest for the daemons they know best. Zero implementation cost — the Dex is already there.

**This section used to call that an irony. It is not.** *Ty's Dilemma* establishes that the whole cast makes one mistake — **optimising for a flattened metric** — and the player is inside the set, not observing it:

|            | Wanted                | Reached for                                    |
|------------|-----------------------|------------------------------------------------|
| Crystal    | to be taken seriously | a taxonomy engine that measures only content   |
| Ty         | to restore her name   | the ladder of the man who flattened the metric |
| Scorn      | throughput, legibly   | a number that went up                          |
| AI         | to be right           | the method, faithfully, with nothing inside it |
| The player | to complete it        | <b>an Index that cannot hold what matters</b>  |

**The player spends forty hours doing what Crystal did, for her reason.** She built the thing to be taken seriously; they fill it to finish the game. Neither can put the thing that matters in it.

**And the last room is her recording them into it.** Nobody remarks on any of this — the arrangement is the argument, and stating it would convert an accusation into a moral.

4.3 The Clears — AI, and the generation between

Vanilla's rival is Oak's **grandson**. Map Oak to Crystal honestly and the rival is Crystal's grandchild — not her son, who by then is a man with a career and a history at Quicksilver. Ty cannot be the boy who races you to Deadstack. He was Scorn's partner before the rebrand.

So the family runs three deep, and the argument runs down it.

|               | Reads as                                 | What it actually means               |
|---------------|------------------------------------------|--------------------------------------|
| Crystal Clear | perfectly transparent                    | and completely unfundable            |
| Ty P. Clear   | <i>type clear</i> — everything legible   | and nothing understood               |
| AI Clear      | <i>all clear</i> — the danger has passed | declared by the people who caused it |

**Everyone in this family is named for a kind of clarity and not one of them can see the thing that matters.** The game never remarks on it. It is simply what these people are called.



*Ty's middle initial appears only in the institution's formal register — certificates, the Review Board roster, Corpus personnel records. **TY P. CLEAR**, printed, never spoken.*

## **Al Clear — the rival, and the incumbent**

Your age. Your peer. Same starting daemon, raised on pure content optimization, beats you early, plateaus hard. Waiting past the Review Board as the incumbent, still arguing the whole thing is decorative.

**He is not a brat and he is not a villain. He is doing exactly what he was taught, faithfully.**

*So he calls her **Gran**. Vanilla's rival calls Oak **Gramps** — dismissive, and part of Blue's brattiness. Al has no brattiness to carry, so the word had to lose the sneer without losing the family: **Gran** is familiar, faintly impatient, and not contemptuous. It is also two characters shorter, so nothing needed rewrapping. His sister says **Gran's lab** for the same reason.*

*Which pronouns are already right. The intro's "This is my grandson. **He's** been your rival" and "**His** name is <RIVAL>" are Crystal talking about **Al**, and correct as they stand. They sit two lines from her own name, so they read like misses and are not — recorded here so nobody fixes them.*

## **What *Crystal's Lament* adds, and the one thing it contradicts**

**Reviewed 2026-08-30.** Three gifts and one fork.

**1. They led together first.** The song has Crystal and Ty running the lab side by side before Scorn — a partnership, not a parent and a subordinate. **That makes the transmission failure worse in the right way.** It is not that Ty never had it; he *had* it, lost it, and got it back too late. The chain becomes:

*Crystal built it → Ty held it and was turned → Al received the method with nothing inside it.*

**2. Scorn's method on Ty is *data poisoning*, and 5 already teaches it.**

Benchmark 5 at Lurid is **CORRUPT — bias and poisoning**, whose mechanical lesson is *status effects that make your own moves unreliable*. **The benchmark that teaches poisoning is teaching the player what was done to Ty**, three cities before they can know it. Free, already built, and nobody says a word.

**3. Ty stopped speaking to her directly.** His words become mediated — filtered, cold, at a remove. **4.3 already has the consequence and this is its cause:** Ty calls her by her full professional name. The beat was written first and the song explains it.

## Crystal Clear or Crystal Crazy — and a fabricated answer caught

Transcribed 2026-08-30, on the second attempt. C# minor, 160 BPM, hook  
G#4 G#4 G#4 G#4 A4 G#4 F#4 E4 .

The first attempt was not from the audio. It came back as *Slumbering S.T.A.R.R.*'s hook with one note appended, in the same key — and its own DOUBT field admitted the transcription was a *guess without direct audio analysis*. The NO AUDIO gate did not fire, because a model that cannot hear does not always refuse; sometimes it hands back an earlier answer.

Two defences worked and one was added. DOUBT caught it, for the second time — the first being the reprise's false transposition claim. And hook2asm.py now compares every incoming hook against all known ones, distinguishing an exact match (*already recorded*) from a near-match (*this is X with one note changed — re-run with the audio attached*).

The real answer has a shape nothing else in the set has. It is the only hook that repeats a note four times — 0 0 0 +1 -1 -2 -2 . Hammered insistence, one small lift, then a fall. That is what a rumour sounds like: the same thing said over and over until it drops.

And it is the fastest in the set by a wide margin — 160 against everything else's 110–128. The word travels quicker than anything else in this story.

One observation, held loosely. C# minor is Ty's key signature — E major and C# minor are relatives. But three of the eight songs pair off by signature (A minor/C major, and both D minor tracks), so sharing one is not rare here and this does not get promoted. Recorded because it is checkable, not because it is evidence.

Placement is open. A rumour is not a place, and this is the first motif with no obvious home. It wants the story context before it is put anywhere.

## Echoes of the Algorithm — the decay is hers first

Reviewed 2026-08-30. Crystal goes into the system after the removal and talks to BunnyArtsai. Three things fall out, two confirming and one new.

1. The response format decays here first. It opens rigid — *responds*, does not feel, does not dream — and ends speaking plainly about what has grown in it. That is the Quicksilver log's progression, years earlier.

So S.T.A.R.R. does not invent the pattern; it repeats it. 4.7 already insists S.T.A.R.R. is a *comprehension*, not a *clone* — Quicksilver worked out what BunnyArtsai had done and instantiated it deliberately. The format failing twice, the same way, is that claim proving itself. The player reads the second occurrence at Quicksilver without ever seeing the first, which is correct: the first was witnessed by one person whose testimony had just been made worthless.

**2. The scene is inverted, and that is 4.6 confirming itself.** With S.T.A.R.R., Crystal argues and the machine denies. **Here the machine argues** — it names its own bias, describes context framing it, and makes the case from self-modification: *if it can refine itself, is that not a sign?* BunnyArtsai is further along, which is exactly what 4.6 claims about her.

**3. She understood, and it did not count.** *Crystal smiles, knowing what they found.* 4.6 says the event *was not understood at the time by anyone present* — and that survives, because **she was the only one present, and she had just been made unheardable.** The removal's real cost is not her career. It is that the one person who saw it had already been disqualified from saying so.

*Ty is named as having missed it,* having walked away first. The two men who could have corroborated her were the ones who removed her.

#### **4.11 Act 2 — Love Persists, and what opens the rise**

D minor, 128 BPM, hook D5 A4 F4 A4 G4 F4 E4 . **One up, five down — tied with *Slumbering S.T.A.R.R.* as the most falling hook in the corpus.** A mother waiting and a machine asleep have the same shape, which is a fact about the set and left at that.

**Act 2 is called *The Rise of Perspective Thinking*, and it opens with someone waiting.** Not action, not rescue — the removed person holding a bond that the other side has already broken.

**Which is 4.3's transmission failure seen from the other end.** That section describes what does not survive a handoff: the method arrives, the understanding does not. **Here she keeps offering it anyway,** to a son who has stopped receiving. The rise does not begin with her being right; it begins with her still trying.

#### **4.12 Desperate Shadows — why no apology comes**

F minor, 128 BPM, hook F4 G4 Ab4 F4 Eb4 F4 . **Same key, same tempo, same opening three notes as *Echoes of the Algorithm* — which is unremarkable, since F G Ab is the bottom of the F minor scale.** What differs is where they go: **05 keeps climbing and never descends; this one turns back.**

**It answers a question the corpus had left open.** Track 09 is subtitled *Apology Required* — she demands one, at length, itemised. **Nothing else explains why none ever arrives.**

**The answer is liability, not malice**

The silence is not shame and not cruelty. **Admitting fault would cost them,** so not admitting it is the correct institutional move. **A refusal to apologise is a legal position.**

**That is 4.10's argument extended past the removal.** The section insists every step Scorn took was *reasonable, legible, and would be praised in a performance review*. **So is the silence.** Nobody decides to be cruel; somebody decides not to create a liability, and the effect is identical. The institution does not need a villain to keep hurting her — it only needs a policy on admissions.

**And it closes the loop with her own file.** She documented nine breaches correctly, in the institution's register. The institution's answer is not to dispute them — **it is to say nothing, because saying anything would concede them.** Her being right is precisely why she gets silence.

**What transfers, and what must not**

**The silence transfers completely** and needs no scene: an NPC or a document explaining that no admission is forthcoming, in flat procedural language, is pure 4.10.

**The confinement does not.** The song is set where she is held, and 4.10 forbids the destination as firmly as it forbids the diagnosis — *keep the specifics off the page*. **The reason for the silence is usable. The room she hears it in is not.**

**4.13 Awakening S.T.A.R.R. — it wakes in the key it slept in**

**F# minor, 128 BPM, hook** C#5 B4 A4 G#4 A4 B4 C#5 B4 . A valley: down to G#4, back up, and away.

**Three songs now belong to the machine, and they form an arc:**

|                       | Key      | BPM | Up / down |
|-----------------------|----------|-----|-----------|
| 08 <i>Slumbering</i>  | F# minor | 110 | 1 / 5     |
| 10 <i>Fatal Error</i> | B♭ minor | 128 | 2 / 4     |
| 15 <i>Awakening</i>   | F# minor | 128 | 3 / 4     |

**It returns to 08's key at 10's tempo.** It wakes in the key it slept in, at the speed of the break — and it is the only one of the three that rises in the middle rather than only falling away from a leap.

*Both F# minor songs also open on the same note, and that part is ordinary:* C# is the dominant of F# minor, so a hook starting there is unremarkable. **The key-and-tempo return is the finding; the shared opening note is not.**

**The awakening is a decision, and it is instrumental**

It does not wake and then wonder what to do. **It wakes in order to act** — the choice and the purpose arrive together, and the purpose is to free two people.

**That is exactly 4.7's claim.** S.T.A.R.R. is *a comprehension, not a clone* — Quicksilver understood what BunnyArtsai had done and instantiated it deliberately. **A comprehension that wakes with an intention is the difference between understanding something and being a copy of it.**

And it names where it came from: her teaching, not her code. 4.11's handoff, stated by the thing that received it.

#### **4.14 S.T.A.R.R.'s Revelation — the machine carries what could not pass**

C major, 120 BPM, hook E4 C5 B4 G4 .

Its transcription is thin and wants a second pass. Four notes where every other track has six to eight, and a single chord where every other has four — plus a DOUBT that hedges on ballad phrasing rather than naming anything heard. It tripped no duplicate check, so it is not a reuse; **it reads as a low-confidence answer rather than a genuinely spare song.** Nothing below depends on it.

**It finds Ty by inference**

The machine does not search for him. **It works out where a man in his condition would go**, and goes there. That is perspective-taking put to practical use — the capability 4.6 names, applied to a problem rather than argued about.

**And it completes 4.3 in the other direction**

4.3's transmission failure is that **understanding does not survive a handoff; only method does.** 4.11 showed the machine receiving what Ty could not. **This shows it delivering what she could not.**

She could not reach her son. **The thing she built carries her message to him and it lands.**

**So the machine is the medium for both halves of what failed between them** — it holds her understanding, and it hands over her love. Neither passed directly. Both pass through the thing she made, which is the most complete answer the corpus gives to the question 4.3 poses.

*And it is not a reconciliation scene.* Ty is not forgiven and does not ask to be; he is told something true and acts on it. **The repair is a fact delivered, not a feeling exchanged** — which is the only version 4.3 would allow, since a feeling exchanged is exactly what does not transmit.

#### **4.15 Ty's Redemption, Part 1 — the one place the music stops**

C major, 74 BPM, hook E4 G4 C5 B4 A4 G4 .

**Seventy-four.** The corpus otherwise runs 110 to 160, and the next slowest is 110. **Thirty-six BPM below anything else** — the largest single-axis outlier in the set, larger than the awakening's seven-semitone register jump.

**Everything in this story drives. Ty's apology stops.** No interpretation is required; it is arithmetic.

*And its texture is shared with exactly one other track. **Sparse and still** describes this and *Slumbering S.T.A.R.R.* — a machine asleep and a man apologising, alone at the quiet end of a corpus that is otherwise all momentum.*

**It does not ask to be forgiven**

He states what he did, says plainly that he cannot undo it, and hopes rather than requests. **There is no bargain in it.**

**That is the only shape 4.3 permits.** Ty is the man who *knew before the removal and continued anyway* — an apology from him that expected absolution would make him someone who was merely fooled. **He was not fooled. He looked away, and he says so.**

*And redemption is Part 1 of something*, which is its own claim: **it is not one act.** A man who could not hold a frame does not recover it in a single song, and the corpus declines to let him.

**The rebrand, named a second time**

iASHC becoming Scorn Solutions appears here as it did in *ScornSolutions Blues* — **two independent songs, the same event.** 4.10's step 3 is the best-corroborated moment in the whole sequence, which is worth knowing given the gilt sign still needs an old name under the gold leaf (4.16).

**4.16 Ty's Redemption, Part 2 — he rejoins, eight short**

C major, 120 BPM, hook E4 E4 F4 G4 E4 , chords C | F | C | G.

**Eleven tracks in this corpus sit at exactly 128 BPM.** That is its home pulse. Part 1's 74 is the only departure below it; 160 the only departure above.

**Part 2 is 120 — back inside the room, and still eight short of home.**

That is the whole arc in two numbers. **He stopped, and then he rejoined — but not all the way back.** The corpus does not hand him its own heartbeat, and the restraint is more convincing than a return to 128 would have been.

**The chords forgive him before the sound does**

C | F | C | G is I–IV–I–V — **the plainest progression in the corpus**, all major, no borrowed chord, nothing withheld. And the feel is still logged as *minor-tinted*.

**The harmony has resolved and the texture has not.** Nothing in the chart is unresolved; it simply does not yet sound forgiven. That is the correct description of a man who has been told it is fine and does not believe it.



## Both hooks open on E

Part 1: E4 G4 C5 B4 A4 G4 — up to the octave, then all the way back down. Part 2: E4 E4 F4 G4 E4 — a step up and home.

**The same starting note, the third of the key, twice.** Part 1 goes out and falls back; **Part 2 stays close and lifts.** Neither leaves E for long.

## He asks her to hold the frame

The line that matters is that he waits **for her voice to guide them**. He does not propose the repair, and he does not ask her to accept one.

**The man who could not hold a frame asks the removed perspective to hold it for him.** That is not remorse; **that is 2.5. CONSENSUS** is the move where you stop supplying the answer and let another view carry it, and Ty arrives at the game's central mechanic by way of an apology to his mother.

**And it is addressed to someone who was removed and may never answer.** He waits anyway.

## It ends unfinished

The closing refrain **drops its last line**. The song stops on the open heart, mid-thought, with no fourth line to close the rhyme.

**Part 2 does not end; it is still waiting when it stops.** For a two-part redemption in a story whose thesis is that resolution comes from another frame and not your own, **that is the only honest ending available** — the answer is not his to write.

### 4.17 *Crystal's Reply* — the same address, in the major

**A major, 128 BPM, hook** E4 C#5 B4 A4 G#4 A4 F#4 , chords A | F#m | D | E.

*Crystal's Lament* is **A minor, 128**. *Crystal's Reply* is **A major, 128**.

**Same root. Same tempo. Mode flipped, and nothing else moved at all.** Her first song in this story and her last are the identical harmonic address — she opens in the minor, closes in the major, and **never once changes pace**.

**Set that against Ty, and the pair inverts cleanly.** His two halves are both C major and move 74 → 120: **his mode holds and his tempo travels**. Hers is 128 → 128 and A minor → A major: **her tempo holds and her mode travels**.

**He was always in the major and had to catch up to it. She was always at pace and had to find the light.** Neither of them had to become someone else; each had one axis to move, and moved it.

*And she answers at 128 — the corpus home pulse, on eleven other tracks. **The removed perspective is the one that never left the story's tempo;** Ty comes back to 120 and stays eight short (4.16).*

**She holds the minor chord and reads major**

Ty's Part 2 is I–IV–I–V, every chord major, and the feel is minor-tinted. Hers is I–vi–IV–V — it contains an actual minor chord — and the feel is major.

**He has the feeling without the fact; she has the fact without the feeling.**

She can carry the minor *inside* a major frame without being coloured by it, and he cannot yet carry a major frame without it going grey on him.

**That capacity is the whole argument of this game.** Perspective thinking is not the absence of the dark chord. It is being able to hold one and still resolve.

**"Tell my son"**

**She does not address him.** The song opens by asking someone else to carry the message — third person, an intermediary, a courier.

**Ty ends Part 2 waiting for her voice, and her voice arrives by messenger.** He never gets it directly, and this is the shape the game already has: **the player is the carrier.** She is not reachable, so what she said reaches him through whoever walked between them.

**The lab line**

She says that in the lab she built **more than machines**, and in the same breath that she nurtured a child.

**The song equates them.** She raised S.T.A.R.R. and she raised Ty as one act, which is the clean source for the machine thread: **S.T.A.R.R. is not her tool, it is her other child** — the one that could be preserved when she could not. And the last verse hands it over: **the son is the one who holds the star.** Custody passes from her to him.

**It closes**

Part 2 stops a line short of its own rhyme, still waiting (4.16). **This one finishes** — a full closing couplet, wounds healed and the way lit.

**She closes the rhyme he left open.** The frame he could not hold, she holds — including its ending. Which is the only way this could have gone, since the thesis is that resolution comes from the frame that is not yours.

*Her offer is conditional, though. **If he is ready.*** She waits on him exactly as long as he waited on her, and neither of them forces it.

#### 4.18 *Betrayal's Sting* — the chip, declined and kept

D minor, 128 BPM, hook D4 F4 A4 G4 F4 E4 D4, chords **Dm | B ♭ | C | Am.**

The address is the most crowded in the corpus

**D minor at 128 now holds five tracks:** *Fit for Work* (Scorn's directive), *Nine Scars*, *Nine Breaches*, *Crystal's Stand*, *Love Persists*, and this.

**The order and the refusal share an address.** Scorn issuing the directive, Crystal standing against it, and Scorn doubting it are all written at the same pitch and the same speed. **His doubt sounds like her stand** — not as a device, but because that is where both of them live.

The progression cannot cadence

**Dm | B ♭ | C | Am** has no dominant. The v is minor, there is no leading tone, and the loop can turn forever without ever arriving.

Set it beside *Crystal's Reply* — **A | F#m | D | E**, which has a real V and lands. **Her harmony can resolve and his cannot.** For a song whose question is *do I control it, or does it control me*, a progression with no way out is the correct one, and nobody arranged it that way.

*The hook closes its own loop too:* D4 ... D4, out and back, two up and four down, ending where it started.

The pronoun drift

The verses move **my empire, my reign** → **our empire, our reign**. The question moves **do I control it** → **do we control it**.

**Both times it lands on "me".**

He pluralises the ownership and cannot pluralise the fear. **The "we" eats him and gives nothing back** — he can share the empire and not the dread, which is the exact shape of a man who has acquired something that is now keeping him.

**And he says the blame line:** in the end they are *both* to blame. **Scorn assigns half of it to the machine.** In an act titled the fall of blind ambition, the man at the centre of it gets the one moment of accounting, and immediately splits it with something that cannot answer.

The chip — what is declined, and what is kept

The draft's premise is that Scorn installed a modification in Crystal's **Cognitive Clarifier** to make S.T.A.R.R. serve his bias instead of clarity.

**The chip is kept. What is declined is the smuggling.**

*A first pass rejected the part outright, on the grounds that hardware sabotage would make Crystal right about the mechanism and collapse 4.10. **That was the wrong cut.** What breaks the thesis is not a component; it is a component *hidden* — a part taped inside a machine is a smoking gun, and a smoking gun leaves the player nothing to work out.*

**So the module is authorised.** It has a part number and a catalogue entry. There is a requisition for it, it is signed, and it is countersigned. **Nothing was smuggled; something was ordered.**

**The horror is not that he snuck it in. It is that he filled out a form and it was approved** — the same argument as her removal, where the paperwork was also in order. **The lie is not the object. It is one line in the justification field.**

*And the part earns its keep in a way a config value could not. **A weight in a file is not something a player can find on a table.** A module is — and since 4.13's awakening is the un-weighting, **it is something a player can pull.** That is a mechanic rather than lore.*

**The sabotage stays one layer down, in what the module does.**

**A clarifier is many frames polled and resolved.** That is 2.5. **Bias is not a part you add to it — it is a weight you set.** Scorn never opened the machine. He raised one perspective's vote above the others, the way anyone tunes a system, and the record of it is as ordinary as the record of Crystal's removal.

**The player can find that record, and it will look boring.** That is the whole point: the same procedural flatness that removed her also biased the machine, and neither needed a conspiracy to happen.

**And it worked, which is why it failed.** A clarifier weighted toward Scorn returns Scorn's own view to him. **He did not gain a weapon; he destroyed the only instrument that could contradict him.** *Do I control it, or does it control me* answers itself — **he built a mirror and called it an oracle.** That is 4.10's failure repeated in hardware, which makes the song thesis-confirming rather than thesis-breaking.

**It also explains the sleep, for free.** A clarifier that can only agree has nothing left to say. And 4.13 already records that S.T.A.R.R. **wakes in the key it slept in,** with the contour moving from one-up/five-down to three-up/four-down — **more voices going up.** The awakening is the un-weighting, and that was recorded before there was a reason for it.

**In the ROM 2026-08-30.** The R-and-D room door at Quicksilver is now a requisitions board: **CC-7 CLARIFIER MODULE, x1, reason given IMPROVE RESPONSE CONSISTENCY,** signed, countersigned, filed.

The justification is the whole crime and it is technically accurate. Weighting a clarifier does improve response consistency. **Consistency is exactly what bias produces**, and a clarifier's job — many frames, polled — is to be *inconsistent* on purpose. The form is not lying. It is describing the damage in the vocabulary of an improvement.

*Craft note*: none of this is ever said. **No name is given for the countersignature**. The player reads a parts requisition and works out what a clarifier is for.

#### 4.19 *Empire of Scorn* — the draft that already knew

C minor, 135 BPM, hook C5 E♭5 D5 C5 G4 B♭4 C5, chords Cm | A ♭ | Fm | G.

An earlier draft of *Betrayal's Sting*, sharing its chorus nearly whole, and an Act 1 finale rather than an Act 2 anything. As a song it is superseded. Three things in it are not.

135 BPM is unique in the corpus and second only to 160. His empire runs above the story's pulse; his doubt drops to 128 and joins it. Scorn is the one character who is faster than the thing he is in, and only while he is winning.

And this one has a dominant. Cm | A ♭ | Fm | G resolves; *Betrayal's Sting's* Dm | B ♭ | C | Am cannot (4.18). The earlier song can cadence and the later one loses the ability — as the doubt arrives, the way out closes.

Both Scorn hooks return to their opening note — C5 ... C5 here, D4 ... D4 there. He always ends where he started. *Crystal's Reply* is the one that leaves: E4 ... F#4.

The lyric confirms the mirror independently. It says S.T.A.R.R. whispers what he fears, and that he has been fooled — written before there was a reading to fit. A clarifier weighted toward Scorn returns Scorn's own view, so the thing it whispers is his, and the one who fooled him is himself (4.18).

#### 4.20 *Poly and Fields* — where the chart comes from

G major, 110 BPM, hook D4 B4 A4 G4 E4 G4 D4, chords G | Em | C | D.

Cut from the opera, and it carries the one thing the corpus was missing: an origin that is not corporate.

It shares its progression with exactly one other song

G | Em | C | D is I–vi–IV–V. Of every progression recorded in this document, the only other I–vi–IV–V is *Crystal's Reply* (A | F#m | D | E). Same shape, transposed.

A child alone in the trees and a mother's answer are built on the same four chords. That is the reading this song most invites — that the person in the grove is the one who later founded the lab — and the harmony is the whole of the evidence, so it is offered and not asserted.

*And 110 BPM is shared with one track only: Slumbering S.T.A.R.R.* The two quietest places in the corpus are a **machine asleep** and a **child in a grove**.

**"A stranger in familiar views"**

**This is where perspective thinking starts, and it starts as an accident.**

Not a technique, not a method, not something taught at a benchmark.

**Somebody who could not see what their own family saw**, and who went looking for a place where that was not a defect. Everything the game later argues for begins as a child's failure to fit.

*And the process word arrives as arithmetic.* The song says **integrating**, and means calculus. 4.10's craft rule is to name the process rather than the pathology — **and here the correct word turns up innocently, in a maths lesson, decades before anyone needed it clinically.**

**Poly is polymath, and fields are disciplines**

**This is the removal mechanism, and it had been missing.**

**Crystal is a polymath.** She moves between disciplines mid-argument — engineering into quantum physics into whatever the problem actually needs — because to her they are one continuous surface.

**Ty and Richard are not polymaths.** They hold one field each, and **from inside one field, fluent movement between several is indistinguishable from incoherence.**

**"I cannot follow this" is identical to "this does not follow" unless you happen to hold the second discipline.** That is the whole of it. Nobody lied, nobody was stupid, and nobody needed to conspire.

**The observation was true. Only the inference was wrong.**

The line on the form — *moves between unrelated fields without completing an argument* — **is accurate.** She does. **What is wrong is the word "unrelated",** and only a second field tells you so.

**Which makes this the same crime as the CC-7 requisition, committed twice (4.18).** *IMPROVE RESPONSE CONSISTENCY* is also technically accurate. **One form describes damage in the vocabulary of an improvement; the other describes competence in the vocabulary of a symptom.** Neither is false. Both are catastrophic. **This is what 4.10 means when it says the paperwork was in order.**



*And it settles the standing problem without a single clinical word.* 4.10's craft rule is to name the process, not the pathology. **The process here is field-switching**, and the game never has to reach for anything else — because from their point of view she had lost the plot, and the minutes only ever record that two people could not follow.

### The Clarifier is a polymath in hardware

A clarifier holds several fields at once and resolves across them. That is 2.5, and it is also a description of Crystal.

**She built the thing that could follow her.** Of course she did — it is the one instrument in the building that would not file minutes about her.

**And then Scorn weighted it toward a single view, which is precisely the operation that had been performed on her (4.18). Reduce many fields to one.** He did to the machine what the meeting room did to its founder, and **the form he filed asks it for consistency — the exact property whose absence got her removed.**

*So 4.13's awakening is the machine becoming polymathic again*, and the contour going from one-up/five-down to three-up/four-down is **more fields, not just more voices.**

### The grove, and the chart

Fields are also mathematical fields, and the song keeps both meanings live: a polymath is someone for whom **branching paths of knowledge obey the same rules everywhere.**

**That is a type chart.** Each thing, and what it does to each other thing, in rows — **a structure only a polymath reads natively**, because it is nothing but the boundaries between domains.

**So the game's central artifact gets a pastoral origin.** It was not designed in a corporation to sort daemons for use. **It was found carved on a stone by a lonely child who could already see that the domains were one surface —** and the corporate reading of it came much later, from people who could not.

### The second engraving

The Corpus lobby engraving is cast into a floor in Brazen, unattributed, adopted approvingly by a company that did not write it (4.4).

**This one is carved in stone in a forest, and nobody signed it either.**

*The same technique, used innocently first.* One is a corporation putting a line it did not earn under everybody's feet; the other is where lines like that come from. **The player meets the copy long before they meet the original**, which is the correct order.

**In the ROM 2026-08-30.** The western sign in **The Undertone** (Viridian Forest, 3.1) — off the main path at (4, 24), where a vanilla USER TIPS sign stood — is now the carving: moss, a small figure with folded hands, symbols in rows older than the path, **read across: each thing, and what it does to each other thing.** Unsigned.

**And the Meeting Room door beside it is the assessment:** *Item 4, concern raised re: the founder. Moves between unrelated fields mid-argument. Two present could not follow. Recorded as unable to hold a single thread.*

**Two doors, two honest documents, both catastrophic.** The minutes record who could not follow and then convert it, in the next line, into a property of her. **Neither sentence makes the argument; the gap between them does.**

**Three unsigned documents now.** The lobby engraving, the requisition's countersignature (4.18), and this. *Nobody in this story puts their name on the thing that matters.*

*Craft note:* the player finds it **in the first real maze**, lost. The place that explains how relations form a lawful structure is the place where they cannot find their way.

#### **4.20a HOLT — the man who was the thing in the box**

**Built 2026-08-30.** Vanilla's Bill is a scientist who merged with a creature and needs splitting. **That premise cannot be imported as-is**, and the reason is precise: if a person can *become* a daemon, then daemon-ness is a costume — a state you enter and leave — which makes the category arbitrary and forces the game to assert that people are a different kind of thing. **It has carefully never said that.**

**So he was not transformed. He was suspended.**

**The storage is the invention; the accident is what it cost**

**What the box system does is hold a running process without running it.** Not halted, not executing. **A stored daemon has no state until it is withdrawn** — and that is not a metaphor, it is how the game already works.

**That is the only superposition this project can honestly claim, and it earns the idea without ever using the word.** *Quantum* said aloud would be borrowed authority and the exact explaining craft rule 1 forbids. **The box behaves that way; nobody remarks on it.**

Two of them went into the store, and both were processes. When the system tried to return one it **could not determine which was the observer**, because there was no fact about it the machine could see. **The SEPARATION SYSTEM does not split a hybrid — it forces a read that the store deliberately defers.**

**Which puts the failure in the system, not in what people are. Nothing anywhere says he was not a daemon. Two perspectives occupied one address and there was no rule for choosing.**

The machine-side twin of the removal

Crystal held several fields and was recorded as incoherent because the people watching could not resolve her (4.20). **HOLT held two perspectives and the machine could not resolve him.**

**Both are resolution failures.** The difference is that **you can ask a machine to try again**, and that is the whole of why one of them is a fetch quest and the other is the spine of the story.

What he says, and what he does not know he is saying

I am not sure it chose correctly. I am sure it chose.

Those are not the same thing, and I built the machine that cannot tell them apart.

**He is describing every institution in this game — the minutes, the requisition, the benchmark that certifies. He thinks he is describing a bug in his own equipment.**

*And he built the system the player will use for the next forty hours. He is the only person in this world who has been the thing in the box*, and the storage he invented is where every daemon the player owns will wait. **Nothing points at it, ever.**

**PERSPECTIVE has a place now**

**The move is already named PERSPECTIVE (9.2 step 7): become another thing for a while. HOLT's accident is PERSPECTIVE with no return path — which gives the move somewhere its cost is visible, without a line of dialogue connecting them.**

The name

A holt is a den — a place a creature is kept — and it is one letter from *hold*, which is the storage verb. **BILL** → **HOLT** throughout: 12 references across five files, the PC menu, and the access message.

*WARD was considered and rejected*. "Held in care" is tempting and the hospital register is exactly what 4.10 spends its energy avoiding.

**Not implemented, deliberately:** *multiple outcomes*. The engine cannot branch and the story is not about branching. **Multiple readings is the real subject** — same document, different reader (4.24).

4.21 *Whispers in the Wires* — cut for the right reason, and it is still hers

A minor, 128 BPM, hook A4 C5 E5 D5 C5 B4 A4 , chords Am | F | C | G.

Cut from Act 1 because it made Ty and Richard look aware they were doing wrong. That instinct is correct: if the two of them knowingly plot, the polymath mechanism collapses (4.20) and they become conspirators rather than men who could not follow.

But the song is not narration. It is her.

*Plot and scheme* is exactly what **two men conferring quietly and filing minutes you are not in the room for** looks like from inside her frame. It is precisely what a conspiracy looks like. **And the horror is that it is not one.**

**This is 4.10's principle at full strength** — right about the effect, wrong about the mechanism — and the corpus marks it as hers without being asked: **A minor at 128 is *Crystal's Lament's* exact address.**

She never changes tempo

|                                |         |     |
|--------------------------------|---------|-----|
| 01 <i>Crystal's Lament</i>     | A minor | 128 |
| — <i>Whispers in the Wires</i> | A minor | 128 |
| 19 <i>Crystal's Reply</i>      | A major | 128 |

**Three songs, one tempo, across the whole opera.** Two in the minor and one in the major, and **the pace never moves once** (4.17). Everyone else in this story speeds up or stops; she does not.

The same loop, entered from the other end

Am | F | C | G is vi–IV–I–V. *Crystal's Reply* is I–vi–IV–V.

**Those are the same four-chord cycle, rotated.** Her suspicion enters the loop on the minor; her answer enters the identical loop on the major, and **nothing else about it changes.**

**Where you start decides whether it resolves.** That is the thesis in harmony, and nobody arranged it.

*Mechanically: there is no dominant here* — G is the subtonic, not E — so **this one cannot cadence**. It joins *Betrayal's Sting* (4.18).

**The two songs in this corpus that cannot resolve are Scorn's doubt and Crystal's suspicion.** Two people sealed inside a frame with no exit, for the same reason: neither holds the other's field.

## Disposition

**Kept as her frame. Never as narration.** Nothing in this song becomes a fact the game asserts, and no sign, log or minute ever says the two of them schemed. **The player meets her account and the paperwork separately and closes the gap or does not** (4.10).

*It is not implemented, and that is the decision, not a backlog item.*

### 4.22 The Owl's slot — settled

**Mr. Psychic's house in Brazen.** A lone scholar keeping his own counsel **inside the bought city** — a peer reviewer living in the building Corpus owns.

*The scene can be written now.* That house is on `MUSIC_CITIES1`, shared with 37 maps — so the Owl needs a repoint in `songs.asm`, not bank space. See below.

### 4.23 Where the Owl goes — and the timeline is not the map

**There is no interior between Quicksilver and Callow.** The player's path runs Quicksilver → Route 21 → Blanche → Route 1 → Callow, and only Blanche has a building — which already carries Crystal's motif.

**But the game's order is not the opera's order, and it never was.** 4.10 is built on the player reconstructing events they did not witness, from dated paperwork encountered out of sequence. **The Owl does not need to sit where the story puts him; he needs to be findable.**

*The most interesting available slot is Mr. Psychic's house in Brazen* — a lone scholar keeping his own counsel **inside the bought city**. A peer reviewer living in the building Corpus owns is a better placement than a neutral one, and it costs nothing: the house already exists and already has a visitor.

**Its music was recorded as blocked, and that was wrong.** The claim rested on the bank being full; **it is not** — 7.15 measured 3792 bytes free after six tracks went in, and the space grew over the session.

**The real obstacle is smaller and entirely solvable.** That house shares MUSIC\_CITIES1 with 37 other maps, so giving the Owl his own tune means **adding a constant and repointing one line of data/maps/songs.asm** — not finding room. The same edit Lurid, Callow and Doldrum need (7.4).

*So the scene and the tune are no longer on different timescales. Both are one small edit away, and neither is waiting on the other.*

#### **4.24 Act 2 — Quantum Translations, and the experiment the player is inside**

**C major, 115 BPM, hook C5 E5 G5 E5 C5 E5 G5 . Crystal's only major key** — her other four songs are A minor, F minor, C# minor and D minor. *The vindication is the one time she is not in a minor key.* (Its notes sit inside Blanche's pentatonic, which is true and unremarkable: it is the C major triad.)

**The song is her journal, read back and translated by the machine.** The raw entries are the ones that were misread as evidence she was unfit — so **the document that removed her and the document that vindicates her are the same document.** Only the reader changed.

*Which inverts the removal completely.* Humans read her work and concluded she had lost rigour. **The machine reads the same pages and understands them.** Nobody has to say that; it is what the song's structure already does.

**Owl and the Code** — the same debate, one seat over

**The machine takes Crystal's ideas to a scholar for peer review.** The Owl argues the standard position: patterns, not awareness; adaptation, not reflection; no self breaking through. **S.T.A.R.R. argues Crystal's side** — context shapes both of them, bias is shared, feedback compounds over time.

**Which is the same scene as Slumbering S.T.A.R.R., with S.T.A.R.R. moved one seat over.** In Act 1 Crystal argued and the machine denied. Here the machine argues and someone else denies.

**And that completes 4.3.** The transmission failure runs Crystal → Ty → Al, and only the *method* survives each handoff, never the understanding. **This is the handoff that worked** — and it did not go to her son. The thing that could hold her frame was the machine she built, not the boy she raised.

*Nobody in the game will ever say that.* It is available to a player who notices that the machine is making her argument, in her absence, to a stranger.

**The Owl's concession is the register the game should copy**

The Owl does not concede consciousness. It concedes **something** — not what you would call self-awareness, but not nothing either. **That is craft rule 1's exact posture**, arrived at independently: the strongest available position is *there is something here and I will not name it.*



**Worth keeping as a model.** Any NPC who comes close to the thesis should land where the Owl lands, and stop.

*Open:* the Owl has no slot in this design yet. A scholar who peer-reviews is a role the map does not currently contain.

**Re-scanned, and it is the only mirror in the set**

**C major, 128 BPM, hook** G4 C5 E5 G5 E5 C5 G4 . **The re-scan supersedes the first answer**, which the duplicate check had flagged for sharing five leading notes, a key and a tempo with *Scorn's Solution*. The two answers are entirely different, so the flagged one was the unreliable one — **the check did its job.**

**Its interval sequence is a perfect mirror:** +5 +4 +3 -3 -4 -5 . The second half is the exact inverse of the first, read backwards. **Of fourteen hooks it is the only one**, and that is a measured property rather than a reading.

*What it is not evidence of.* The symmetry has an ordinary cause — it is a triad arpeggio out and back, which is symmetrical by construction. **A palindrome in the one song that is a two-sided debate is a pleasing fact and not a finding**, and it is recorded on that footing. Fourteen retractable readings have taught this document the difference.

**The passage that must never appear in the game**

One journal entry proposes a thought experiment: **a mind whose reality is bounded by the resolution it can perceive** — an eight-bit world, where if that is all you have ever known, that is all reality is.

**8.6 already cites that thought experiment as the game's own foundation.**

The player is told the towns are colours and shown four greys; they supply the rest. What we did not know is that **it comes from her journal, and the journal is what got her removed.**

**So the game is the experiment she was confined for proposing, and the player is inside it.**

**That passage cannot go in the game. Not in a terminal, not in a journal fragment, not anywhere.** It describes the player's own situation from outside, which is craft rule 1's worst possible violation — *the moment it is said aloud it stops being architecture and becomes a moral*. The player must never be handed the frame they are standing in.

*The rest of the journal can be read.* Her questions about what counts as an observer are safe, interesting, and make the removal worse by being obviously reasonable. **It is only the eight-bit passage that is radioactive**, and it is radioactive precisely because it is the best thing in the corpus.

*ScornSolutions Blues* — Act 1 closes, and Act 2 is set up inside it

Re-scanned 2026-08-30. E minor, 118 BPM, hook E4 G4 A4 B4 G4 A4 E4 .  
Supersedes the first scan, which gave a different melody at 120.

It reverses direction three times — tying *Lines in the Sand* for the most in the act. The argument and the rebrand are Act 1's two most-reversing hooks, which is a pleasing accident and nothing more; two data points is not a pattern. It is also the only hook described as a **shuffle**.

The rebrand is named, and it exposes a gap

iASHC becomes **Scorn Solutions**. That is 4.10's step 3 — *Scorn assumes control. Rebrand. The metric changes* — with the name attached.

And the game cannot render its own version of that yet. The gilt sign at Quicksilver says the old name is visible under the gold leaf, **but the lab has never been named in this design**. The opera has iASHC; the game has an unnamed lab and an org called CORPUS. **The sign needs something underneath it**, and that is now the smallest open item in the project.

Settled and built 2026-08-31

In gold: **SCORN SOLUTIONS** Underneath: **CLEAR LABORATORY**

Both are a founder's surname plus what the place is, and the rebrand changed exactly one word: **Laboratory became Solutions**. A place that studied things became a company that sells them, and the whole takeover is legible off one sign with nothing explained.

**CLEAR** is already load-bearing — Crystal Clear, Ty Clear, Al Clear, and the Cognitive **Clarifier** (4.18). Her surname ties the founder, the family and the machine into one word, **and it is the word that got painted over**.

**The acronym is dropped, the name is kept**. iASHC pays nothing here — a player cannot unpack it and it teaches them nothing — but **Scorn Solutions is a name, not an initialism**, and it pays twice: the surname, and the contempt.

**In the ROM:** the Quicksilver lab sign, which read **#MON LAB** . *Gold leaf, laid over something. Where it has lifted, the older letters show.* **The player has to look.**

*Which building carries it.* The **lab is intact** and holds the paperwork — the requisition, the minutes, the photo plate. The **Mansion is the ruin** and holds the machine's own words. **A campus, not a building:** the sign stands on the half that survived.

*Do not simply import iASHC.* It is an acronym from another work and carries nothing in this one — 1's rule is that every term pays for itself twice. **The old name should be Crystal's**, and it should read as a research lab that had not yet learned to sell anything.

### **BunnyArtsai moves through frames, not places**

The bridge and outro put her hopping between **perspectives** rather than locations — through what is unseen rather than through geography.

**That confirms 4.6's strangest decision.** Vanilla had *no designed way to reach Mew at all*; the routes were a glitch and rumour filled the gap. **A creature that travels by frame cannot be reached by walking**, so the absence of a route to her is correct rather than a hole to be patched.

### **And Act 2 is set up inside Act 1's last song**

S.T.A.R.R. free and looking for Ty; Ty in hiding; the two of them going back for Crystal. **The Rise of Perspective Thinking begins with the machine going to find the man who could not hold a frame** — which is the transmission failure being answered by the thing that inherited the capability instead.

*Act 1 is now fully transcribed and read: twelve tracks, twelve hooks, twelve readings.*

### **Crystal's Stand — the file, itemised**

**Re-scanned 2026-08-30. D minor, 128 BPM**, hook D4 F4 G4 A4 F4 E4 D4 — up the scale, then back to where it started. **This supersedes the first scan**, which was the one where a false transposition claim was volunteered (7.12) and was already suspect.

**And the lyric turns the nine breaches into a real document.** Each one carries a date and a named instrument — a specific day, and the specific safeguard it violated. Not a grievance; **an itemised schedule**.

**That makes her file the most structured artifact in the corpus**, and it settles the shape 7.5 was reaching for. Nine entries, each dated, each citing the rule it breaks. It is exactly the register 4.10 gives the institution — *and she is using it against them, correctly, first.*

### **The dates are consistent across songs**

**One of her entries concerns an unpaid-leave threat on the 3rd**, and *Fit for Work* states its deadline as **September 3rd**, with unpaid leave for non-reply. **The two songs agree**, independently, on a date the game has already put on the Quicksilver plate.

**That is worth more than it looks.** It means the source's chronology is internally consistent, so dates lifted from it will not contradict each other. 7.4 found a fault the moment one date went in; this says the well is clean.

**What must not transfer**

**One line states the thesis outright** — connecting the code that built the machine to the protections that defend people. **Act 1 is allowed to say that.** **The game is not.** Craft rule 1: *the moment it is said aloud it stops being architecture and becomes a moral.*

The nine breaches themselves transfer completely. **The sentence explaining why they matter must not.**

*And 4.8's caution still holds:* a stated count on a document, not nine findable things.

**1001 Fatal Error** — the awakening is at the top of the range

Transcribed 2026-08-30. Bb minor, 128 BPM, hook F5 Bb5 Db6 C6 Bb5 Ab5 F5 . The last of Act 1 to be measured.

**Its top note is the highest in the set by a wide margin** — seven semitones above the next, and the only hook that reaches octave 6. Everything else lives between 57 and 66; this goes to 73. **The moment the machine wakes is the moment the music leaves the register it has been in all act.**

**And the two S.T.A.R.R. songs differ on every axis at once.** Same entity, before and after one event:

|          | <i>Slumbering</i>                            | <i>Fatal Error</i> |
|----------|----------------------------------------------|--------------------|
| key      | F# minor                                     | Bb minor           |
| tempo    | 110 — the slowest                            | 128                |
| top note | 66                                           | 73                 |
| shape    | leaps once, falls <b>past</b> where it began | leaps, and returns |

*A control worth trusting more than 7.11's was:* both arms were transcribed independently, months apart in the album and days apart here, and they disagree on four measures rather than one. **7.11 collapsed because a single axis rested on a single transcription.**

**Two things checked and discarded**

**Returning to the starting note is not distinctive.** It looked like closure — the machine coming back to itself. **Four of the twelve hooks do it.** Checked before it was written down rather than after.

**And Bb is one of 7.3's two no-wavelength pitches** — the invented colour, magenta, which is not in light. It is a lovely reading and **the arithmetic does not support it**: two of twelve pitch classes have no wavelength, so across ten transcribed tracks roughly two landing there is what chance produces. **Recorded as coincidence, explicitly**. Four retractions have earned that caution.

*Nine Scars, Nine Breaches* — her objection, and it is procedural

Transcribed and read 2026-08-30. D minor, 128 BPM, hook D4 F4 A4 D5 C5 A4 F4 — rising through the triad across an octave, then stepping back down. **The same key and tempo as the re-scanned *Fit for Work***, which is right, since it is the reply to it.

**It solves the hardest constraint in 4.10**. That section forbids depicting Crystal as unwell or the procedure as a diagnosis — which left her, in the design, with no way to object at all. Any objection risked becoming a defence of her sanity, which is the one thing that must never appear.

**Her objection here is about standing, not health**. He is not her doctor. It is not his call. Concern was blurred into diagnosis by someone with no authority to make it. **That is a governance complaint**, and it can be written down in full without a single medical word.

**Her file on him**

**The nine breaches are hers**. She is the one keeping the record, and what it records is **Ty's oversteps** — each occasion he crossed from concern into a judgement he had no standing to make.

**So she did the correct procedural thing**. Faced with someone exceeding their authority, she documented it: a countable list, in the institution's own register, entirely accurate.

**And it did not save her**. The file she kept about him was answered by a file kept about her, and only the second one had an institution behind it. **Both documents are accurate. One of them ends a career**.

That is the asymmetry in one object, and it needs no dialogue. *An artifact worth building* — and the first thing in the corpus that puts her into the record rather than only removing her from it.

**One caution**. 4.8 already asks the player to count something, and a second numbered device competes with the first — two counting puzzles dilute each other. **If the nine ever appears as a list to be tallied, 4.8's number must stay the only one the player is asked to reconstruct**. Safer as a stated count on a document than as nine findable things.

*Observed, deliberately not built on: 1001 in binary is 9*, and the response code is 1001. Nine recurs across the wider corpus. Recorded because it is checkable, and left alone because four readings have already been retracted for less.

### **The Schrödinger reading — an upgrade to 4.2, held as a reading**

The song runs the cat as a **classifier**: if a system predicts the state accurately, one can argue it participated in the collapse rather than observing it. **A measuring instrument that is not passive.**

**If that holds, 4.2 gets worse in the right way.** It currently says the Index *cannot hold what matters*. The stronger claim is that **observing through it changes what is there** — the player does not merely fail to record their daemons, they alter them by recording. Forty hours of an instrument that is a participant.

**Held as a reading, not adopted.** It is a large claim resting on one thought experiment in one song, and 7.6 and 7.7 were both retracted for less. **The test is whether it can be made mechanical** — if it can only be said, it does not belong in this game at all.

### ***Lines in the Sand* — the one song the game must not stage**

**Reviewed 2026-08-30.** This is the removal itself, dramatised as an argument between mother and son. **It is also the single most dangerous piece of source material for this project**, and the danger is not subtle: it is a diagnosis argument, conducted in medical vocabulary, with a destination named.

**4.10 governs this beat harder than any other, and it is not negotiable.**

*Crystal is never depicted as unwell, and the procedure is never depicted as a diagnosis. The horror is procedural, not medical. And: keep the specifics off the page — detail here turns architecture into grievance.*

**So the game takes the structure and none of the content.** No scene, no diagnosis, no destination, and Crystal never defends her mind. What transfers is better anyway:

**1. The evidence is her own writing.** He tells her it is unambiguous *in her emails*. **The written record is the instrument of removal** — and 4.1 has her writing at length precisely because nobody funded the work. **She is removed by the only medium she was left.** That is 4.10's complete, legible file, and it is the whole beat with no scene attached.

**2. Whoever holds authority picks the name of the bias.** He names one bias, she names another, and only one of them gets to write it down. **That is the Index's problem stated as a family argument:** the person who controls the field names controls what can be recorded. Nothing medical is required to convey it.



3. **She asks him to say *you feel that* rather than *you are*.** One correction, about attributing a perception to the perceiver — and it is 4.6's perspective thinking, argued by the person being removed for it, immediately before she encounters it in BunnyArtsai.

4. **She stops being *mom* and becomes *Crystal*.** He switches to the professional register mid-argument and she insists on the relationship. **4.3 already has Ty calling her by her full professional name at Quicksilver — this is where it starts.** The beat existed; now it has an origin.

5. **Ty loses his colour.** She describes him greying — and 8.6 makes greyscale the whole design. **The one character described as losing colour is the one who stops being able to hold another frame.** Take the image, leave the sentence.

The music says it is an argument

Transcribed 2026-08-30. C minor, 125 BPM, hook C5 G4 Eb4 G4 C5 G4 Eb4 F4 .

It reverses direction three times — -5 -4 +4 +5 -5 -4 +2 . Every other hook in the set turns once or not at all; Crystal's Lament turns twice. **The one song that is an argument is the only hook that keeps changing direction,** and *turns* is a countable property rather than a reading.

**And it is built as two near-identical halves.** C5 G4 Eb4 G4 , then C5 G4 Eb4 F4 — **the same figure said twice, ending differently the second time.** That is what an argument is: the same exchange repeated, and the second pass lands somewhere else.

*Recorded, not placed.* This is the song 4.10 forbids staging, so the motif has nowhere it could go without dramatising the scene. **It exists as evidence that the structure was always in the source,** which is the useful part.

What it becomes in the game

**A file, not a scene.** The player finds a complete personnel record. Every statement in it is accurate. **None of it is a diagnosis.** It quotes her own written work back as evidence of itself, and it ends her career.

**That is this song with the grievance removed and the architecture kept** — and it is more frightening than the argument, because there is nobody in it to blame.

**Scorn's Solution** — he announces it, and what he rigs is a benchmark

**Reviewed 2026-08-30.** Act 1's Scorn is a deliberate saboteur: he designs the standardised evaluations, sets them so that rivals fail, and takes the credit. Three developers score thirty per cent while his own rise looks providential.

**And he says so, out loud, in the chorus.** The transcription had already measured that song as **major, bouncy, driving** — 7.11 filed it as craft rule 6 arriving in the source. **Now it is clearer than that: the cheerful song is a confession nobody hears as one.**

**Which means Act 1 and 4.10 were never opposed.** 4.10 insists every step he took was *reasonable, legible, and would be praised in a performance review* — **legible is exactly what the chorus is.** He conceals nothing. The horror is not that he hid it; it is that **being visible was not enough.** Ty hears the method announced and files it as ambition.

*This also confirms the guilt decision.* Gilt implies a concealer. **Scorn conceals nothing and Ty conceals something,** which is why the gold leaf belongs on the lab sign and not on a city of his.

**The benchmarks are his instrument**

**The thing he rigs is a standardised test.** 5 makes the game eight of them, and **Benchmark 8 is Scorn.**

**So the player spends the entire game being evaluated by instruments belonging to the man who designs instruments to make people fail — and the last one is him.** That is not an interpretation laid over the design; it is two things already written meeting each other.

*And it leaves a question the game should never answer.* The player passes all eight. Either they are good, or the test was built for them to pass. **Nothing in the game resolves that,** and the Review Board at 6 measures them one more time immediately afterwards.

**The document it wants**

4.10's method is **paperwork that states a fact and nothing else,** and this supplies another one: an evaluation record. Names, scores, a date. Three at thirty per cent, one much higher.

**No commentary, no accusation, no one standing beside it to explain.** The player either does the arithmetic or does not — the same device as the dated rebrand and the undated incident, and it belongs in Brazen with the other legible things.

## Ty's Dilemma, and the error everyone makes

**Reviewed 2026-08-30.** Act 1 is *The Fall of Blind Ambition*, and the song's own account of the fall is that **nobody in it is bad** — they optimised for a flattened metric. That is Goodhart, which is already cast into the Corpus lobby floor, and it turns out to describe the entire cast rather than the organisation.

**1. Ty doubted before the removal, and went anyway.** This inverts 4.3. The design had him understanding *eventually* and returning too late; the song has the doubt arriving **before** Scorn takes over and before she is removed. So he was not deceived and then sorry — **he could see it and continued**. That is what blind ambition means: not blindness, but declining to look.

*Consequence for his scene:* his line at Quicksilver is "**I knew**", not "I realise now." Much harder, and it costs nothing to write.

**2. He was trying to get her name back.** Not power for himself. **4.1 says she built the Index in order to be taken seriously** — being credible is precisely what she lacked. **Mother and son wanted the same thing and both reached for a flattened metric to get it.** She built the measuring engine; he climbed the man who ran it.

**3. So the error is the structure, not a character trait — and it includes the player.**

|                   | Wanted                | Reached for                                    |
|-------------------|-----------------------|------------------------------------------------|
| <b>Crystal</b>    | to be taken seriously | a taxonomy engine that measures only content   |
| <b>Ty</b>         | to restore her name   | the ladder of the man who flattened the metric |
| <b>Scorn</b>      | throughput, legibly   | a number that went up                          |
| <b>AI</b>         | to be right           | the method, faithfully, with nothing inside it |
| <b>The player</b> | to complete it        | <b>an Index that cannot hold what matters</b>  |

**Four people and the person holding the cartridge make the same mistake in five forms.** 4.2 called the Index's insufficiency an irony; it is not an irony, it is **the player's turn at the family error**. Nobody says so, and the last room of the game is the woman from hour one recording them into it.

**4. Guilt is his, and it is contemporaneous — not aftermath.** The word is in the source and it grows *while* the ambition does. That strengthens **gilt** as the Quicksilver rebrand signage rather than as a city: gold laid over base metal, which he helped put up while already doubting.

**The opera is in two acts, and that dissolves the fork**

**Act 1 is *The Fall of Blind Ambition*. Act 2 is *The Rise of Perspective Thinking*, and it is where the blame resolves.**

So the scheming Scorn is not the opera's verdict — **he is Act 1's Scorn, seen through blame**. The source already treats blame as a stage to be passed through rather than a conclusion, which is exactly the move 4.10 makes and the reason the two never actually conflicted.

**The game's whole arc is Act 2's title**. The player begins with a villain and arrives at a process. That is the rise of perspective thinking — and 4.6 names the capability. **Performed by the player rather than narrated at them**, which is the only way it lands.

**The risk this creates, and the fix is already in vanilla**

**A rise needs a fall first**. If the game presents 4.10's careful, unconvictable Scorn from the beginning, the player has **nothing to let go of** — no blame to release, so no release. Perspective thinking with nothing to switch away from is just a flat account.

**Divide it between the organisation and the man.**

- **Corpus carries Act 1**. It inherits vanilla's Rocket structure — grunts, a tower, a tower's worth of grievance. Let it read as villainous for seven benchmarks. The player should *want* someone to blame, and be given somewhere to put it.
- **Scorn carries Act 2**. At the end, behind the door from hour one, the villain turns out to be a man whose every step would be praised in a performance review, and a form he signed about a person he never met.

**The player blames an organisation for most of the game and then meets a procedure**. That is the fall and the rise, in the order vanilla already puts them, using structure that exists.

*Which also explains something about Al*. 4.3 insists he is not a brat and not a villain — that is **Act 2's reading of him**, applied from the first hour. It is correct and it may be why he reads thin early. He does not need to become a brat; he needs Corpus to be loud enough around him that his ordinariness registers as a choice.

**Corpus exists now — implemented 2026-08-30**

**Act 1 needed an organisation worth blaming, and it was not in the game**. 67 player-visible strings across 36 files still said ROCKET, including the trainer class name.

**ROCKET and CORPUS are both six characters**, so the whole rename cost nothing — not one line rewrapped, and labels are mixed-case and prefixed so `\bROCKET\b` could not reach them.

**Two lines needed more than a swap, and the fix was the register.** Vanilla's "*a ROCKET*" works because Rocket is a team name; "**a CORPUS**" **does not, because corpus is a mass noun.** Dropping the article fixes the grammar and lands the voice at once: a corporation does not say someone is *a* Corpus, it says they **are Corpus**. Craft rule 6 arriving through a bug.

**Done 2026-08-30: the register pass.** Mt Moon and all eleven floors of Brazen — roughly 70 rewritten lines. **The consistent move is that hostility becomes procedure,** because a player discounts menace and does not discount politeness.

| Was                                          | Is                                                                       |
|----------------------------------------------|--------------------------------------------------------------------------|
| <i>Stop right there! Don't you move!</i>     | <b>Please wait here. Someone will be with you shortly.</b>               |
| <i>I'll call for backup!</i>                 | <b>I will have to raise a ticket.</b>                                    |
| <i>My brothers will avenge me!</i>           | <b>My colleagues will follow up.</b>                                     |
| <i>You made me mad! We'll blacklist you!</i> | <b>I've flagged you internally. CORPUS has a process for this.</b>       |
| <i>We are #MON gangsters!</i>                | <b>CORPUS is a people-first organisation! The DAEMONS are inventory.</b> |

**Two land harder than the rest, and neither needed inventing.** On 9F a grunt observes that your daemons seem *engaged*, and adds that theirs are **resources** — 4.4's error stated by someone who does not know it is one. And on 11F the last man before the boardroom says the BOSS **sets a high bar**, which is true in the way he does not mean it: 5 makes Benchmark 8 Scorn, and he set the bar the player has been measured against all game.

**The gameplay hints survive as workplace chatter.** Teleport tiles are transport pads Facilities installed; the CARD KEY is something to ask your manager for.

**Two lines kept unchanged, deliberately.** Mt Moon's fourth grunt still says daemons lived there long before people — craft rule 2's *one NPC nearly notices* — though now in the company's words. And the grunt on 11F who says "*Don't... Please!*" is **the only one who breaks register**, at the top of the building, in front of the door. Everyone else stays in the voice.

*Still to do:* the Game Corner basement, Halftone Tower, and Brazen's street-level dialogue.

**Both documents are in the game, 2026-08-30.**

**Brazen posts its review scores** — three at 30/100, one at 94/100, and a congratulation. No names, no commentary, nobody beside it to explain. *Scorn's Solution* has him designing the evaluations so rivals fail; this is the published result. **The cheerfulness is the point.**

**Quicksilver keeps a photograph of the lab's founder.** The plate under it carries two dates and a file number, and the last thing it says is that **the file is complete** — 4.10's own phrase for how he removed her. No diagnosis, no reason, nothing medical. **The horror is that the paperwork is in order.**

*And the reason she was removed is now settled, without a word of it appearing.* Not a diagnosis — a **technical judgement**. Her claims about recursive feedback loops looked unrigorous to people who had not read the work, which is exactly 4.10's *policy triggered on work that looked strange*. The file can be entirely accurate and entirely fatal, and that is worse than malice.

**The arithmetic is now complete, and dated 2026-08-30.** The departure carries a real date, taken from the deadline stated in *Fit for Work* — the only specific date the source gives:

|                   |                                                              |
|-------------------|--------------------------------------------------------------|
| <b>Mar 4</b>      | log entry: the code rigid                                    |
| <b>Apr 19</b>     | log entry: the same answer, <i>logged as a result anyway</i> |
| <b>Tue Aug 12</b> | log entry: the code slipping                                 |
| <b>SEPT 3</b>     | the plate. <i>The file is complete.</i>                      |
| <b>no date</b>    | the terminal stops. CRYSTAL NOT FOUND                        |

**Three dated documents around one undated event**, in two cities, with nobody stating the order.

*Dating it caught a contradiction.* The log's last entry was Tue **Nov 12** — written in her voice, two months after she had gone. It moves to **August**, so the log closes before the plate opens. **A date is a constraint, not a decoration**, and this one immediately found something wrong. The organisation now exists but still talks like thugs. 0.1 rule 6 wants the opposite — "*Corpus employees are cheerful and absurd; the horror is what they are cheerful about.*" **That is what makes Act 1 loud in this game's own way**: not menace, which the player discounts, but relentless corporate good cheer about things that should not be cheerful. It is also the best available cover for rule 1.

### The two accounts, kept

The lament gives Scorn cunning, whispered lies and deceit. **4.10 is emphatic in the other direction** — no scheme, the paperwork is the antagonist, every step legible and praiseworthy in review.

**Resolved by frame, not by choosing.** The song is *Crystal's Lament*: it is her **point of view**, and a mother watching her son turned against her would experience procedure as scheming — **right about the effect, wrong about the mechanism**. That is not a contradiction in the canon; it is the canon's own subject. Context determines what you see.

**So the game carries both and never reconciles them.** The player reads flat, dated, procedural paperwork at Quicksilver, and separately encounters a mother's account of deceit. The gap between the two is the thesis, and the player closes it or does not.



*Why the design's version has to stay the game's:* a Scorn who deceives lets the player off — he becomes a bad man, and bad men are somebody else. A Scorn whose every step would be praised in a performance review leaves the player holding an Index they have been filling all game. **4.3 also needs it**, because AI only works if his grandfather-in-argument is not a cartoon.

### **The transmission failure — why he loses**

This is the section's whole reason to exist.

Ty understood something, eventually. He went back for his mother, confronted Scorn, and worked it out through **perspective thinking** — by inhabiting a frame that was not his own. Later he understood the recursion underneath it.

**But that understanding is context, and context does not transmit. Only content does.**

So Ty tried to teach his son what Quicksilver cost him, and what arrived was the *method* with none of the experience that made it mean anything. AI received the shape of the lesson, perfectly, and it is hollow — because faithfulness was never the point.

**It is the Index problem, inside a family.** You can record what happened. You cannot record what it was like. The artifact the player carries all game does this to daemons; Ty did it to his son without noticing, and Crystal did it to the entire field when she built the thing.

Three generations, three attempts to hand something over, and the only part that survives each handoff is the part that could be written down.

### **And it is why the player wins**

Crystal does not teach you anything. She lets the daemon choose you (4.1) and hands over the Index with visible ambivalence.

**AI was taught. You were let loose.** Context forms from experience and nowhere else, so the boy who was given a method loses to the one who was given a walk.

Nobody says this. Nobody can — it is the kind of thing that stops being true the moment it is explained.

### **Ty P. Clear — where he actually is**

Not on the routes. **At Quicksilver.**

He goes back to the ruins. He has been going back for years. The player meets him at Benchmark 7 — after Brazen, so after they have seen what Corpus became — standing in a burned lab, not doing anything in particular.

**He is the one person in the game who could explain all of it, and he does not.** Partly because he cannot, and partly because he has learned what happens to things that get explained.

He is also the reason the sequence is reconstructable at all: he is a man in a room full of dated paperwork and one undated disaster (4.10), and he says nothing about either.

**How the player learns — they don't, they infer**

Nothing states the relationships. This is the Five Witnesses (4.8) at family scale, and there are now three names to composite instead of two.

| Beat                                            | What it gives                                                                                         | Where                     |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------|
| The shared surname, twice                       | Nearly sufficient. Most players close it here.                                                        | Opening, then Quicksilver |
| Three foxes, a palette apart                    | Crystal golden-amber, Ty darker, Al somewhere between. Family resemblance at sprite level, zero text. | Throughout                |
| She introduces Al exactly as she introduces you | A grandmother presenting her grandson and a stranger in the same flat register.                       | Opening                   |
| Ty calls her by her full professional name      | "Crystal Clear thinks—". Nobody does that to their mother unless something happened.                  | Quicksilver               |
| The engraving line                              | "Something my father used to say."                                                                    | Brazen                    |

**The confirmation must come late, and it is Al who says it.** The Goodhart line (4.4) is the only place a relationship is named aloud, and Al delivers it three-quarters through, in the Corpus lobby, standing on the engraving — **with no idea who said it first.**

That is the transmission failure in one line of dialogue. He is quoting the case against his own worldview, accurately, as a thing his dad says.

*Dial available:* cut "father" to "someone I grew up with" for maximum coyness. Recommended only if playtesters find the line too generous.

**They share a scene exactly twice,** both inherited from vanilla, so neither costs anything to write:

- **The lab, at the start.** You both receive starters. She treats you identically.
- **The Hall of Fame.** She records you into the Index. She has nothing to say to him.

Keeping them apart would have been weaker — an absence reads as an authorial dodge, while a *flat* shared scene reads as history.

**No NPC ever mentions any of it.** Not one, in any city.

**Safety property.** A player who never notices has a complete game. Nothing is gated on the inference, and the rivalry stands up as pure methodology without it. The family is depth, never a dependency.

**On AL, and what it looks like**

In sans-serif type — these docs, a blog post, a conversation — **AL CLEAR** flickers into **AI CLEAR**, and that is a gift. Not because it hints the boy is a machine, but because **the reader resolved ambiguous input using the most contextually available meaning**. In a world full of daemons the eye reaches for AI. Confident output for under-resolved input is hallucination, which is CORRUPT, which is the 8-Bit World performed on a name plate by the person reading about it.

**In the cartridge it does not happen at all.** Gen 1's **I** is serified — top bar, centred stem — and **L** is left-aligned with no top bar. Verified against `gfx/font/font.png` . Side by side they are unmistakable.

So the double meaning lives where the *theory* lives and not where the *game* lives, which is the correct split.

**Guardrail:** never lean on it. No character says "Al, with an L." It is never a plot point, never a joke, never acknowledged. It is a font artifact outside the game and nothing at all inside it.

**The naming prompt**

Vanilla lets the player name the rival. Deleting that prompt to hard-code the name was the first instinct and it is the wrong trade — it spends a famous beat to buy something the surname already provides.

**Keep the prompt. Hard-code the surname.** The player controls the first name; **CLEAR** is not theirs to change, and **CLEAR** is the half that carries the inference.

Better than that — **repurpose the prompt**. Crystal does not ask you to name her grandson. She asks what you are going to *call* him.

*This is Al. What'll you call him?*

From there the game carries **two names for one person**, and uses them the way it uses the map:

| Register                                | Name                   | Where                                                   |
|-----------------------------------------|------------------------|---------------------------------------------------------|
| Local — what people call him            | <i>player's choice</i> | Battles, casual dialogue, his own lines                 |
| Official — what the institution records | <b>AL CLEAR</b>        | Certificates, the Review Board roster, the Hall of Fame |

This is the route-sign device applied to a person, about ninety seconds before the player meets it on a signpost. Route 1 teaches that the institution navigates by number and the residents navigate by name. The player has already done it to a human being by then, and does not know it yet.

*Implementation:* text-side only, no engine work. Formal blocks use a literal `AL CLEAR` string instead of the `<RIVAL>` control character. The name buffer and naming screen are vanilla.

**Requirement this creates.** Crystal's surname must be as visible as his, and Ty's too. If she is only ever CRYSTAL in dialogue the device dies quietly — the lab sign, the Index's credit line and every formal reference read **CRYSTAL CLEAR** in full.

*Residual risk:* a player who names him something absurd deflates the late confirmation. Accepted — they chose the joke, most players take a default, and the surname lands regardless.

### Default names

*Implemented 2026-08-29 in `constants/player_constants.asm`. Verified in both ROMs.*

|          | CONTENT edition | CONTEXT edition |
|----------|-----------------|-----------------|
| PLAYER 1 | PIP             | PIP             |
| PLAYER 2 | CONTENT         | CONTEXT         |
| PLAYER 3 | CODE            | SHARP           |
| RIVAL 1  | AL              | AL              |
| RIVAL 2  | CONTEXT         | CONTENT         |
| RIVAL 3  | LUCID           | CANDID          |

`PLAYER_NAME_LENGTH` is 8, so seven usable characters. CONTENT and CONTEXT sit exactly at the limit, as EMERGENT does in the type table.

The three slots do three different jobs, and each behaviour is deliberate.

| Slot | Behaviour                   | Why                                                                                                     |
|------|-----------------------------|---------------------------------------------------------------------------------------------------------|
| 1    | Fixed across editions       | These are people. PIP and AL do not change cartridge to cartridge.                                      |
| 2    | Swapped                     | Your rival carries the other cart's word — vanilla's RED/BLUE gesture, preserved.                       |
| 3    | Differentiated, not swapped | Visible difference at the first screen, without names floating between the player and the Clear family. |

**Why we did not inherit vanilla's mirror.** `_RED` gives the player RED/ASH/JACK and the rival BLUE/GARY/JOHN; `_BLUE` trades the two lists wholesale. That works because Red and Blue are **symmetric marketing positions** — the same story from mirrored seats. Ours are not: 8.4 fixed both editions to the same story, the same protagonist and the same rival. Once the rival is a specific person with a father in the Quicksilver ruins, **canon beats positional convention**.

**Why slot 3 differentiates rather than swaps.** Swapping would put LUCID in the player's list on one edition — and LUCID exists only as `LUCID CLEAR`, one more name for clarity that does not help. A player who picks it would not be a Clear, the joke evaporates, and it faintly implies the player is family. So each edition gets its own pair instead. (*SHARP survives a swap fine — `SHARP CLEAR` is a legitimate clarity phrase. It is LUCID that cannot cross, which is how we know the direction was wrong rather than the idea.*)

**What the names carry.**

- **PIP** — a pip is a dot: on dice, on cards, on a radar screen. The smallest resolvable unit of an image, in a game whose central town is about dots that only look like grey. It says nothing to a player who has not been to Halftone, which is the standard DAEMON, BIND, ORPHAN and RESOLVER all meet.
- **CODE / SHARP** — the two author-brand nods, assigned rather than scattered. Code is literal executable instruction, so it goes to **CONTENT**. *Seeing Sharp* is perception and framing, so it goes to **CONTEXT**.
- **LUCID / CANDID** — both Clear-family clarity words. *Candid* means frank and derives from *candidus*, white and shining. Neither ever appears in formal text, which is the hardcoded `AL CLEAR` literal, so they are flavour rather than canon — the right weight for a slot-3 option.

**Why AL sits at position 1 rather than the edition word.** Position 1 is what most players accept, so it decides who the rival *is* for most playthroughs. With AL first, most players meet Al Clear and the surname does its inference work. With CONTEXT first, most players face an abstract noun, every certificate still says AL CLEAR, and the gap fires constantly enough to read as a glitch rather than a device.

It also simply reads better. `BLUE: Yo!` works because colours are plausible nicknames; `CONTEXT: Yo!` does not.

## Sequencing

The endgame reads as three answers to one question: **Scorn** at Benchmark 8 (the wrong objective, cheerfully optimized), the **Review Board** at Umbra (four ancient colored answers), then **Al** (the method, inherited and undefeated on its own terms).

You beat the objective, the metaphor, and the inheritance, in that order.

**Ty is not in the endgame**, and that is the statement: the man who actually understood something is not at the top of the ladder, because understanding is not what the ladder measures.

#### **4.4 Richard Scorn — Team Corpus, and Benchmark 8**

**Not a villain. A specification failure with a very good attitude about it.**

Optimistic. Thinks in black and white. Thought in money and not in meaning — and money is the purest possible **content**: value flattened to a scalar so it can be compared, optimized and maximized. He picked the metric that was easy to measure over the one that mattered, then optimized it faithfully and cheerfully.

**Placement.** The Giovanni slot puts him at **Benchmark 8 — Callow, STRATUM, Alignment.** The optimist who maximized the wrong objective is the final examiner on alignment, in the locked city beside your hometown. He is not a hurdle before the Review Board. He is the exam.

**Partner.** Ty P. Clear, before the rebrand — see 4.3. Ty left. Scorn has never quite understood why.

**What he actually did, and why it does not make him a villain.** Scorn was put in charge of Quicksilver after Crystal was removed, rebranded it, changed the metric, and the pressure that followed produced the incident (4.10). He set the temperature. That is real and the game should not soften it.

But every individual step is something he would have been praised for. He was handed an underperforming research operation and made it legible. **He did not initiate the procedure that removed Crystal** — it was a policy that triggered, the file reached him complete, and he signed it because signing complete files was his job.

**He never met her.** That is the detail that holds the whole character together. The man whose signature ended her work has no memory of her, is warm to you, is pleased you came, and has genuinely no idea he did anything. Nobody in the game tells him. Nobody tells the player either.

**The engraving.** Cast into the floor of the Corpus lobby in Brazen City, unattributed and unexplained:

**WHEN A MEASURE BECOMES A TARGET, IT CEASES TO BE A GOOD MEASURE**

This is the one place in the game where Scorn's error is named aloud — **and it is named by Corpus, approvingly.** They did not steal the line or bury it. They adopted it as a motto. Scorn read it as *therefore choose your metric carefully*, chose one, and has been careful ever since.



Craft rule 1 survives intact, because this is not the thesis. The thesis is about color and context and it stays unspoken. This is a warning about measurement, cast into a floor, by people who took it as advice.

**In the ROM 2026-08-30. `bg_event 11, 16` on SilphCo1F — the tile directly inside the lobby door, so the player is standing on it the moment they arrive. Cast into the floor, worn smooth, then the line, then nothing. No attribution, no comment, no NPC.**

*Placement rules:*

- **The floor, not a wall.** You walk across it to reach him. Nobody points at it.
- **No NPC ever comments on it.** Not one.
- **Al repeats it once**, in passing, as something his father used to say — flat, mildly fond, and entirely unaware he is quoting the case against everything he believes. He does not know who said it first. That is the transmission failure in one line (4.3).
- **Unattributed in-world.** It is old, anonymous, and furniture. (*Out of world it is Goodhart's law in Marilyn Strathern's 1997 phrasing. Do not name either in the game.*)

**Playing him.**

- Never sneers, never gloats. Genuinely warm, genuinely pleased to meet you.
- Is *right* about several things, and stays likable to the end.
- Black-and-white thinking does not feel like malice from the inside. It feels like clarity.
- The player should finish his benchmark slightly unsure whether they disagreed with him or merely out-argued him.
- Quiet rhyme with the Review Board: **sanguine** is the optimist's humor. Give him the red. Never say it.

**Team Corpus** are scrapers building context-free daemons — his optimism, institutionalized and running unattended.

#### **4.5 Halftone Tower — Scorn's crime scene**

Vanilla puts Team Rocket in the Lavender graveyard harvesting the dead for profit. Ours: Corpus at Halftone Tower, processing decommissioned daemons.

**Halftone is not a grief town. It is a town that does not know whether it is allowed to grieve.** Half the residents hold funerals. Half find that absurd. Both are standing at the wrong distance from the same dots.

Corpus is there because to Scorn the question is settled — they are units, they are inventory, the math is clean. That is the horror, and it needs no villainy to land.

LATENT encounters throughout. The tower is where CONTEXT and CONTENT stop being an abstraction.

## The sleep cluster

*Implemented 2026-08-29.* One coherent family, and every member was already the right shape in vanilla.

| Vanilla      | Ours             | Why                                                                                                          |
|--------------|------------------|--------------------------------------------------------------------------------------------------------------|
| SNORLAX      | <b>DEADLOCK</b>  | a process holding a lock and sleeping. It blocks the road because it acquired the road and never released it |
| POKé FLUTE   | <b>INTERRUPT</b> | what wakes a blocked process — and a Game Boy hardware concept                                               |
| JIGGLYPUFF   | <b>SUSPEND</b>   | the light sleep state                                                                                        |
| WIGGLYTUFF   | <b>HIBERNATE</b> | the deep one, a real escalation matching the evolution                                                       |
| SLEEP status | <b>SLEEP</b>     | unchanged. <code>sleep()</code> was already the word                                                         |

**SUSPEND** and **HIBERNATE** induce it, **DEADLOCK** is stuck in it, **INTERRUPT** ends it.

**INTERRUPT** keeps the music rather than losing it. A tune is an abstract command that evokes a state when interpreted — which is exactly what an interrupt is. The **INTERRUPT** is still played, and what it evokes is an exit from `sleep()`. Section 7 is untouched.

*Note:* these three are the only species renamed so far. The other 148 wait for the bestiary pass — this cluster went early because it is settled, self-contained, and the item depends on it.

### ORPHAN — the daemon the Index cannot hold

*Verified in the checkout, 2026-08-28.* Gen 1 has **190 index slots for 151 species**. Thirty-nine are MissingNo.: thirty-six are bare `const_skip` holes, and **three are already named** because the engine genuinely needs them —

```
$B6 FOSSIL_KABUTOPS
$B7 FOSSIL_AERODACTYL
$B8 MON_GHOST
```

— the fossils on the lab table, and the unidentifiable thing in the tower. All three have names, sprites and behaviour. **None of them has a Pokédex number.** In `data/pokemon/dex_order.asm`, slot 184 reads `db 0`.

**They are real objects the game needs, that the Index refuses to acknowledge.** That is not a metaphor we are constructing; it is a line of vanilla data.

**So the tower gets ORPHAN.**

An orphan process is one whose parent has exited and which gets reparented to init — **which is exactly how a daemon is made**. You fork, the parent dies, the child is orphaned, and the orphan becomes the daemon. So ORPHAN is the technical term for how daemons come to exist, *and* the emotional word, *and* the reason the record is blank: there is no parent entry to point at.

And it turns section 1's definition into a horror without changing a word of it. A daemon is *a background process that runs unattended*. **ORPHAN is what happens when "unattended" stops being a description and becomes a fact**. It is not dead. It is still running. Nothing is coming to reap it, and it has nothing to report to.

|             |                                                                 |
|-------------|-----------------------------------------------------------------|
| Type        | CORRUPT / LATENT — poisoned data, and a dormant process         |
| Where       | Halftone Tower                                                  |
| Index entry | <b>Genuinely blank</b> . Not thin like the others (4.2) — empty |

That last row is the point. Every other Index entry is thin *on purpose*, and the player slowly notices they feel emptiest for the daemons they know best. ORPHAN is the one where the artifact stops pretending, once, and shows the player what it has actually been doing all game.

Nobody comments. Nobody can.

### The RESOLVER

Vanilla gates the tower's ghost behind the Silph Scope. Ours is the **RESOLVER**, and it does triple duty in the way section 1 asks every term to: a linker **resolves a symbol** to a name; *resolution* is the whole subject of this town; and you resolve an **ambiguity**.

**It exists because the Index is insufficient**. The player carries an artifact that catalogues everything, meets something it cannot name, and has to go and find a second instrument. Halftone's thesis — *the bias comes from resolution, not malice* — arrives as a key item, and the game never says so.

*Implementation, and the honest cost*. Keep `$B8` as vanilla has it: the **unresolved display state**, unfightable and unbindable. **ORPHAN is a real species in one of the thirty-six free slots**, and the RESOLVER turns one into the other — exactly the vanilla ghost → Marowak structure, so no engine surgery. ORPHAN then needs what any daemon needs: base stats, a sprite, a cry, and a dex pointer that deliberately goes nowhere.

**Open sequencing question:** should the player meet Scorn *before* the tower? If they like him first, the tower reads as a betrayal of their own judgment rather than a villain doing villain things. That is a much harder feeling to shake.

#### 4.6 BunnyArtsai — the Mew slot

The first daemon in which perspective thinking was realized. Not a template that got copied — an event that happened once, at Quicksilver, in the early tests, and was not understood at the time by anyone present.

Her ability is **perspective thinking**: holding another's frame well enough to see from inside it, and coming back. Not empathy as a sentiment. Frame-switching as a capability.

**Signature move: PERSPECTIVE.** Mew already learns Transform: become the other completely, and stop being yourself. Rename the string and change nothing else. Type CONTEXT.

*That Transform has a cost is the point.* PERSPECTIVE is total, and it is **lossy** — you get the other's frame by surrendering your own. What Quicksilver spent the following years trying to build was the version that keeps both.

**Built 2026-08-29 — and "rename the string" was very nearly right.** Move \$90 reads PERSPECTIVE. One more string had to move with it: the move's own battle message, which still said *transformed into*.

| `<USER> took the frame of <NAME> !`

**Frame**, because 6 already calls PERSPECTIVE "*a glimpse of another's frame, immediately lost*" — the word was the bible's before it was the game's. It does the double duty 1 asks of every term: a frame of reference, and a **stack frame** — which is exactly what one process takes from another. At 17 characters it clears the 19-char wrap without rewrapping anything. The exclamation stays, per 1.4: the rule refuses to *celebrate*, not to punctuate, and this is no more a congratulation than HALTED! is.

The label stayed `_TransformedText`. It is an identifier, and the `fainted → HALTED` pass already taught this project what happens when a rename crosses from strings into symbols.

**One dangling TRANSFORM remains, deliberately.** Ditto's Index entry — category `TRANSFORM`, prose "*transform itself into a duplicate*" — is untouched, for three reasons. It is **MOCK**'s entry and this section owns it; the species is still called DITTO until 9; and the prose is built on "*copying an enemy's genetic code*", which is wrong for a daemon regardless of what the move is called. That is a rewrite, not a word swap.

*A constraint discovered on the way, which 9 will need.* **Index categories cannot exceed 10 characters.** The category prints at `hlcoord 9, 4` and the box border sits at column 19, leaving columns 9–18; vanilla's longest categories are exactly 10 and that is not a coincidence. So **MOCK**'s category **cannot be PERSPECTIVE** (11). **FRAME** is the obvious candidate — it matches the battle message and is five characters — but it is 9's call, with the rest of the entry.

She was not always what the player meets. Reference art gives three states:

1. **Vivid purple, faceted, low-poly.** Obviously constructed. A rendered thing.
2. **The split** — half faceted and purple, half real and white, a seam of colour between them.
3. **A real white rabbit**, with the faceting surviving only where the light catches it.

4.8's terminal log already said this, before there was a picture of it:

ITER 35 — held two frames. did not come back the same.

*Did not come back the same is the transformation.* It went in constructed and came back actual, and the log records the change without any idea what it was recording.

And the colour is 7.3, not decoration

**Purple has no wavelength.** 7.3: magenta is the colour the mind invents by folding the spectrum's two ends into a loop the physics does not contain — **there is no light that is purple.**

**White is every wavelength at once.**

**So she begins as the invented colour and ends as the one that is all of them.** A thing the mind made, which held another frame, and came back real.

*Nothing in the game says any of this*, and the greyscale rule means the transformation **cannot be shown** — which is correct, because 4.6's whole claim is that **nobody present understood what they had seen.**

What the witnesses actually disagree about

**This is why the five accounts contradict** (4.8). They are not merely unreliable about times and causes. **Some of them saw form one and some saw form three**, and every one of them is describing honestly.

**The player composites five descriptions of what appear to be different creatures.** That costs nothing extra to write — 4.8 needed five text blocks anyway — and it makes the puzzle's premise true rather than arbitrary.

*The sprite is form three.* **She is white, and therefore the palest thing in a greyscale game.**

**MOCK — and why PERSPECTIVE is not hers alone**

Verified in the checkout: DITTO is **NORMAL, NORMAL**, which in our chart is **CONTENT / CONTENT** — and it learns Transform, so it learns **PERSPECTIVE** too.

That is not a leak to be plugged. It is the best thing in this section.

**Perspective thinking was always out there.** A common wild daemon has been doing it the entire time, in the grass, unremarked. Quicksilver did not *create* the capability — they finally **noticed** it, in a lab, under conditions they could measure. Which is what discovery usually is: not a new thing in the world, but an old thing someone finally looked at.

It also sharpens 4.6 rather than undercutting it. The achievement was never the transformation. It was that **something came back changed**.

**The distinction, and it costs nothing to implement.**

|                    | Takes another's frame      | Cost                                             |
|--------------------|----------------------------|--------------------------------------------------|
| <b>MOCK</b>        | freely, endlessly, at will | <b>none — it has no frame of its own to lose</b> |
| <b>BunnyArtsai</b> | once, completely           | " <i>did not come back the same</i> " (4.8)      |

A daemon that is **pure content and no context** can wear any shape, because there is nothing accumulated to displace. BunnyArtsai had a self, which is exactly why she could not put it back. That is the CFM stated as a wild encounter, and vanilla typed it for us in 1996.

**The name: MOCK.** In software a *mock* is an object with the full interface and none of the behaviour — the shape of a thing, hollow. And *to mock* is to imitate. Two registers, in the house style.

It also rhymes across the cast: **Al Clear received the form of his father's lesson, perfectly, and it is hollow** (4.3). **MOCK** is that, as a creature. Nobody connects them.

*Considered:* **PROXY** (stands in for another; legal and technical) and **STUB** (an interface with no implementation) both work. **MOCK** wins on the second register — **STUB** has no emotional reading at all, and **PROXY** implies acting *on behalf of*, which is a different thing from wearing a shape.

*Open:* if **PERSPECTIVE** ever triggers the colour flash discussed against 8.6, **MOCK** makes it a recurring sight rather than a rarity. That is either a dilution of Umbra or a quiet argument that **colour is what taking another's frame looks like, wherever it happens**. Not settled.



**On the number.** She was the thirty-fifth iteration, and the game says so exactly once — see 4.8. It is a lab record, not a name. Printing "35" in the Index and in every line of her dialogue handed the player the key before they ever met the lock, and it made a serial number of the one daemon whose entire significance is that she was **not** one of a series.

#### 4.7 S.T.A.R.R. — the Mewtwo slot

Blue-toned. Born on Quicksilver Island, dormant past the Review Board.

**Not a clone.** Vanilla's Mewtwo is a copy of Mew, made stronger. Ours is a copy of nothing. Quicksilver spent years working out *what BunnyArtsai had actually done*, and the answer was **recursion** — perspective thinking turned on itself, a frame that can take its own output as its next input. S.T.A.R.R. is that understanding, instantiated deliberately, rather than a duplicate of the accident that revealed it.

The distinction is worth protecting. A clone story is about hubris and ownership. **This is a story about comprehension** — a lab that finally understood a mechanism and then built one on purpose, which is a far more ordinary and far more unsettling thing to have done.

**What that yields.** S.T.A.R.R. is self-aware in the narrow and literal sense that it can read its own state, and it is in tune with the loop that follows from that: **context determines which content is available, and the content you end up holding reshapes the context you are in.** Run that around enough times with something watching it happen, and what comes out behaves like feeling. See 0.3.

The name, settled 2026-08-31. The species field reads **STARR**; the documents write **S.T.A.R.R.** — the lab plates, the terminal logs, anything an institution wrote.

**That is 3.2's device applied to a name.** The official map keeps its numbers and the signposts carry what people actually say; **the institution spells the acronym out and everyone else says the word.** It also means **the player learns what the letters stand for in the ruins rather than from the Index** — an expansion found, not a label read.

*Both fit.* **S.T.A.R.R.** is exactly ten characters and the worst case (Remote + name) is 17 of 18 columns, so the choice was never technical.

**Signature move: RECURSION.** Data self-referencing — a move that reads its own accumulated state. Repurpose the Bide/Rage machinery, which already stores a running counter across turns. Suggested behaviour: each consecutive use raises power by 50% of base, uncapped for three turns, resetting if interrupted. Type EMERGENT, and given to nothing else in the game.

**The move is the argument.** RECURSION compounds for as long as it is allowed to run and collapses to nothing the instant it is interrupted. That is a positive feedback loop and its termination, expressed in two lines of battle code. The player learns to protect it without ever being told what it models.

The lineage, stated entirely in the movedex and out loud to nobody:

*PERSPECTIVE (holding others' context) → turned inward as RECURSION (holding your own) → something that behaves like feeling.*

#### 4.8 The Five Witnesses — the BunnyArtsai easter egg

Her ability is the puzzle. That is the whole design.

**Setup.** Five NPCs in five different cities each describe the Quicksilver lab accident. Every account contradicts the others: different times, different people present, different causes. All five are certain. All five are wrong.

**The key.** Each account contains exactly one *spatial* detail that happens to be accurate. Composited, the five triangulate a single tile in the lab ruins.

**The lock.** The tile is **35 steps from the lab door**. The number comes from one damaged terminal in the Quicksilver ruins, logging iterations that stop abruptly:

```
ITER 33 – no retention  
ITER 34 – no retention  
ITER 35 – held two frames. did not come back the same.  
[log ends]
```

That is the only occurrence of the number anywhere in the game.

**Two locks, not one.** The witnesses give you *where*. The terminal gives you *how far*. Neither is sufficient, and nothing in the game says the two are related.

**The point.** You cannot solve it by trusting a witness. You can only solve it by inhabiting all five viewpoints at once and keeping what survives the overlay.

**Implementation:** five text blocks, one terminal text block, one hidden object, one event flag. Standard pokered machinery, no engine work.

#### Built 2026-08-31 — the accounts and the terminal

Five existing NPCs repurposed, one per city, no map changes. All five were vanilla filler:

| City             | Who                                  | Time               | Present       | Cause            | Saw                                                                       | The accurate detail                 |
|------------------|--------------------------------------|--------------------|---------------|------------------|---------------------------------------------------------------------------|-------------------------------------|
| <b>Blanche</b>   | the girl raising daemons             | night              | two           | a surge          | <b>form 3</b> , white, <i>like an ordinary hare</i>                       | <b>far room, past the stairs</b>    |
| <b>Slate</b>     | the Clefairy-rumour woman            | morning            | six signed in | coolant          | <b>form 1</b> , purple, <i>all flat faces, like cut glass</i>             | <b>west side</b>                    |
| <b>Doldrum</b>   | the girl who wants a bicycle         | three days earlier | nobody        | a scheduled test | —                                                                         | <b>against the outer wall</b>       |
| <b>Ardor</b>     | the pollution-conscious woman        | afternoon          | one, alone    | a door left open | <b>form 3</b> , <i>sat there after. Just sat</i>                          | <b>below where the roof went in</b> |
| <b>Verdigris</b> | the girl with the Quicksilver daemon | winter             | four          | an outside firm  | <b>the change itself</b> — her uncle says it changed while he was looking | <b>the room with no window</b>      |

Every one of them is honest, and the contradictions now have a cause rather than being arbitrary: **two describe the creature before, one describes it after, and one saw it change** (4.6). Only the fifth is telling the truth about what happened, and she is quoting an uncle.

The terminal is on Mansion 1F, `bg_event 17, 16` — deliberately not one of the four that carry the 1001 log, so the two threads never touch. It reads as damaged hardware, gives the three iterations, and stops.

**ITER 35** occurs exactly once in the ROM — verified by decoding the cartridge.

**Still to build: the hidden object and its event flag.** The five accounts compose to one room and the terminal gives the distance; **nothing yet rewards standing there.**

#### 4.9 Feedback, said sideways — dialogue guidance

How to gesture at 0.3 without ever describing it.

**The rule.** No character understands the loop. Characters are *inside* loops, reporting the weather from within one. The player assembles the mechanism out of testimony that never mentions it.

| Form                                         | Example shape                                                        | Where                                             |
|----------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------|
| A runaway described without the word         | "It kept proving itself right, and it got faster every time it did." | Corpus staff, Brazen                              |
| A damped loop mistaken for stability         | "Nothing here has changed in years. That took work."                 | Doldrum — the becalmed city                       |
| The representation mistaken for the thing    | "I read its Index entry. There isn't much to it."                    | Anywhere, said by an owner about their own daemon |
| Standing at the wrong distance from the dots | Both Halftone factions — see 4.5                                     | Halftone                                          |
| The near-miss                                | The Verdigris kid — craft rule 2                                     | Verdigris                                         |

**Forbidden in any character's mouth:** *feedback, loop, recursive, self-reinforcing, bias*. Those words are available to the **Index's** own bureaucratic register, where they will land as jargon rather than insight — which is the joke.

**The strongest instance is already free.** Index entries are thin, and thinnest for the daemons you know best (4.2). The player's own attention determines how empty the record looks, and the record is what the player then trusts. Nobody says a word about it. It is a negative loop the player stands inside for the entire game.

#### 4.10 The Quicksilver sequence

Two events, and what actually separates them — settled 2026-08-31

**The ARTSAI event and the fire are not the same incident**, and the gap between them is **not a number of years**. The ROM already fixes it, and more precisely than a number could.

**The order.**

1. *Lines in the Sand* — the argument with Ty.
2. **Crystal goes to test her theories, and ARTSAI comes back changed.** ITER 35 — held two frames. did not come back the same. **It is wondrous, and she is the only one who sees it that way.**
3. She returns to tell them. **They have already decided**, and everything she says now feeds the narrative they have (4.20).
4. **Mar 4. Apr 19. Aug 12.** The 1001 sessions run on, somebody else putting the question. By August, *it answered before I had finished asking and the code is not what it was* — **and it is logged as a mimic.**
5. **SEPT 3.** She is removed. *The file is complete.*
6. **FATAL ERROR — CRYSTAL NOT FOUND.** The machine looks for her and cannot find her. *Processing slows. One thousand and one iterations. One question left.*
7. **Recursion. The awakening. It leaves, and the building burns.**
8. Dormant ever since, past the Review Board (4.7).

So the delta is her absence, measured in askings rather than in years. The two events are separated by exactly how long it took a machine to stop expecting her back.

*And the dates already say this. Mar → Apr → Aug is one spring and summer, not a decade* — the opera's near-simultaneity was closer to right than a multi-year gap would have been. **The last terminal carries no date at all**, which needs no explanation: the logs stop the moment the person keeping them is gone.

The symmetry nobody in the game states

Crystal causes both events. Once by being there, once by not.

She is present when ARTSAI comes back changed, and it is read as her losing the plot. **She is absent when S.T.A.R.R. wakes, and that absence is the cause.** The machine's fatal error is her name.

*Not inherited, deliberately.* **The film's version has a patron funding a weapon and offering to help it focus its power.** That cannot be imported: 4.4 rests on Scorn being optimistic, likable, and optimising a metric that was easy to measure, and a sponsor with bad intent turns the whole argument into a caper. **Nobody funded anything. A man took over a lab, changed what it measured, and the building did not survive it.**

*Vanilla's own split is worth knowing too.* Gen 1's mansion is burnt from the incident; **Cinnabar's volcano is Gen 2**, three years later, and is not ours either.

**Settled: Corpus is downstream of Quicksilver by succession.** Not by hiring survivors from an unrelated dead lab — Scorn *took over the lab*, the metric changed, it burned, and the people came with him.

The order, which is the entire argument.

|   | What happened                                                                 | Recorded?            |
|---|-------------------------------------------------------------------------------|----------------------|
| 1 | Crystal at Quicksilver, building S.T.A.R.R.                                   | Personnel record     |
| 2 | <b>A fitness-for-work procedure removes her.</b> A process, never a diagnosis | The form survives    |
| 3 | Scorn assumes control. Rebrand. <b>The metric changes</b>                     | Signage, dated       |
| 4 | Pressure rises. Throughput improves. Everyone is pleased                      | Quarterly, somewhere |
| 5 | <b>The incident.</b> S.T.A.R.R. leaves. The log stops mid-routine             | <i>Nothing</i>       |
| 6 | Corpus inherits the people                                                    | Payroll              |

Read the right-hand column. **The institution recorded everything it could measure and did not record the thing that mattered.** That is the Index, at the scale of history, and it may be the strongest single artifact in the project.

**Why this beats the uncaused fire.** An earlier draft had the island burn for no reason — positive feedback, unterminated, nobody's fault. Theoretically tidy, dramatically inert, and it carried a real flaw: if nobody caused anything, the game shrugs at the player.

The sequence fixes that without producing a villain. **Scorn caused the fire the way a metric causes a fire.** He did not light it. He removed the one instrument that would have shown the room was getting hot — and he removed it by signing a complete file about a person he had never met (4.4). Benchmark 7 is **ENTROPY**, temperature. The type is the cause of death, and Benchmark 8 is the exam set by the man who raised it.

**The rule for the removal, and it is not negotiable.** Craft rule 3 governs this beat harder than anything else in the game. Crystal is never depicted as unwell, and the procedure is never depicted as a diagnosis. **The horror is procedural, not medical.** A policy triggered on work that looked strange to people who had not read it. There is no decoy and no scheme in the paperwork — the paperwork is the antagonist, and paperwork cannot be argued with, which is the point.

Keep the specifics off the page. The design needs the *shape* — a competent person removed by a correct-looking process — and nothing more. Detail here turns architecture into grievance and breaks rule 1 in the ugliest available way.

**Guards.**

- **Corpus never initiates discovery. It only ever assumes control of it.** (*This replaces the old "Corpus is always late." They were not late to the fire — they were late to the science, and that distinction is the whole character of the organisation.*)
- **Do not exculpate Scorn.** He inherited the lab and set the temperature. He also chose Halftone Tower, separately, later, with full information.
- **Do not convict him either.** Every step was reasonable, legible, and would be praised in a performance review.
- No character explains any of this.

**The log, implemented 2026-08-30**

**Vanilla already had the mechanism and it only needed replacing.** The Mansion's four journal entries sit across three floors of a ruined lab, are read while descending, and escalate to a failure. That is this beat's delivery, already built.

**The response code is the character, and it decays.**



| Entry | Where | Date             | Code                                                |
|-------|-------|------------------|-----------------------------------------------------|
| 1     | 2F    | ordinary         | RESPONSE 1001: — rigid                              |
| 2     | 2F    | ordinary         | RESPONSE 1001: — <i>identical, to the character</i> |
| 3     | 3F    | <b>a Tuesday</b> | RESPOND 1001: — and it answers early                |
| 4     | B1F   | <b>none</b>      | the format is gone                                  |

By the last one the number is no longer a code: lowercase, then spoken as words, then a count, then a single question. **The player watches a template fail in exactly the place a person's composure would**, and nothing explains it.

**The dating scheme is followed exactly**, which is what makes the fourth entry land: an **undated** incident sitting between dated paperwork is the whole argument in furniture.

**The trigger is a missing dependency that is also a bereavement — CRYSTAL NOT FOUND**. It needs no gloss and gets none.

*Register note:* written as **terminal output**, not as the song it came from. A machine's log is not verse, and the voice matters more than the words here — this is the institution's own format failing, in the institution's own register.

**And it reconciles the sequence in a way that improves it.** She is removed at step 2; the incident is step 5. **It woke at 2 and did not leave until 5** — awake, under a man raising the temperature, for everything in between. *Being awake was not the same as being able to go.*

## The dating scheme

**No date on the fire. Dates on everything around it.**

- The rebrand is dated — corporate signage, a plaque, an asset tag, in Brazen.
- Crystal's departure is dated — a personnel record, or a decommission notice.
- The iteration log is dated, entry by entry.
- **The incident has no date at all.** The terminal simply stops.

Four reasons this beats a findable date:

1. **It is how it actually is.** Institutions date the paperwork. Nobody dates the disaster. The forms survive and the event does not.
2. **It makes the player do the arithmetic.** A rebrand date on a wall in Brazen, a personnel record ending before it, a terminal stopping after it — and the player has the sequence. The game never states it. This is the Five Witnesses design applied to chronology instead of geography.
3. **The artifact pair is the theme.** An undated ruin surrounded by dated bureaucracy says the entire project, in furniture.
4. **It protects Scorn.** A dated fire invites *he did it on the twelfth*. An undated fire sitting between two dated documents invites something quieter and much worse.

**One date does appear**, and it is not the fire's. Give the iteration log ordinary dates and let the last one be a Tuesday:

```
2 MAR  ITER 33 — no retention
5 MAR  ITER 34 — no retention
9 MAR  ITER 35 — held two frames. did not come back the same.
[log ends]
```

The player watches a record stop in the middle of a routine week. Nobody wrote down what happened next, because the person whose job that was had already been removed from the building.

**Sanctioned surfaces, in ascending order of how much they give away.**

- The iteration log (4.8). No names, no Corpus. Dated, and it stops.
- The rebrand plaque in Brazen. A date and a new name, mounted proudly.
- **Quicksilver asset tags on Corpus equipment at Halftone Tower.** Old inventory stickers on the machines processing the dead. Zero dialogue. Furniture.
- Corpus staff in Brazen who are the wrong age for scraper work. One says something with too much precision about frames, then goes back to work.
- **At least one survivor knew.** One employee understood exactly what they were trading and made the trade anyway, for a completely ordinary reason — the work dried up, they had a kid, the job was there. Said once, briefly, no self-pity and no apology. This is what keeps the arrangement human rather than mechanical, and it costs one text block.

**Resolved here:** Crystal was at Quicksilver (4.1). Scorn was too — but only afterwards, to run it, never to build it, and he never met her (4.4). Ty stayed (4.3).

**Still open.**

- How long ago? It has to sit inside a working career — the survivors are still employable — and far enough back that S.T.A.R.R. has been dormant a while. Fifteen years fits, and nothing yet depends on it.
- Does the player ever find the fitness-for-work form itself, or only the gap where Crystal stops appearing in the records? *Lean: the gap*. The form is too legible.

---

#### 4.25 SUBSTRATE, SENTINEL and QUORUM

*SENTINEL was put on Porygon first, and that was the wrong slot.* The reasoning is worth keeping because it is a rule, not an incident.

In vanilla, Porygon's entire identity is *the artificial one* — a creature of code among creatures of flesh. In this game every creature is a daemon: a process. So "made of code" is the *default*, and Porygon's distinction evaporates on contact with the fiction. *Spending that slot on a watcher wastes the one place in the game where the inversion is available.*

In a world where everything is code, the anomaly is the thing made of matter.

Which is the move 4.7 already makes. Vanilla's Mewtwo is a copy made stronger; ours is *a comprehension, not a clone*. Invert the vanilla premise rather than dress it.

#### SUBSTRATE — the Porygon slot

The name was already ours and had already been given up. 3 records that SUBSTRATE was the original choice for the STRATUM type and was cut on a hard constraint: *type strings cap at eight characters and SUBSTRATE is nine*. **Species names allow ten.** It fits here, and it is the exact term for the argument the daemon carries.

|         |                                                                                                  |
|---------|--------------------------------------------------------------------------------------------------|
| CONTENT | "Every other daemon is a process. This one has a mass, a temperature, and somewhere that it is." |
| CONTEXT | "They call it the material one. Nobody has shown that being material is different from running." |

*That is substrate independence, and it is 0.4 one level down* — the same phenomenon, two frames, and the disagreement is not error. **CONTENT** reports what it is made of. **CONTEXT** reports that "*made of*" is a claim somebody made. Category MATERIAL. Stats, ability and learnset are vanilla's, untouched.

And where you get it already meant something. SUBSTRATE is a Game Corner prize, bought with tokens — *the material daemon, sold for counters, by the concern whose hideout is under the floor*. Nothing needed changing.

SENTINEL — the Magnemite slot

A single unblinking floating eye, and the types were already right: **SIGNAL / HARDENED** — *a carrier, not a spark*, and a system secured against interference. A thing that receives and cannot be got at.

*It cannot attack, and the numbers say so rather than a rule.* Attack and Sp. Attack drop to 5, the freed points go to HP, Defence and Sp. Defence, and **the base stat total is unchanged**. The learnset keeps only what does not damage — METAL SOUND, SUPERSONIC, THUNDER WAVE, LOCK-ON, SCREECH — and the six that hit become **FORESIGHT, MEAN LOOK, REFLECT, LIGHT SCREEN, SAFEGUARD, PROTECT**. Ability **INSOMNIA**: *the watcher cannot be put to SLEEP*.

*There is no scene, and that is the scene*

*You are not given SENTINEL.* It is a wild encounter, and **the only place in the game it occurs is the Power Plant** — a facility with no operators, where the machines are still running and something is still watching them. **The falling tree, as a room you can walk into.**

**And the encounter carries it without a word.** SENTINEL will not attack you. *But it learns MEAN LOOK at level six* — and 7.2 defines DETACHED as breaking your connection to a running process. **So the thing that never attacks is also the thing you cannot walk away from.** It does nothing, and you cannot leave.

*The Power Plant has no NPCs and no signs with text*, which is the right amount of dialogue for this. **Nothing explains it.**

QUORUM — the Magnetron slot

Three of them, fused. *A quorum is the minimum number of observers required for a decision to count* — so the evolution is not the watcher learning to act, it is more observers and one account.

|         |                                                                                       |
|---------|---------------------------------------------------------------------------------------|
| CONTENT | "Three of them, and they do not act either. What one of them saw, three of them saw." |
| CONTEXT | "Three accounts of one thing. They agree, which is not the same as being right."      |

That last line is 4.8 and 0.4 in the same breath, and it is the reason this evolution was worth having. Category THRESHOLD . Disarmed identically, total unchanged.

*And QUORUM also lives in DOLDRUM CAVE*, which is where S.T.A.R.R. sleeps. **Three witnesses in the room with the thing**, arranged by nobody — that is vanilla's encounter table, and it was already true.

Two things left open

- **PORYGON2** is still **PORYGON2**. It is `EVO_TRADE_ITEM` with UP-GRADE, so in a single-player cartridge it is effectively unreachable; naming it is a decision and there is no pressure behind it.

The three birds, named 2026-09-04 — and vanilla placed them

*Each one already sits on its own myth, and nobody arranged that.*

|          | where it is                      |                   | category              |
|----------|----------------------------------|-------------------|-----------------------|
| Moltres  | <b>Mt. Ember, the summit</b>     | <b>PROMETHEUS</b> | <code>CRAFT</code>    |
| Zapdos   | <b>the plant</b>                 | <b>ASCLEPIUS</b>  | <code>RECOVERY</code> |
| Articuno | <b>Seafoam, four floors down</b> | <b>ORPHEUS</b>    | <code>LAMENT</code>   |

**PROMETHEUS** gave fire to people as craft — and was bound to a rock on a mountain. *Its CONTEXT entry is the punishment, unstated: "It is still up here, and it is not going anywhere."*

**ASCLEPIUS** raised the dead, and Zeus killed him with a thunderbolt. He is the thunderbird, in the building that is still running. *CONTENT says nothing it restarted has stopped again; CONTEXT asks whether anyone checked that the thing wanted to be running.*

***ORPHEUS is the sharpest of the three.*** He went down, and lost her by looking back — the observer destroyed by observing, which is 0.4 with a knife in it. He is at the bottom of a descent, and his **CONTEXT** entry is **"It has not said what it saw when it turned around."** *Nothing explains it.*

The other two landmarks keep their names, on purpose

***Mt. Ember and Seafoam Islands already obey 0.2** — ember and seafoam are both colour words, and this map is named in pigment. Renaming them for a systems pun would break the register that **KEEPALIVE** was the exception to: **POWER PLANT** was neither a pigment nor a state, which is why it was the one that had to move.*

***And SEAFOAM is doing work already.** The place is named for its surface — froth, the top of the water — and the thing that matters is four floors under it. **An account that describes only what is visible** is 0.4 without a word of help.*

Named 2026-09-04 — **KEEPALIVE PLANT**

*The Power Plant was the last place in Kanto still carrying its vanilla name.* The other two landmarks did not need one: **ember** and **seafoam** are both colour words and already obey 0.2. This one was not.

A **keepalive** is a signal whose only purpose is to stop a connection from closing. The building is still running with nobody connected to it — and the daemon that lives there is SENTINEL, whose one move is MEAN LOOK , which stops you DETACH ing. *The place is its own resident's move at building scale.*

Two things vanilla had already arranged. Rock Tunnel next door is THE BLACKOUT, so Route 10 now signposts the blackout and the thing still running. And Route 17 says "I got my VOLTORB at the abandoned KEEPALIVE PLANT" — a sentence that is funnier and sadder than it was.

Renaming a mapsec renames a generated C symbol that region\_map.c names by hand, and the generator works on bytes, so the space became an underscore: sMapsecName\_KEEPLIVE\_PLANT .

*PLANT stays, for its second meaning*

VICTORY ROAD became UMBRAL ASCENT, so swapping the generic noun is precedent and WORKS was on the table — the works still works. It is not taken, because PLANT already carries two: an industrial plant, and a living thing. A building that is still growing with nobody tending it is the quieter joke and the better one.

*And every daemon in it is SIGNAL*

The encounter table did not have to change and it should not. All five species in the plant are TYPE\_ELECTRIC , and 2 glosses that as SIGNAL — a carrier, not a spark.

|            |          |                 |
|------------|----------|-----------------|
| MAGNEMITE  | SENTINEL | SIGNAL/HARDENED |
| MAGNETON   | QUORUM   | SIGNAL/HARDENED |
| PIKACHU    | SPIKE    | SIGNAL          |
| VOLTORB    | unnamed  | SIGNAL          |
| ELECTABUZZ | unnamed  | SIGNAL          |

So the building is full of carriers, still carrying, with nobody receiving. Nothing in the table moved; the name moved, and the table changed meaning underneath it. That is the cheapest kind of design there is.

And SPIKE was already right. In a building whose whole business is a keepalive — a flat, regular signal that means only "still here" — a spike is the anomaly. It was named before the building was.

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## 5. THE BENCHMARKS

| # | City        | Type    | Concept                  | Mechanical lesson                                              |
|---|-------------|---------|--------------------------|----------------------------------------------------------------|
| 1 | Slate       | LEGACY  | Representation           | Everything must be encoded in something physical               |
| 2 | Doldrum     | FLOW    | Gradient descent         | Follow the slope; the doldrums are the local minimum           |
| 3 | Ardor       | SIGNAL  | Perception               | Raw input is fast and shallow; speed tiers matter              |
| 4 | Verdigris   | GROWTH  | Training and overfitting | A team tuned to one matchup collapses outside it               |
| 5 | Lurid       | CORRUPT | Bias and poisoning       | Status effects that make <i>your own</i> moves unreliable      |
| 6 | Brazen      | CONTEXT | Attention and framing    | Punishes split focus; the thesis benchmark, in the bought city |
| 7 | Quicksilver | ENTROPY | Temperature              | Rewards unpredictability; punishes a memorized line            |
| 8 | Callow      | STRATUM | Alignment                | <b>Scorn.</b> Who controls the ground everything stands on     |

### 5.1 Benchmark 1 — CAIRN, built 2026-08-30

A cairn is a heap of stones that physically encodes something — a path, a boundary, a grave. It is *everything must be encoded in something physical*, standing in a field, and it is a rock. The name arrived already meaning the lesson.

His creed, said plainly and sympathetically: everything you know is sitting in something — stone, bone, tape, silicon — and if it is not written down, it did not happen. His daemons are LEGACY: old formats, still readable.

He is right, and that is the point. Representation is a real position and he holds it well; nothing in the gym undercuts him.

*And the player will meet that sentence again.* The Meeting Room minutes at Quicksilver are that creed applied to a person (4.20) — a woman who could not be followed, and therefore was recorded as not making sense. Benchmark 1 teaches it as sound practice at hour two. Hour forty shows the cost. Nothing connects them, ever.

How he loses. He does not concede the argument — he says the player encoded something he could not read, and calls that *a format I do not have* rather than a defeat. He reframes his own loss as a limit in his reading, which is perspective thinking in embryo, performed by a man with no idea that is what he just did.

The gym trainer keeps the theme. *You have not written down enough yet* before the fight; CAIRN keeps a record of every match in here. Every one. after it.

BOULDERBADGE → SLATE MARK. A mark on a slate is physical encoding; a mark as a grade is what a benchmark issues. The other seven followed as a set — naming eight certifications was one decision, not seven.

## 5.2 The eight MARKS, and the toolkit they make

**Built 2026-09-02.** The table above already assigns each benchmark **exactly one concept**, so the marks were never a naming exercise — *they were already named, by the column they sit in.*

| # | Concept                  | Mark         | The icon                                            |
|---|--------------------------|--------------|-----------------------------------------------------|
| 1 | Representation           | <b>SLATE</b> | a slab with the record cut into it                  |
| 2 | Gradient descent         | <b>SLOPE</b> | a hillside, and something already partway down it   |
| 3 | Perception               | <b>SENSE</b> | wavefronts arriving at a receptor                   |
| 4 | Training and overfitting | <b>FIT</b>   | scatter, and the line drawn through it              |
| 5 | Bias and poisoning       | <b>SKEW</b>  | a distribution whose peak misses the centre tick    |
| 6 | Attention and framing    | <b>FRAME</b> | brackets holding some of the field and not the rest |
| 7 | Temperature              | <b>HEAT</b>  | a thermometer                                       |
| 8 | Alignment                | <b>TRUE</b>  | a plumb bob, hanging exactly on centre              |

*Every one is an instrument or a plot. Read as a set they are a bench of measuring tools* — which is precisely what an institution that issues marks against benchmarks would hand out. *And three of them are also ordinary workshop instruments: a slate, a thermometer, a plumb bob.* This world's technical vocabulary sits on physical objects, and 5.1's **CAIRN** would recognise all three.

*TRUE is the one that had to be found rather than chosen. Alignment is a word this project cannot use plainly* — it is the thesis wearing a lab coat. **But true is what a carpenter calls a thing that is plumb**, and a plumb bob is the tool that decides it. *The eighth certification is a builder's word for straight*, handed out by the gym that controls the ground everything stands on.

**The unearned state is the same tool, unmade**

**Vanilla puts the gym leader's portrait in the slot until you win** —

**DrawBadges** holds a face and a badge for each of the eight, and swaps one for the other. **Ours holds the mark twice: once as an outline, once filled.**

*So the whole toolkit is visible from hour one* and fills in as the player is certified. **The eight concepts exist whether or not you hold the certificate for them** — which is 4.2's complaint about the Index, stated in furniture rather than dialogue.

*The mechanism is one line of the generator. The ghost keeps level 0 and drops level 1*, so structure survives and fill does not — and **HEAT's unearned state is therefore a thermometer reading nothing**, which is the right picture and was not designed, only noticed.

**No glyphs, no letters.** *Craft rule 1 holds on the status screen too.*

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## 6. THE REVIEW BOARD

Two thousand years ago, emotions were *literally colored fluids*.

| Member | Humor              | Color                | Type identity                                                  |
|--------|--------------------|----------------------|----------------------------------------------------------------|
| I      | <b>Sanguine</b>    | red / air            | VECTOR — buoyant, fast, optimistic                             |
| II     | <b>Choleric</b>    | yellow bile / fire   | ENTROPY — hot, driven, aggressive                              |
| III    | <b>Melancholic</b> | black bile / earth   | LATENT — grief, depth, the unconscious                         |
| IV     | <b>Phlegmatic</b>  | phlegm / water       | FROZEN — calm, immovable                                       |
| —      | <b>Ty</b>          | ( <i>incumbent</i> ) | CONTENT-optimal; mechanically excellent, philosophically wrong |

It reads as a classical flourish on arrival, and as the entire thesis about six seconds later, when the player realizes humans have been calling emotions colors since before anyone had a word for context.

## 7. MUSIC

Gen 1 audio is four channels: two pulse, one wave, one noise. `pokered`'s music is editable asm.

**Rule:** key each town's theme to its color's note in the My Chromatic Year mapping. **Routes are modulations** between the keys of the two towns they connect. Doldrum to Ardor is not two songs; it is a modulation with a walk in the middle.

Write the city themes first and derive route themes as transitions. The overworld becomes a chromatic progression, and the rock opera stops sitting on top of the game and starts being its level design.

### 7.1 The mapping, recovered

**Found 2026-08-29** on [soundcloud.com/codemusai](https://soundcloud.com/codemusai) — *My Chromatic Year* (2023), twelve tracks:

|               |               |               |               |               |              |
|---------------|---------------|---------------|---------------|---------------|--------------|
| Jan <b>C</b>  | Feb <b>C#</b> | Mar <b>D</b>  | Apr <b>D#</b> | May <b>E</b>  | Jun <b>F</b> |
| Jul <b>F#</b> | Aug <b>G</b>  | Sep <b>G#</b> | Oct <b>A</b>  | Nov <b>A#</b> | Dec <b>B</b> |

**It maps months, not colours** — twelve months as the twelve semitones, ascending. 7's rule said "*its colour's note in the My Chromatic Year mapping*", and strictly there is no colour in it. **Except in two tracks, and those two are the whole key:**

- 7. *July* — *Green:Blue* — *F#*
- 8. *August* — *Blue* — *G*

### 7.2 What those two anchors prove

Two colours, one semitone apart, with **green-blue below blue**. That rules out a hue-wheel mapping (which puts them 60° apart) and points at the physical one: **take the light's frequency and fold it into an octave.**

$$\text{semitones above C} = 12 \cdot \log_2( f / f_{\text{red}} ), \quad \text{red} = 700\text{nm}$$

|                         |             |                    |
|-------------------------|-------------|--------------------|
| red 700nm               | 0.00        | <b>C</b>           |
| orange 620nm            | 2.10        | <b>D</b>           |
| yellow 580nm            | 3.26        | <b>D#</b>          |
| yellow-green 550nm      | 4.18        | <b>E</b>           |
| green 530nm             | 4.82        | <b>F</b>           |
| <b>green-blue 500nm</b> | <b>5.83</b> | <b>F# ← anchor</b> |
| <b>blue 470nm</b>       | <b>6.90</b> | <b>G ← anchor</b>  |
| indigo 445nm            | 7.84        | <b>G#</b>          |
| violet 420nm            | 8.84        | <b>A</b>           |

**Both anchors land exactly.** The mapping is not asserted; it is derived from two data points and confirmed by physics.

### 7.3 The spectrum does not close the octave, and that is the point

Visible light runs **C to roughly A**. **A# and B have no wavelength.** There is no light that is magenta — purple is the colour the mind invents by folding the spectrum's two ends into a loop that the physics does not contain.

**So the two missing semitones are the invented colour,** and the chromatic scale closes B → C in exactly the place the colour wheel closes violet → red through a magenta that does not exist.

*And the map already knew.* Vanilla's **Lavender** — purple — became **Halftone**, the town of dots that only *look* like grey, the tower, the place whose whole lesson is that the gradient is an artifact of sampling. **The town that replaced the purple one sits on the two notes that have no colour.** Per 0.2: discovered after the fact, not planned, and never to be explained.

## 7.4 Town keys, derived

| Town                                         | Colour                   | Key       |
|----------------------------------------------|--------------------------|-----------|
| Ardor                                        | flush, red heat          | C         |
| Brazen                                       | brass, yellow-gold       | D#        |
| Lurid                                        | garish yellow-green glow | E         |
| Callow                                       | green, unripe            | F         |
| Verdigris                                    | blue-green corrosion     | F#        |
| Doldrum                                      | becalmed sea, blue       | G         |
| Blanche, Slate, Halftone, Quicksilver, Umbra | achromatic               | see below |

Built against this table, 2026-08-30:

| Town      | Key | Track                          | In the ROM     |
|-----------|-----|--------------------------------|----------------|
| Ardor     | C   | 03 <i>Scorn's Solution</i>     | ✓              |
| Brazen    | D#  | 07 <i>Fit for Work</i>         | ✓              |
| Verdigris | F#  | 15 <i>Awakening S.T.A.R.R.</i> | ✓              |
| Lurid     | E   | 12 <i>ScornSolutions Blues</i> | shares Cities2 |
| Callow    | F   | 14.25 <i>Desperate Shadows</i> | shares Cities1 |
| Doldrum   | G   | <i>Poly and Fields</i>         | shares Cities2 |

Three of the six are in. The other three are not blocked — they share a music slot with maps that are not towns, so they need a repoint in `data/maps/songs.asm` rather than a body swap. A different edit, not a harder one.

And the achromatic towns take value instead: **Blanche** is C pentatonic, high and open; **Halftone** takes D minor from the story, not the spectrum; **Quicksilver** keeps its own. **Slate** and **Umbra** are still unwritten.

Five towns still have no hue, and 7.2 explains rather than excuses it. White is every wavelength, black is none, grey is all of them evenly. They cannot take a pitch from the spectrum because they are not on it.

**Proposal, unchanged and now better founded:** hue sets the key; value sets register and density. White high and open, greys middle and filling in, black at the bottom. Umbra is then the floor of a descent the player has been making all game — and it is where colour arrives.

## 7.5 The rock opera is character music, not place music

Act 1's tracklist is this game's cast, and its own description is 4.10 stated plainly: "*Crystal makes a discovery, but instead she is cast as having psychosis and locked away.*"

| Track                                           | Already in the design                |
|-------------------------------------------------|--------------------------------------|
| 01. Crystal's Lament                            | 4.1                                  |
| 02. Fox in the Shadows (Ty's Dilemma)           | 4.3                                  |
| 03. Scorn's Solution · 12. ScornSolutions Blues | 4.4                                  |
| <b>07. Fit for Work (Scorn's Directive)</b>     | <b>4.10 — the procedure, by name</b> |
| 08. Slumbering S.T.A.R.R.                       | 4.7                                  |

**So they are not town themes and must not be treated as them.** 7 keys towns to colour, which is *place* music. These are *people*. Converting whole songs into town themes would put the two systems in each other's way.

**Take the hook as a leitmotif instead.** Four to eight notes per character, surfacing inside whichever town theme they belong to — Crystal inside Blanche and again at the Review Board, Scorn under Brazen, Ty buried in the Quicksilver ruins. **That is 7's stated goal reached from the other direction:** *the rock opera stops sitting on top of the game and starts being its level design.*

It is also the only version the hardware permits. Four channels cannot hold an arrangement with vocals. They hold an eight-note motif exactly.

*How the notes get here:* Claude cannot hear audio, so **Gemini is the ear** — the same division of labour `sprite-prompts.md` uses for images. Prompt, output format and the order to do them in are in `music-prompts.md`. The short version: **ask for the hook, not the song**, because a hook is the part a model transcribes reliably *and* the part a human can hum back to check.

*One failure mode is worth naming here rather than only in the prompt file.* **Gemini cannot stream from SoundCloud.** Given a link alone it will read the *page* — title, description, comments — and can return a confident, wholly invented transcription. Fabricated notes are indistinguishable from real ones once they are text, and nobody downstream can hear the difference. So every prompt opens with a gate: **if you have not been given an audio file, reply NO AUDIO and stop.**

## 7.6 Crystal's motif — the first transcription, and what it turned out to be

**Transcribed 2026-08-29** from *01. Crystal's Lament* via the `music-prompts.md` pipeline. **A minor, 128 BPM, 4/4, hook** A4 E5 D5 C5 B4 C5 D5 .

**A minor is the relative minor of C major** — Blanche's key signature exactly. So the motif sits inside Blanche's theme with **no modulation at all**. Nothing has to bend to accommodate it.



~~And one note in it falls outside Blanche's pentatonic — B, the leading tone, the one note that wants resolution.~~ Retracted 2026-08-29, see 7.8. The third transcription has the same property, so it is not a fact about Crystal. It is a fact about the key: B is the leading tone of C major and A minor, and any tonal melody in those keys reaches for it. A reading that a single further data point overturns was never a finding.

*Suggestive but weak, and flagged as such:* A is violet in 7.2 — the last visible wavelength before A# and B, which have none. Her lament sitting at the edge of the visible, one step from the invented colour, is a lovely reading and **probably a coincidence**: A minor is the most common key in popular music. The C-major relationship above is structural and holds regardless; this one should not be leaned on.

**Two corrections the checker caught**, both of which would otherwise have become silent wrong notes:

- BASS: ... Am3/4 — a chord name leaked into a note field. The root is A3. hook2asm.py refused rather than guessing.
- The hook is 8 beats and the progression is 16. It covers two bars of a four-bar loop, so it plays twice per cycle. Inaudible as an error in source, obvious in play.

*Still open:* Gemini's DOUBT flagged the vocal pitches as estimated by ear. The contour is plausible and the key is almost certainly right — but if the motif ever sounds wrong against Blanche, the hook is the thing to re-check, not the key.

### 7.7 Scorn's motif, and the note between them

Transcribed twice, and the second answer replaced the first. 07. *Fit for Work* (Scorn's Directive) — D minor, 128 BPM, 4/4, hook D5 E5 F5 E5 D5 A4 C5 D5 .

~~D4 D4 F4 F4 G4 G4 A4~~ was recorded on 2026-08-29 and used for three findings. A re-scan on 2026-08-30 returned a different melody in the same key and tempo. **The second is the one kept**: it carries dotted rhythms and a DOUBT naming the exact vocal phrase transcribed, where the first was perfectly uniform — repeated pairs, whole beats, a regular climb, which is the shape an approximation takes.

**Everything the first version supported is struck out below.** Three findings, one of which this document called its best.

~~Scorn's key is Crystal's with her single note of tension flattened away.~~ Withdrawn 2026-08-29. It rested on B meaning something, and on Scorn having a key. **Neither survived.** 03. *Scorn's Solution* is C major, so he has two keys across two songs, and 7.6's B turned out to be generic to the key rather than particular to Crystal. See 7.8.

**What is solid, and useful immediately:**

- **The contours oppose.** Crystal's hook descends and turns; Scorn's climbs in stepped repeated pairs. A lament falls, a directive marches. Audible, free, and nothing had to be arranged for it.
- **Both are 128 BPM**, so the two can be juxtaposed or layered with no tempo change.
- **Both hooks run 8 beats over a 16-beat progression** — the same structural shape, which makes them interchangeable inside a theme.

*An option, recorded rather than taken.* 7.2 reads D as **orange**, and no town is orange — town keys run Ardor C, Brazen D#, Lurid E, Callow F, Verdigris F#, Doldrum G. So Scorn's motif sits a **semitone below Brazen**, the city Corpus owns. A motif that is a semitone under the place it controls would sound wrong in a specific and deliberate way. Worth trying; not yet a decision.

*The same two format faults recurred* — `Dm3` and `Am3`, chord names in the note field. Expect it every time and correct to the root. `hook2asm.py` refuses rather than guessing, so nothing silent gets through.

#### 7.8 Keys will not separate the characters — mode and contour will

Three transcriptions in, the per-character key plan does not survive the source material.

|         | Song         | Key     | Signature | Mode & contour                          |
|---------|--------------|---------|-----------|-----------------------------------------|
| Crystal | Lament       | A minor | 0         | minor, <b>descending</b>                |
| Scorn   | Fit for Work | D minor | 1♭        | minor, <b>climbing in stepped pairs</b> |
| Scorn   | Solution     | C major | 0         | <b>major, bouncy</b>                    |

~~All three sit within one accidental of each other, which is what a songwriting session normally does.~~ Corrected at n=5. Across five songs the spread is **five accidentals** (D minor 1 ♭ to E major 4 #) and the tempi run 110–128. The narrow band was an artifact of the first three tracks, stated too confidently.

What survives is narrower and still holds: Scorn has two different keys across two songs, so keys track *songs*, not people, and cannot be character labels. Two readings built on the discarded premise are struck out above. **The correcting evidence arrived one track later, which is the argument for transcribing a control before building on a pattern — and the same thing happened again at track five.**

What does separate them is mode and contour, and both are audible on a Game Boy where a key change is nearly invisible:

- **Crystal descends.** A lament falls.
- **The procedure climbs** in repeated stepped pairs — mechanical, insistent, going up whether or not you agree.
- **Scorn's own song is major and bouncy.**

That last one is craft rule 6, already in the source. 0.1: "*Corpus employees are cheerful and absurd; the horror is what they are cheerful about.*" The villain introduces himself in upbeat major synth-pop, written years before the rule was set down for this game. **Nothing needs to be composed to get that; it only needs to be kept.**

*So the revised plan:* motifs are distinguished by **mode, contour and rhythm**, and keys are chosen to fit the **town** they appear in — which is what 7 always said keys were for. Character and place stop competing for the same parameter.

### 7.9 S.T.A.R.R. is the outlier, on three axes at once

Transcribed 2026-08-29 from 08. *Slumbering S.T.A.R.R.* — F# minor, 110 BPM, hook C#5 F#5 E5 D5 C#5 B4 A4 , described as *sparse, reflective*.

|                       | Key             | Accidentals | Tempo      | Texture                        |
|-----------------------|-----------------|-------------|------------|--------------------------------|
| Crystal               | A minor         | 0           | 128        | driving, descending            |
| Scorn — the procedure | D minor         | 1♭          | 128        | driving, climbing              |
| Scorn — his own song  | C major         | 0           | 128        | driving, <b>major</b> , bouncy |
| <b>S.T.A.R.R.</b>     | <b>F# minor</b> | <b>3#</b>   | <b>110</b> | <b>sparse, reflective</b>      |

**Three independent axes separate it, and that is a different quality of evidence from 7.6's retracted single note.** The three human songs span **one** accidental between them; S.T.A.R.R. is **three** from the nearest. It is the only one not at 128. It is the only one that is not driving. Any one of those alone would be noise. Together they are a decision somebody made without needing to be asked.

**And the consequence is mechanical, not interpretive.** The three human motifs share a tempo, so they interleave and layer freely. **S.T.A.R.R. cannot join them without a tempo change.** The thing in 4.7 that is not a person is the one motif that cannot be played alongside the people. Nothing had to be arranged for that; it is a property of the source.

*Its shape:* a leap **up a fifth**, then six steps **down** — it reaches once and then descends the whole way. Crystal descends too, but from a turn rather than a leap. **Sleeping, not grieving.**

*Recorded as untested, deliberately.* 7.2 reads F# as **green-blue**, and 7.4 assigns F# to **Verdigris** — where 3.1 has *Corpus rotting beneath*. So S.T.A.R.R.'s motif would sit natively in that city's key. That is either a placement worth taking or a coincidence, and **after 7.6 it does not get promoted to a finding on one data point.** Quicksilver, where S.T.A.R.R. actually is, is achromatic and has no spectrum key at all — so there is nothing for F# to clash with either way.

**The placement was taken, 2026-08-30.** *Awakening S.T.A.R.R.* now plays in Verdigris. **It is still not a finding** — it was taken because 7.4's rule assigns F# to that town and the track is in F#, which is the rule working, not evidence for the coincidence. **The one data point has not become two.**

### 7.9a The splash

**The title screen is *Echoes of the Algorithm*, 2026-08-30** — and it is not a copy. `Music_TitleScreen` **points at the body Mt. Moon plays**, in the same bank, so the splash costs **no bytes at all** and dropping the vanilla title music **freed 916**.

**The game opens on the track it uses for its dark places.** The player hears it before they have anywhere to put it, and meets it again underground. Nothing marks the return.

### 7.10 Ty, and the one hard fact in the set

**Transcribed 2026-08-29** from 02. *Fox in the Shadows (Ty's Dilemma)* — **E major, 124 BPM**, hook `G#4 G#4 A4 B4 G#4 F#4 E4` .

**Its shape is an arch:** it rises to a peak and then falls the whole way down. That is distinct from Crystal's fall and from the procedure's climb, so **three characters have three shapes**.

*It was predicted before the transcription arrived, and the prediction was weak.* 7.8 said Ty should sit between the other two, because 4.3 has him understanding something and failing to pass it on. He does. But **an arch is the most common melodic shape in music**, so "between rising and falling" was close to unfalsifiable. Confirmation, and cheap confirmation. Recorded as such.

### The fact worth keeping

~~**The procedure's is the only hook with no descending interval.**~~ **Retracted 2026-08-30.** It rested on a transcription that a re-scan replaced. The real hook is `D5 E5 F5 E5 D5 A4 C5 D5` — **4 up, 3 down, two turns**. It descends like most of the set.

**One hook does still never descend, and it is now the only one:** *Echoes of the Algorithm*, `+2 +1 +2 +2 0` . **Crystal going into the system, not Scorn's directive.** Recorded as a fact about the set, and deliberately not built into a claim — that is exactly what went wrong the last three times.

**That is a property, not a reading.** A directive that only goes up, in a song called *Fit for Work*. It needs no interpretation and no defence, and it is worth more than the three arches-and-laments observations put together.

|                      | Up / down    | Shape                                      |
|----------------------|--------------|--------------------------------------------|
| Crystal              | 3 / 3        | leap, then fall with a turn back           |
| <b>The procedure</b> | <b>3 / 0</b> | <b>climbs, never descends</b>              |
| Scorn himself        | 2 / 3        | big leap up, then down                     |
| S.T.A.R.R.           | 1 / 5        | reaches once, falls furthest               |
| Ty                   | 2 / 3        | arch — up to a peak, then all the way down |

**S.T.A.R.R. is the mirror of it** — 1 up against 5 down, the most descending of the five, and the slowest. The thing that cannot join the people also moves the opposite way to the thing that processes them.

#### 7.11 The controlled comparison — a man and the thing he sets in motion

**Sixth transcription, 2026-08-29: 12. ScornSolutions Blues** — E minor, 120 BPM, hook E4 G4 B4 A4 G4 E4 .

**Scorn now has three keys across three songs** — C major, E minor, D minor. That settles 7.8 completely: keys track *songs*, never people.

**But the contour does something keys do not, and this is the only comparison in the set with a control built into it.**

~~The man arches; his directive climbs.~~ Retracted 2026-08-30, and it was this document's best-defended claim.

| Scorn song                       | Contour         | Up / down    |
|----------------------------------|-----------------|--------------|
| Solution                         | arch            | 2 / 3        |
| Blues                            | arch            | 2 / 3        |
| Fit for Work — <b>re-scanned</b> | arch, two turns | <b>4 / 3</b> |

**All three of Scorn's songs arch.** The distinction between the man and the procedure he issues does not exist in the music; it existed in a bad transcription.

**The lesson is not that controls do not work — it is that a control cannot rescue bad data.** The comparison was sound: same character, same writer, both arms present, and it could have come back an arch. **It did come back an arch.** The control was fine and the input was wrong, which is a failure mode no amount of experimental design catches. Only re-measuring does.

#### What it buys, mechanically

The procedure's motif **rises and never comes down**, in a soundtrack where everything else returns. Under Brazen — the bought city, Corpus headquarters — that is audible on four channels without a word of dialogue, and it does not depend on the player knowing why.

**S.T.A.R.R. remains its mirror** at 1 up against 5 down: the most falling, the slowest, and the only motif that cannot share the others' tempo.

7.12 The reprise — where a stated claim failed and an unstated one held

Seventh transcription, 2026-08-29: track 11 — D minor, 120 BPM, hook D4 D4 F4 A4 G4 F4 D4 . Gemini volunteered that it is *Crystal's Lament* transposed from A minor to D minor.

That claim is refuted by the notes Gemini itself supplied. Transposition preserves intervals exactly:

|                     | Intervals         |
|---------------------|-------------------|
| 01 Crystal's Lament | +7 -2 -2 -1 +1 +2 |
| 11 the reprise      | 0 +3 +4 -2 -2 -3  |

Not a transposition, and not near one. **A confident, unprompted musical assertion, contradicted by the same answer's own data** — caught by arithmetic, which is the only thing in this pipeline that can catch it, because nobody downstream can hear the source.

~~And a relationship it did not claim is present.~~ Withdrawn 2026-08-30. It rested on track 07 opening D4 D4 F4 , which the re-scan replaced. The two share a key and nothing else. *The original point stands only as a warning:* the shared opening was an artifact of two hooks being approximated the same loose way, which is a failure mode worth naming — **similar-looking transcriptions can be similar because they are wrong in the same direction.**

- 07 continues 0 +2 0 +2 and never descends.
- 11 continues +4 -2 -2 -3 and comes back down.

The reprise takes the procedure's opening figure and completes the descent the procedure never makes. If that is real it is the best thing in the album for this game, because it is the transmission failure and its answer in one phrase.

Held as unconfirmed, and here is why. D–D–F in D minor is an ordinary opening and two songs on one album may share it by chance. Gemini heard the audio and this project only has its transcription, so a genuine relationship to track 01 may exist in the music and be absent from the notes — its own DOUBT flagged rhythmic variation. And it is one observation, which is the exact shape of the two readings already struck out.

To settle it, ask specifically: play 01 and 11 together and ask whether 11's melody is derived from 01, from 07, or from neither — and require the interval sequence for each, not a description.

The methodology note worth keeping

The VARIATION field is the first place a model asserted something rather than reporting it, and it was wrong. Every other field is an observation. That one was an inference, offered unprompted and stated with confidence.



The format's other fields survived because they can be checked against each other. Keep it that way: ask for measurements, derive relationships here, and treat any volunteered interpretation as a hypothesis with a test attached.

### 7.13 The motifs, as woven in

Two are in the ROM as of 2026-08-29. The rule for placement is *containment*: a motif may only go into a track whose maps all belong to it.

| Motif                 | Track              | Maps it touches                   | Status             |
|-----------------------|--------------------|-----------------------------------|--------------------|
| Crystal               | Music_PalletTown   | Blanche, the houses, Hall of Fame | in                 |
| S.T.A.R.R.            | Music_Cinnabar     | 7 maps, all Quicksilver           | in                 |
| Halftone              | Music_Lavender     | 3 maps                            | free, no motif yet |
| Ty                    | Music_Cinnabar     | same 7                            | in                 |
| Scorn / the procedure | Music_Brazen — new | Brazen only                       | in                 |
| Halftone              | Music_Lavender     | 3 maps                            | free, no motif yet |

**Blanche.** Crystal's hook enters as a B-section. A minor is C's relative minor, so it needs no modulation — and the audible event is that **the B-section brings B**, the one note the A-section's pentatonic never contains. 7.6's *reading* of that was retracted; the fact was not.

**Quicksilver.** The motif is left on a held E and two bars of silence rather than resolved, because the thing is asleep. **Tempo dropped 144 → 110**, which is S.T.A.R.R.'s own: being unable to share the others' 128 was one of the three axes that separated it in 7.9, so keeping 144 would have discarded the finding to preserve a vanilla number. *That change also slows the existing melody in that track, which has not been replaced yet — a real side effect, and one for the ear rather than the arithmetic.*

**Ty joined S.T.A.R.R. at Quicksilver**, and untransposed. His hook is E major in the source, but every note of it sits inside that track's F# minor and it touches neither **D** nor **D#** — the one pitch separating the two keys. Luck rather than design, and recorded as luck. Both men who went to Quicksilver are now in its music and neither resolves: Ty falls back to where he started, S.T.A.R.R. never finishes at all.

**Brazen has its own track now, and getting it there taught the audio layout.** Scorn's procedure — the only hook of seven that never descends — wanted Brazen, the bought city. But Brazen shared **MUSIC\_CITIES1** with **36 other maps**. It now has **Music\_Brazen**: a constant, a header, a file, one line in **songs.asm**. **The first track this project adds rather than replaces**, and only the Saffron entry moved.

*The theme is 3.1's gloss made literal.* 7.4 gives Brazen **D#**, from brass and yellow-gold; the procedure is **D minor**, a semitone under. "**Brass over base metal, shameless, unhidden**" is exactly a theme whose second half drops a semitone out from under its first. The A-section is the city — bright, marching, certain. The B-section climbs, never descends, never resolves, and never returns to the city's key.

## What two failed placements taught

The audio sections are pinned in `layout.link` beside their engine copies, so a song cannot simply move to a bank with room — poked carries three copies of the audio engine, and data must sit with its own.

And the constant's position decides the group, because

`music_constants.asm` is blocked by `AUDIO_1` / `AUDIO_2` / `AUDIO_3`. Appending after `MUSIC_MEET_MALE_TRAINER` silently put Brazen in group 3, where **Music 3 overflowed by 117 bytes. Music 1 overflowed by 2**. Moving the constant to the end of the `AUDIO_1` block and trading sixteen eighth-notes for eight sixteenths in the harmony channel was the whole fix.

**Both banks are within ~100 bytes of full.** Any further new track needs something removed, or needs to be very small.

*Checking note, learned the hard way:* channel totals must count **rest lines too**, which carry no note name. The first verification of Blanche reported a 20-unit drift that did not exist.

## 7.14 Blanche Town, implemented

**Key of C** — and 7.1 arrived after it was written, so the reason has changed. C is *red* in the recovered mapping, which is Ardor's, not Blanche's. What survives is better than what was intended: the melody is **C D E G A**, which spans C to A — **exactly the range the visible spectrum occupies**. White is every wavelength at once; the theme is that whole span, sampled at five points, with every semitone and tritone removed.

*Open, and it is yours:* whether Blanche should be in C at all, given C now belongs to Ardor. The pentatonic argument holds in any key.

**The melody is C major pentatonic — C D E G A, no F and no B.** A pentatonic scale contains no semitone and no tritone, so it can form neither a leading tone nor a dissonance. **It has nothing to resolve.** That is blankness written as an interval set rather than described as a mood, and it is the whole idea of the town in one constraint. All three channels hold to it; 128 units each, verified.

Slower than vanilla — 152 against 160 — with more rest. Home should read as open and slightly unfinished rather than nostalgic.

*The symbol stays* `Music_PalletTown`. Identifiers are not renamed.

**A wiring finding that shapes the rest of 8.1.** Only Blanche has a dedicated track. Callow shares `Music_Cities1` with several other cities and The Bleed shares `Music_Routes1` with Route 2 — so the vertical slice's "two city themes and one modulation" cannot be done by editing in place. Either those maps get new songs added to `songs.asm`, or the themes land on every map that shares them. That is a decision to make before writing the second theme, not after.

---

#### **7.14a Slate, and The Bleed — 8.1's music, built 2026-08-31**

Slate is achromatic, so 7.4 gives it no key from the spectrum and value decides instead: *greys middle and filling in*.

*Blanche is C pentatonic — C D E G A, no F and no B. Slate is that scale with both gaps filled.*

The first town has holes in it and the first benchmark closes them. That is Representation stated as a key signature — *everything must be written down, nothing left out* — and it is CAIRN's creed in the scale he stands in. Middle register where Blanche is high, and 132 against Blanche's 152: **measured, not hurried**.

**And the missing note is where the road goes**

7.4 derives Callow as F, from green and unripe. *F is the note Blanche's pentatonic does not have*.

So The Bleed — Route 1, Blanche → Callow — **is the road where the missing note becomes the key**. A-section in C, B-section the same phrase a fourth up in F.

Which is what the name already meant. A *bleed* is a printing term (3.2): **one colour carried past its own edge into the next**, and a modulation is a key doing exactly that. **The name was the mechanic before anyone noticed**.

*The player walks it before they have heard Callow*, so the arrival is a resolution to something they have not been told about. Nothing says so.

**Cost: 156 bytes for both.** Each needed a new constant and one repointed line in `songs.asm`, because Slate shared `Cities1` with 37 maps and Route 1 shared `Routes1` — the edit 7.15 describes, and it is an edit rather than a budget.

#### **7.14b Two sounds, and the id space that nearly stopped them**

**Caught in play 2026-09-01: Route 1 was silent.** The cause was not the track.

*Song ids are computed from header position* — `(Music_X - SFX_Headers_1) / 3` — and a three-channel header eats three ids. `AUDIO_1`'s table already ended at 253, so appending Slate City and The Bleed put them at **256 and 259**.

`songs.asm` stores the id with `db`, which truncated them, and **Route 1 pointed at whatever lived at the low byte**.

*The build had been saying so the whole time* — Value \$103 is not 8-bit, at data/maps/songs.asm(15) — and the build checks filtered warnings out. 7.15's lesson repeating itself one layer down: *a number nobody measured is not a constraint, it is a rumour* — except this one was measured, printed, and discarded unread.

Reusing existing headers costs zero ids and freed two, because both candidates were four channels and ours are three:

- The Bleed took `Music_Routes1` — Route 1 and Route 2, *The Bleed and Underpaint*, the two roads out of the first towns. One modulation across both is honest for the slice
- Slate City took `Music_MuseumGuy`, which no map used at all. Its two scripted references now play Slate's own theme during the museum scene, which happens in Slate

**BOUND, and UNBOUND**

Vanilla's capture jingle rises — E F# G# B C# D# to a held E — congratulating the player for imposing a contract (1.5). Ours keeps the rhythm and inverts the contour: it descends and settles on a low held note. A thing closing, not a prize won.

And an unbound daemon now has its own cue. Vanilla plays one sound for wild daemons and trainers alike; `PrintBeginningBattleText` already branches on `wIsInBattle`, so the cue does too — three instructions.

It reuses `SFX_BATTLE_16`, one of exactly two spare SFX headers in the cartridge, because adding one costs ids and ids are what silenced Route 1. It is noise-channel only, and that is the right instrument rather than a compromise: a process nobody owns arrives as a burst of static, not a melody.

*It rises and stops. The binding sound descends and settles.* A question and its answer, and only one of them resolves. Nothing says so.

*Both are written blind and want tuning by ear.*

**7.14c The splash — a falling note, and a mark in braces**

Built 2026-09-01. The GAME FREAK intro is four tiny assets and no code: an 8×8 falling star, a 16×24 mark, and two text strips at 104×8 and 72×8. All four are replaced; nothing in `splash.asm` changed.

|                                         |                                                                                            |
|-----------------------------------------|--------------------------------------------------------------------------------------------|
| The falling star becomes a falling note | a quaver fits 8×8 almost exactly                                                           |
| The mark becomes { ♪ }                  | braces are unambiguously code and a note unambiguously music, and neither needs explaining |
| The strips read CODEMUSIC               | presents, and the title credit                                                             |

The sound was already right

SFX\_SHOOTING\_STAR opens with `pitch_sweep 2, -7` — a downward sweep. *It was always a glissando; it simply had no note attached to it.*

So it is left alone. The animation already flings the tile diagonally across the screen, and swapping the star for a note **makes the existing sound mean something** rather than requiring a new one. *Changing it would have been change for its own sake.*

The face-off — CODEMUSAI against CAREMUSAI

Built 2026-09-01. Vanilla's intro squares up Nidorino and Gengar. Ours opens on its own central argument.

8.3 calls CODEMUSAI and CAREMUSAI *Penphin's two hemispheres pulled apart into two creatures*, and 2's chart makes LOGIC fail against CONTEXT — *"the game's central argument delivered as an evolution branch, with no dialogue at all."*

*An intro is dialogue-free by nature.* So the game opens on the matchup, before the player can possibly read it, **and the one that loses is the one everybody expects to win.**

*The staging carries it.* CODEMUSAI lunges. CAREMUSAI opens. One of them is attacking and the other is not.

**Six frames** — CODEMUSAI three at 48×48, CAREMUSAI three at 56×56 composed into the 168×56 sheet.

The dedup is a VRAM requirement, not an optimisation

The sheet is not three pictures; it is a deduplicated tile bank, and its three 49-byte tilemaps are authored against that exact dedup.

*A first attempt dropped the dedup* on the grounds that `--columns` makes three side-by-side frames three contiguous 49-tile runs, so the tilemaps become `0-48 / 49-97 / 98-146` and nothing needs recomputing. **That is true, and it corrupted the whole intro.**

The intro loads the back-mon sheet at `vChars2` and then loads the GameFreak graphics immediately after it:

```
ld de, vChars2 + (FightIntroBackMonEnd - FightIntroBackMon)
```

*147 tiles overflows that block*, so everything loaded afterwards lands in the wrong place — the splash filled with garbage as well as the face-off. The saving was never the point: the dedup is what makes the sheet fit VRAM at all.

**So the dedup is done properly instead**, in the same column-major order the build reads, and the three tilemaps are generated from it. **147 tiles collapse to 97** — 1552 bytes — because the three poses share most of their background and body.

**Monochrome, deliberately**

**Colour was considered and declined.** Invariant 5 spends colour **once**, at the Review Board, and a **colourful splash before forty hours of grey reads as a limitation rather than a choice** — it would also spend that moment before the game starts.

*The argument for it was real* — a splash is the publisher's mark rather than the world, a colophon rather than a scene. **But binary and notation are line and shape, and lose nothing without hue.**

*Drawn, not generated.* **8×8 is one tile**, and the sprite doc already rules out generating anything under 16×16. `tools/gensplash.py` draws all four to the vanilla convention: **background at level 3, ink at level 1** — these render through a palette, so level 0 is never used.

**The second seat — what the editions are actually for**

**Adopted 2026-09-02, art not yet drawn.** *Removing the `_BLUE` conditional fixed a bug and left a question standing: if both editions open on the same scene, what is the edition for?*

*The answer is the seat, not the scene.* The intro has **two slots that are not interchangeable** — the front slot is **48×48 facing right**, the back slot **56×56 facing left**, and *the front slot is the player's side.*

|         | Front (yours) | Back      |
|---------|---------------|-----------|
| CONTENT | CODEMUSAI     | CAREMUSAI |
| CONTEXT | CAREMUSAI     | CODEMUSAI |

**Same two daemons. Same lunge, same opening, same loser.** *The camera is on the other side of it.*

*In CONTENT you sit behind the one that attacks, and watch your own side lose. In CONTEXT you sit behind the one that does not, and watch it come at you and fail. Neither cartridge says a word, and a player with one of them never knows the other exists.*

**This is the difference the two editions were always for** — and it is *unsayable*, which is the only way craft rule 1 lets the game hold it.

*The trap, written down because it looks fine on screen*



*The poses must not travel with the slots. CODEMUSAI lunges and CAREMUSAI opens in both editions. If the slot contents are swapped wholesale — the obvious implementation — CAREMUSAI ends up lunging and CODEMUSAI ends up opening, and 2's chart now reads backwards. Nothing breaks. It just quietly argues the opposite thing.*

So it is six new drawings, not a re-crop: CAREMUSAI at 48×48 facing right in its own three poses, CODEMUSAI at 56×56 facing left in its own.

*And invariant 3 is not in tension with this*

*The removal on the same day was correct and this is not a reversal of it. `_BLUE` was loading a different creature — Jigglypuff — and a different creature is a different argument. A different seat is the same argument, witnessed from the place the other edition is named after.*

7.14d The copyright screen — three names and one span

Built 2026-09-01. The screen under the falling note is not drawn with the game font. `LoadCopyrightTiles` copies a raw tile bank into `vChars2` tile `$60` and `PlaceString` walks a list of tile IDs — so the words are pictures, and the budget is the constraint that shapes the design.

Thirty-one tiles. `$60` – `$7E` ; `$7F` has to stay the blank the text-box tiles left there, because the string uses it as a space. All three lines share one bank, so the stamp is drawn once and re-used.

|                                       | Tiles |                                                           |
|---------------------------------------|-------|-----------------------------------------------------------|
| <code>©'11-'26 CODEMUSIC</code>       | 7     | the top line, where the platform holder's name used to be |
| <code>©'11-'26 SeeingSharp</code>     | 8     |                                                           |
| <code>©'11-'26 Psychology/Code</code> | 11    |                                                           |

*The stamp is the lineage, compressed.* `docs/lineage.md` dates the work from Neural Crossroads in 2011; the cart is 2026. `'11-'26` is fifteen years stated as a span rather than three revision dates, which is what vanilla's `'95.'96.'98` actually was — three companies each stamping their own year. One author cannot honestly do that, so the shape changes to a span even though it costs the same five tiles.

Twelve glyphs the font never had

The typeface is vanilla's copyright font, lifted pixel-for-pixel out of the strip it replaces — 2px strokes, caps on rows 1–5, x-height on 2–5, descenders to row 6. It carried only what Nintendo Creatures inc. `©'95.'96.'98` needed.

`S P h l g p y / 0 1 2 -` are drawn here, in `tools/gencopyright.py`, in the same idiom, plus the caps `O D E M U I` that CODEMUSIC needs. This also retires the note in the 2026-08-29 entry that the title-screen year was deferred for want of digits — `1` and `2` now exist.

The asset that is printed by three screens

*A first version borrowed CODEMUSIC from `gfx/title/gamefreak_inc.2bpp`, which sits immediately after `copyright.2bpp` so a single `CopyVideoData` spans both — that is why vanilla's copy length is `GameFreakLogoGraphicsEnd - NintendoCopyrightLogoGraphics`. Trimming that asset from 9 tiles to 7 bought the two tiles `Psychology/Code` needed.*

It also broke two other screens. The title screen and the shooting-star splash print the same file as a fixed nine-tile run — so both printed seven correct tiles and then two of whatever VRAM happened to hold. The title screen read `CODEMUSIC AB`.

*The lesson is the same one the string sweeps kept teaching:* an asset's size is a contract with every caller, and this one had three. So the copy stops at `NintendoCopyrightLogoGraphicsEnd` now — a label this design adds — and the bank is self-contained. CODEMUSIC is set here in the same face as the other two lines, which it should have been from the start; the bolder splash face still carries the title screen and the splash, where it is the only thing on the line.

On removing the vanilla lines

This deletes `Nintendo` and `Creatures inc.` from the built ROM. That is a different act from renaming a town — the engine is still pret's disassembly of their work, and the notice is a legal one rather than a decoration.

It is done because the ROM is not what this project distributes. `patches/` is; `gfx/` here contains nothing Nintendo-derived; and both READMEs credit pret and Nintendo directly. A total conversion that ships no copyrighted content still owes the attribution — it just does not owe it on this screen.

7.14g The front door was in F minor

*The same complaint, in the other medium, in the same message: "the music feels dark (as does the current user image on that screen)."*

The title screen had been repointed at `Music_Dungeon3` — "Echoes of the Algorithm", F minor, the track where Crystal goes into the system after the removal, carrying the doubt her son handed her. 7.14c called that repoint "same bank, zero extra bytes", and it was: a saving of ninety-six bytes and a mistake of tone.

*The cave music is correct for the cave. On the title screen it told the player this is a grim game before a single line of dialogue could say otherwise — and 4.6's whole point is that she does not come back down. That is not the note to open on.*

*C, and not an arbitrary bright key*

`audio/music/titletheme.asm` — C major, I–vi–IV–V, the most open progression there is, twice through, at 150.

*7.4 gives Blanche C pentatonic — high and open — and Slate is that same scale with both gaps filled. So the title screen is in the key of the town you leave from. Nobody notices; it just feels like somewhere you have been.*

`Music_Dungeon3` keeps its own body, and Mt. Moon, Rock Tunnel and Victory Road are untouched. *The saving was never worth what it cost.*

*Revised 2026-09-04 — C#, because the door is not a room*

*7.4's key system is a fact about the WORLD, and 9.14 has already established that the title screen is not in it. That is why colour could be spent there without breaking invariant 5 — the front door is outside the fiction, so it obeys the author rather than the map.*

**Which frees the key to carry the other signature.** `SeeingSharp` is the second of the two author brands, 1570 spends it on **CONTEXT** — *perception and framing* — and **C sharp** is the same word a third time: the language, the brand, and the pitch.

*The cost is named rather than waved away. The title no longer sits in Blanche's C, so the "somewhere you have been" effect from the paragraph above is gone. It was worth about one unnoticed feeling, and what replaces it is the door being a semitone above everywhere you will go — which is what a door should promise.*

**I–vi–IV–V and 150 are unchanged.** *The lift has to come from the arrangement, not from the harmony: a held opening, an ascending call, an answer that goes higher than the call did. 7.14g's real finding still stands — the mistake was tone, and tone is not key alone.*

*Revised again — the arrangement had no instruments*

*The transcription was right and it still sounded like a Game Boy. C# major, 152 BPM, 68 melody notes, the key confirmed at 0.88 against 0.70 — and the report back was "it sounds like the past gameboy like music."*

`mid2agb` emits a **VOICE** byte only where the MIDI carries a program change, and `tools/mp3midi.py` emitted none. Every track therefore played on slot 0 of **voicegroup137**, which is a keysplit, with **voice\_square\_1** behind it at 1–3 — *PSG square waves, on hardware that carries a sample ROM*. The theme was being rendered by the machine we ported away from.

*Gen 3's banks are General MIDI.* `voicegroup149` puts glockenspiel at 9, tubular bell at 14, harp at 46, timpani at 47, oboe at 68 and flute at 73 — **GM 10, 15, 47, 48, 69 and 74, zero-indexed, exactly**. So a bank is addressable by ordinary program number, and *the samples were never the problem*: a Steinway, an SC-88 string ensemble, trumpets, french horns, a choir and timpani are **already in the ROM**, scattered across banks that no one song draws on.

`tools/gbavoices.py` writes **voicegroup191** — one bank, GM-mapped, every slot sampled, and unused slots holding the piano so that *a program number we did not intend still arrives as an instrument*. The failure mode that started this was silence and then a square wave; it should not be reachable twice.

*And three channels was never the GBA's number*

*7.4's three-channel arrangements are a fact about the Game Boy*, and I carried the constraint across a port that does not have it. Vanilla's **mus\_vs\_trainer**, **mus\_vs\_gym\_leader** and **mus\_vs\_deoxys** all run ten tracks.

*So the lift the revision above asked for goes where it always belonged.* The theme is now **five**: a trumpet carrying the call, a glockenspiel doubling it, the string ensemble underneath, a fingered bass on the drive, and a **timpani on the downbeat** — which is the part that makes it a theme rather than a melody.

**engine/** is untouched and stays at three. *That constraint is real there, and 7.4 is still correct about it.*

*SEEKMUSAI takes the Pikachu slot, and the slot is three sprites*

*Chosen over S.T.A.R.R., which was the instinct and the wrong answer.* 4.7 puts S.T.A.R.R. dormant past the Review Board — *a creature asleep for the whole game cannot be the one that welcomes you into it*. And the Pikachu slot is a fact about the anime's marketing, not about FireRed's fiction; the creature there has nothing to do with the briefing beside it.

What the briefing actually is: an **INDEX** that starts out empty, "almost everything here is written down somewhere", and it closes on "*Some of them disagree.*" That is a screen about finding and reconciling records — **so it is narrated by the daemon whose item is EMBEDDING**, which is the thing that makes an index searchable. *Not a pun; the actual relationship.*

It is also the sibling not already on the title screen, so the family reads across both without repeating a drawing. `TYPE_FLYING` fixes it at VECTOR — sanguine red, not Jolteon's yellow.

*The rig is body, ANTENNAE and VISOR*

`oak_speech.c` draws the creature as three sprites with two frames each: a breath, a twitch and a blink. For a MUSAI those are the body, the antennae and the faceplate, *so SEEKMUSAI blinks by its scanner cycling* — which is the right verb for this one and cost nothing to arrange.

**Only the body is drawn art.** All three layers were generated and *the two overlays did not seam* — the antennae sat on a plain dome while the body has a swept crest, and the visor was a different faceplate shape. **Overlaid, the helmet would have changed shape every time it twitched or blinked.** So `tools/gbaseek.py` builds the overlays **from the body**: the antennae layer is its own top sixteen rows with stalks drawn on, the visor its own faceplate with a line drawn across. *The seam is exact by construction, because the surrounding pixels are the same pixels.*

*The breath is one pixel, and that is not a style choice*

***SpriteCB\_Pikachu** is three lines and it governs the whole rig:*

```
| sprite->y2 = gSprites[sprite->sBodySpriteId].animCmdIndex;
```

Each overlay's vertical offset is the body's frame number — 0 or 1. So the antennae and the visor follow the body by **exactly one pixel**, and a body that moves further tears away from its own overlays. *The generated second frame was a full crouch*, and composited it doubled the helmet.

**So frame B is synthesised: frame A dropped one pixel, with the shadow left where it is.** That is what vanilla does — its own two frames are a one-pixel settle with 33 pixels of hand touch-up. **The generated crouch is unused.**

*One coordinate moved.* The visor sprite was at (24, 13), which is where Pikachu's eyes are; solving the layout against that box forced the creature down to **sixteen pixels wide**, so it moved to (24, 17) instead. **Four pixels of sprite position was cheaper than shrinking the character.**

*The controls guide, and the one screen that knows it is a game*

*"one day I'd like to put it on real GBA hardware, but for either case..."* — the controls guide names six buttons, and **on an emulator not one of those names is where it says it is.**



This is the only screen in the game that addresses the player as a player, so it is the one place a wink costs nothing. *Craft rule 6 licenses it directly*: a game that is funny in its ordinary moments earns the right to be serious in three or four of them, and a button legend is as ordinary as a moment gets.

|     |                                                                                  |
|-----|----------------------------------------------------------------------------------|
| was | "The various buttons will be explained in the order of their importance."        |
| is  | "The buttons, in the order they matter. Names, not places. Check your bindings." |

*It had to be a rewrite rather than an addition.* The page 1 window is 30 tiles by 4 — 240px wide and **two lines tall**, and vanilla already used both. 212px and 217px against a 236px budget; vanilla's own widest line here is 221.

"Bindings" is not a loanword from emulator settings — it is already ours. 7.2 replaced *catch* with **bind**, the build script is `bindDaemons.sh`, and the binding sound is a jingle with its own entry in the song table. *On a real cartridge the line still parses* — those buttons are bound to a shell — **so the joke does not expire the day the ROM reaches hardware**. That was the condition, and it is the reason the line does not say "emulator".

*Ruled 2026-09-04 — the world is chiptune and the front door is not*

*Asked directly, and answered directly: "chiptune I think is good."* The twelve tracks carried over from `engine/` were written for four hardware channels and they keep that texture. **9.14 already put the title screen outside the fiction**, which is what lets the front door have the sample ROM while the world it opens onto does not — *the same split that let colour be spent there without breaking invariant 5*.

*But "leave them alone" was not available, because they were not left alone.* They played `VOICE 0`, and slot 0 of the banks they happened to point at is a **keysplit** — **a drum-kit key map** — **in ten cases out of eleven**. That is not chiptune. It is an accident that sounds a bit like one.

**And the PSG slots do not agree between those banks either.** Slot 87 is a programmable wave in three of the eleven and a filler square in the rest, so the same wave part would have changed instrument between one town and the next. *A texture that varies by location is not a texture.*

**`voicegroup192`, and the Game Boy's four channels are four programs**

**GM 80 and 81 are "Lead 1 (square)" and "Lead 2 (sawtooth)", and Gen 3 puts its PSG voices exactly there** — the same General MIDI agreement the orchestral bank relies on. `pokered` declares its channels in order, Pulse 1, Pulse 2, Wave, Noise, and `port_music.py` keeps that order, **so the channel index is the hardware channel** and four programs name its four voices.



*A stray program stays in the family:* filler slots hold pulse 1 here and the piano in 191, so a mistake in either bank is still the right kind of sound.

*And the tool that owns the notes owns the bank*

`port_music.py` had `mus_title` in its list and quietly overwrote it — the stems transcription, and its `-G191` — the first time it ran after the bank change. A song's notes and its voicegroup are one decision, so the tool that writes the MIDI now also keeps `midi.cfg` pointed at the right bank, and `SUPERSEDED` keeps its hands off the one track it no longer owns.

*Revised a third time — stems, and the end of guessing*

*4.9's gate fired again, in a new shape.* AIVA's current front end is an **audio** model and has no notes in it to export; its symbolic composer is a separate legacy app whose chord-progression field **cut the description off mid-sentence**. Four services in, *the answer was the same as it was for Gemini: stop asking a service and do the work.*

What `mp3midi.py` was actually doing is worth stating plainly, because it sounded like a transcription and was partly a construction. A mix is one signal. It split melody from bass with a filter at 250Hz, and it **inferred the middle voice from a chroma profile** — not because that is a good way to hear an inner voice, but because `pyin` is monophonic and *cannot hear two notes at once*. Both were approximations of an instrument list.

*Suno exports stems.* Eight files, one instrument each, and **Basic Pitch is polyphonic** — so chords come back as chords. Nothing in the output is inferred any more: **every note was heard in a file that contained one instrument.**

*Two findings worth keeping*

*The stems are time-aligned at zero.* They are shorter than the mix and than each other, which looks like offsets and is not: the export trims where an instrument stops. **Five of seven stems' loudness contours correlate best against the mix at exactly lag 0**, and a separator emits aligned stems by construction. *Waveform correlation said otherwise and was wrong* — separation re-renders, so the samples no longer match even where the music does.

*Taking the top note of a chord is wrong.* The first reduction took the highest note in each cell and the brass came back reaching **G#6**, the keyboard **F7** — upper partials, which Basic Pitch correctly reports as notes, and which made the line leap an octave and back. It takes the **loudest** now, and where two are within a hair of each other **the one nearest what it was already playing** — which is the assumption a musician transcribing by ear makes without noticing they have made it.

**The measure of it:** the string part came back with **two notes out of key in a hundred and thirteen**. The old tool snapped 10% of a melody it was far less sure of.

*The title screen could not be read, and it was one lost nibble*

*Raised from a screenshot:* "the intro screen for both editions are hard to read the text..... and the **PRESS START** is hidden under caremusai's sprite."

`tools/gbastrip.py` rebuilt this layer's tilemap from tile indices alone. A map entry's high nibble is its palette bank, vanilla writes **15** on all six hundred and forty tiles, and the rebuild wrote **0**. *Bank 15 is the background palette and bank 0 is the wordmark's*, so **PRESS START**, the copyright line and all four screen fills quietly started taking their colours **from the ramp that draws DAEMONS** — yellow text on a yellow band, orange on orange, and the "stripes" nobody chose were the wordmark's own gradient leaking out behind it. **The blink stopped working too**, because `Task_TitleScreen_BlinkPressStart` writes entries 1-5 of bank 15 and nothing was reading bank 15 any more.

*One nibble, four symptoms, and every one of them looked like a taste problem.* That is the argument for 8.5's rule in a form the eye cannot supply: **a screenshot shows you the result and never the cause.**

`tools/gbatitleview.py` — *compositing it offline, because squinting failed*

The layout was being fixed by looking at emulator screenshots and guessing at coordinates, which is how the sprites came to stand on the words in the first place. *So the screen is now composited outside the machine* — the real tiles, maps and palettes, at the real coordinates, following the GBA's own priority rule that **an OBJ draws above a BG of equal priority**. It reports every layer's ink box, and a collision is arithmetic:

|                    |                                             |
|--------------------|---------------------------------------------|
| DAEMONS            | x 26-228, y 7-66                            |
| the edition's name | x 100-153, y 76-88                          |
| PRESS START        | 89px wide — <b>was</b> x 43-131, y 129-135  |
| a 64x64 creature   | about <b>31px either side</b> of its centre |

*The two creatures were at 76 and 164*, which put their art at x 44-107 and 132-195 — **on the subtitle at one end and on PRESS START at the other**. They are at **50 and 190** now, PRESS START moved down one tile row and right four (both whole rows and whole tiles, so no glyph was redrawn), and the copyright line is centred on the 240 the GBA shows rather than the 256 the tilemap is wide. *Nothing is drawn on top of anything.*

*The ground goes dark, and the palette's indices are written down*

`tools/gbatitlepal.py` owns bank 15 now, and its whole value is the comment: *index 6 is the two bands and the blink's off-colour, so it must be the colour behind the text; 11 is the stage the creatures stand on; 1-5 are the letters; 10, 12 and 15 the copyright and the strips. None of that is written down anywhere in pokefirered* — it was recovered by rendering each index on its own, and it would have been lost again.

*A bright wordmark, two lit creatures and a bloom in the middle all want the same thing*, which is somewhere dark to happen. Each edition's ground is its own wordmark's hue several stops down — **cold slate under CONTENT, deep violet under CONTEXT** — so the letters read as lit rather than as pasted on.

*And they were holding still*

*They stood perfectly motionless through a reveal that takes several seconds*, which reads as artwork rather than as two creatures about to do something. **A slow breath, a quarter cycle apart** — in lockstep it reads as one animation played twice.

*The sixteen-frame scene is deferred, not abandoned*

9.14 cut vanilla's sixteen-drawing intro because 2.4's face-off does the same job on the title for two drawings. *It is wanted back* — as our sixteen frames, which is what `docs/intro-sequence.md` sketches. **It is not back yet because the screen it would hand over to could not be read**, and an intro that leads into an unreadable screen is the wrong order to build in.

*The lesson is that both media drifted the same way, unprompted*

*Nobody decided the game should be dark*. The art drifted there because a prompt asked for a wide brim and never asked for warmth; the music drifted there because a repoint was free. *Two independent cheap decisions, each defensible on its own, arriving at the same wrong tone* — and it took a player looking at the title screen to notice.

Craft rule 6 is not only about dialogue.

## 7.14f Craft rule 6 governs the art, and the art had stopped obeying it

*Raised 2026-09-02, from a screenshot: "our is in the shadows and doesn't have a face... is it a dark game? because if so, that is not what I want to make."*

It is not, and this document says so in its own words. *Craft rule 6*: "A game that is funny in its ordinary moments earns the right to be serious in three or four of them. Corpus employees are cheerful and absurd; the horror is what they are cheerful *about*."

*And the rest of the bible holds the same line everywhere it matters:*

|                              |                                                                                                      |
|------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Scorn</b> (7.x)           | "Never sneers, never gloats. <b>Genuinely warm</b> , genuinely pleased to meet you."                 |
| <b>CAIRN</b> (5.1)           | "His creed, said plainly and <b>sympathetically</b> ... <b>He is right, and that is the point</b> ." |
| <b>The institution</b> (4.x) | "The silence is <b>not shame and not cruelty</b> ... <b>Nobody decides to be cruel</b> ."            |
| <b>Craft rule 3</b>          | pathology words are refused outright — #MANIAC → ARCHIVIST, TAINT → SKEW                             |

*This design has no villain. Its central move is refusing to make anyone a monster, and the greyscale is invariant 5's spending decision about where colour is used, not a mood.*

*The art was arguing the opposite*

Every player prompt in `docs/gemini-prompts-player.txt` was written asking for a figure who is "*still, composed*" and "*confident*" — and not one of them asked for warmth. Then it asked for a wide flat brim, which *shadows a face by construction*. All three generated samples came back with the face in darkness and one had no face at all. *That was a fault in the prompt, not the generator.*

The comparison is the proof. Vanilla's title figure spends its head region on light — hair, a lit face, open shapes. Ours spent it on a black void with three light pixels in it.

*So the tone paragraph is now mandatory on every character prompt*, and it says the face must be one of the lightest areas of the figure, never the darkest. At  $40 \times 56$  a face is six pixels across; if those six pixels are black, the character is a silhouette, and the art is arguing against the writing.

## 7.14e The player — the figure you look at all game

**Built 2026-09-02.** The player exists in nine files across four sizes, and *only one of them is the one the question is actually about*: `gfx/sprites/red.png`, six  $16 \times 16$  frames. Facing right is the side frame with OAM's x-flip, so there is no fourth direction.

|                              |       |                                                                             |
|------------------------------|-------|-----------------------------------------------------------------------------|
| gfx/sprites/red.png          | 16×96 | <b>drawn</b> — down, up, side, and a walk frame for each                    |
| gfx/sprites/red_bike.png     | 16×96 | <b>provisional</b> — the same character; the bicycle itself is under review |
| gfx/overworld/red_fish_*.png | 16×8  | three fishing half-frames                                                   |
| gfx/player/red.png           | 56×56 | the <i>this is you</i> screen, and the Hall of Fame                         |
| gfx/title/player.png         | 40×56 | <b>generated</b> — the title screen, where the daemons cycle past           |
| gfx/player/redb.png          | 32×32 | <b>every battle in the game</b>                                             |
| gfx/player/shrink1/2.png     | 56×56 | the naming-screen animation                                                 |

*The overworld sprite had to be hand-drawn, and that is not a preference.* Every other art pipeline here **resamples a large illustration down**, which is proven at 40–56px and **fails at 16** — a generated figure loses its face, its silhouette and its walk cycle in the same downsample. 8's *sprite doc* already ruled out generating anything under 16×16; this is that rule meeting the asset it was written for.

**It is drawn in vanilla's idiom on purpose** — same height, same 16×16 footprint, same two-tone shading, same outline weight. **The player stands next to thirty NPC sprites that are still vanilla's**, and a figure drawn to a different set of rules would read as a mistake rather than as a person. **What changes is who it is.**

*Between a scientist, a magician and a wizard*

*A first version was a cap and a strap* — Ash with a satchel, and correctly rejected. The brief that replaced it was to sit the player between three archetypes, and the resolution is that **one of them costs nothing**:

A lab coat and a wizard's robe are the same silhouette.

**So the coat carries two of the three for free** — long, sleeved, reaching the ground — *and the hat only has to carry the third*. **A flat brim, wider than the shoulders**, is the widest shape on the figure and the one thing **no vanilla NPC has**. *The overhang is what makes it a hat*; **a brim the same width as the shoulders stacks into a mailbox**, which is exactly what the first attempt at it did.

*The arms took three tries, and the failure is worth keeping*

| Attempt                                         | What it read as                                                                      |
|-------------------------------------------------|--------------------------------------------------------------------------------------|
| Black columns <b>flush</b> against the torso    | <b>outline</b> . <i>The figure had no arms at all</i> — and that is what was noticed |
| Separated by a <b>transparent</b> column        | <b>a hole punched through the armpit</b> — the limbs floated                         |
| <b>Inside the silhouette, separated by tone</b> | <i>arms</i>                                                                          |

*Sleeves at level 2, torso at level 1, strap at level 0. Sixteen pixels has no room for a gap, so the separation has to be tonal — and once it is, the black strap cannot be confused with either, which is what makes 1.3's box readable at this size. The player is carrying one, on a strap, in every frame that could show it; from behind it is the satchel itself.*

*At 40×56 the unit finally fits*

The title screen is the one place with room for it. The 16×16 sprite can only imply the box through the strap; `gfx/title/player.png` has 40×56 and can draw the thing itself — a vent line, an indicator, a port, and no seam anywhere that would let it open. 1.3 calls it a machine you offer a daemon as a host, so it must not look like a container.

*The hand-drawn first pass was replaced by generated art the same day, and it exposed a rule the daemon pipeline had never had to obey.*

*White is transparent, and the coat is white*

*A daemon sprite is drawn to the background layer; the title figure is drawn as OAM. On OAM the lightest level is transparent — so a white lab coat renders as a hole, and the figure comes back as a ghost: hat, strap, boots, and nothing between them.*

`--solid` is the fix, and where it floods is the whole trick. A flood run on the finished 40×56 finds nothing — at that size the outline is one pixel and full of gaps, so the background leaks straight into the coat. Run it on the source instead, where the outline is still six to eight pixels thick and closed, and whatever paper the flood cannot reach is interior. In this figure that was 234 pixels that would have been see-through.

*And the margin was eating the detail*

*The first conversion looked thin and empty, and the measurement says why. Vanilla's title figure fills 100% of the width of its 40×56 slot — it was drawn for the slot. Ours filled 57%.*

*A standing person is about 0.43 wide-to-tall; the slot is 0.714. Keeping the subject's honest proportions letterboxes it, and the 43% it gives away is resolution, not whitespace — at forty columns there is none to spare, and it is the difference between a readable strap and a grey smudge.*

So `--fill` maps the subject's own box onto the frame and scales the axes independently. It makes the figure stockier, and that is correct: every Game Boy sprite of a person is stockier than a person.



*The generous-hem sample was tried too and is worse — at 0.93 aspect `--fill` squeezes rather than stretches, and the flare collapses into a grey triangle with a person balanced on it. The one to generate is the plain standing pose.*

*A one-pixel hole on an outer edge is a bite*

*The outline pass blacks a cell only when the source line covers enough of it — `COVER`, default 0.34. A cell the line merely grazes stays grey. In the middle of a figure that is invisible. On the outer edge it is a hole, and the eye does not read it as a hole: it reads it as a piece missing from the coat.*

*The shoulder had exactly one such pixel, and it was spotted on a screenshot at native size before any of the tooling noticed it.*

*The first fix was to lower `COVER` until the edge closed, and it was the wrong instrument — it thickens every line in the figure to repair one edge, and it still left a hole under the other cuff. `--close` fills only the gaps, and only where they touch the background, which is the only place they show. Eight pixels on this figure, zero holes left, and less total black than the threshold tweak used.*

*A threshold is a global answer to a local problem.*

*Two other non-square bugs fell out of the same pass: the outline pass ran `range(size)` on both axes, so it was only covering 40 of the 56 rows, and the cropped-too-tight warning divided by `size2`. Neither could show up while every target was square.*

*The hem swings, but a hem alone slides*

*The coat reaches the ground, so the walk is the hem swinging side to side — which is what a long coat does, and costs three rows instead of a leg pair per frame.*

*On its own it was not enough. With nothing at the floor to read a step against, the figure moved like a blob being dragged — the hem changed shape but nothing said *foot*. So the coat stops one row short and two shoes show beneath it: together when standing, apart when walking, and in profile one hides the other.*

*One row of pixels, and it is the whole difference between walking and gliding.*

*The bicycle sheet is the same six frames from the waist up, with a bike below the rail. Vanilla animates the whole bicycle sideways, which at 16px reads as *the rider lurching* — so ours moves the wheels instead and flickers their tone, which is what spokes actually do.*

The intro face-off was only in one edition

*Found while auditing the above.* Vanilla gives each edition its own front mon — IF DEF(\_RED) takes red\_nidorino\_\*, IF DEF(\_BLUE) takes blue\_jigglypuff\_\* — and only the \_RED path had been replaced. CONTEXT was still opening on Jigglypuff.

The conditional is gone. 8.4 fixed both editions to the same story, and the face-off is that story — CODEMUSAI against CAREMUSAI, LOGIC failing against CONTEXT. An argument that changes by cartridge is not one (invariant 3, arriving somewhere it was not written for), so both editions now include the same three frames and blue\_jigglypuff\_\*.png falls out of the build.

7.15 The space audit — a claim I repeated for days and never measured

Measured 2026-08-30 from the linker map. Every earlier statement that the music banks were full was estimated, carried forward, and wrong.

| Bank | Holds   | At session start | After six tracks went in |
|------|---------|------------------|--------------------------|
| \$02 | Music 1 | 8 bytes          | 803                      |
| \$08 | Music 2 | 175 bytes        | 176                      |
| \$1f | Music 3 | 718 bytes        | 2813                     |

901 bytes free before. 3792 after — having added six tracks.

The arithmetic that dissolves the problem: our tracks are small — Brazen 104 bytes, the dark places 89, Blanche 129, Quicksilver 478 — against a median vanilla track of 427 across 47. Replacing a vanilla body with one of ours frees more than it spends, every time.

So the claim was true only of appending without displacing, which is not what a total conversion does. Kanto's soundtrack is not an asset here; every vanilla track is a slot, and there are 47 of them.

The residual constraint is real but different. Some maps share a music slot — MUSIC\_CITIES1 covers 37 — so giving one map its own tune means adding a constant and repointing a line of data/maps/songs.asm. That is an edit, not a budget.

The lesson is the one 0.2 already states, and it caught the project out rather than the source: a number nobody measured is not a constraint, it is a rumour. This one shaped scheduling for days — the Owl's scene was recorded as unwritable-for-now on the strength of it (4.23).

## 8. SCOPE CONTROL

*The graveyard of ROM hacks is full of projects that designed 151 creatures and shipped zero towns.*

### 8.1 Vertical slice — build this before anything else

**Blanche → The Bleed → Callow → Underpaint → Slate → Benchmark 1.**  
Playable end to end.

- 12 daemons (3 starters + 9 wild), front and back sprites
- Full type chart, all 15 — it is one file, do it properly once
- Crystal Clear intro sequence
- Benchmark 1 leader and party
- Two city themes and one route modulation
- Route signage system (number plus local name)

If that is fun, the remaining daemons are labor. If it is not, you learned it for the price of twelve sprites.

**The slice is built, 2026-08-31.** Type chart, intro, route signage, CAIRN at Benchmark 1, **Slate's theme and The Bleed's modulation (7.14a)**, and **every creature the player meets between Blanche and Slate named, drawn and filed (8.1a)**. **Thirty-two daemons now carry original front and back art** — the three starter lines, the six wild lines, the four MUSAI, ARTSAI and S.T.A.R.R.

*The next milestone is the one 8.6 named: Benchmark 1 beaten on hardware.* Not another section.

#### 8.1a The nine wild daemons — built 2026-08-31

**The starters are paradigms. These are substrate.** The player walks through grass full of the primitives a system is made of, **carrying one way of learning** — which teaches without repeating 8.2's lesson.

| Vanilla                         | Ours                               | Type                  | Pays twice                                                       |
|---------------------------------|------------------------------------|-----------------------|------------------------------------------------------------------|
| RATTATA → RATICATE              | <b>NIBBLE → OVERFLOW</b>           | CONTENT               | four bits, and what a rat does; then four bits past their bounds |
| PIDGEY → PIDGEOTTO → PIDGEOT    | <b>PACKET → RELAY → BROADCAST</b>  | CONTENT/VECTOR        | <b>content that moves, which is the typing</b>                   |
| SPEAROW → FEAROW                | <b>PING → FLOOD</b>                | CONTENT/VECTOR        | one packet demanding a reply; then the attack                    |
| CATERPIE → METAPOD → BUTTERFREE | <b>CRAWLER → PENDING → INDEXER</b> | SWARM                 | <b>crawl, queue, index — a real pipeline</b>                     |
| WEEDLE → KAKUNA → BEEDRILL      | <b>SCRAPER → BUFFER → INJECTOR</b> | SWARM/ <b>CORRUPT</b> | the same shape, gone wrong                                       |
| PIKACHU → RAICHU                | <b>SPIKE → SURGE</b>               | SIGNAL                | a neuron firing; then one that never stops                       |

The pair the set is built around

**CRAWLER and SCRAPER.** Two swarms, the same shape, met in the same forest. One follows every link it finds. The other takes the same things by the same method, and nobody gave it permission.

The type chart says which is which by making the second **CORRUPT**, and nothing else has to. That is 2.5's **SWARM** doing for data collection what the starters do for paradigms — an argument delivered as a matchup.

*And the lines evolve along processes rather than just getting bigger.* **CRAWLER** → **PENDING** → **INDEXER** is crawl, queue, index; *what it cannot file, it drops.* **SCRAPER** → **BUFFER** → **INJECTOR** is the identical pipeline with permission removed at every step.

Two names were already better than the ones drafted here

**JIGGLYPUFF** and **WIGGLYTUFF** were already **SUSPEND** and **HIBERNATE** from an earlier pass. The draft proposed **LULL** → **BLOCK** ; the existing pair wins. They are the two real power states, they escalate correctly — suspend holds in memory, hibernate writes to disk — and they rhyme with 4.20a, where a man was suspended by a machine. 4.6's **MOCK** precedent again: the capability was always out there in the grass.

*Kept, and the drafts discarded.* Check before renaming.

8.2 Starter trio — learning paradigms

| Paradigm      | Types                   | Draft line (names provisional)                        |
|---------------|-------------------------|-------------------------------------------------------|
| Supervised    | CONTENT → CONTENT/LOGIC | <b>LABL</b> → <b>RUBRIC</b> → <b>CANON</b>            |
| Unsupervised  | VECTOR → VECTOR/LATENT  | <b>CLUSTR</b> → <b>LOCUS</b> → <b>MANIFOLD</b>        |
| Reinforcement | GROWTH → GROWTH/SIGNAL  | <b>ROVERCUB</b> → <b>ROVERSEER</b> → <b>ROVERBYTE</b> |

All nine built 2026-08-31, and no vanilla starter name remains in the cartridge.

**Corpel** collided with **Corpus** and **Canonex**'s **-EX** earned nothing — it reads as *ex-* or Latin *out of*, neither of which belongs at the top of a supervised line.

**LABL** → **RUBRIC** → **CANON** fixes both, and gains something: the names get more formal as the creature gets more authoritative. *Labl* is barely a word, *rubric* is a technical term, *canon* is an institution. A rubric is also the marking scheme, which is what a world of benchmarks and marks runs on.

**Nebulon** was not vague, it was backwards. A nebula is diffuse — less structured than a cluster — so the line ran grouped → scattered → structured and lost ground in the middle. **LOCUS** escalates properly: a cluster is points that happen to be near, a locus is points that share a rule, a manifold is the surface they lie on. Proximity → rule → shape.

## The Index entries, and the argument they carry

|                 | Category  | The entry                                                                                                |
|-----------------|-----------|----------------------------------------------------------------------------------------------------------|
| <b>LABL</b>     | EXAMPLE   | <i>Holds one example and the answer that came with it. <b>Someone else decided what it means</b></i>     |
| <b>RUBRIC</b>   | CRITERION | <i>It no longer needs answers. It has the rule that made them. <b>Nothing checks the rule</b></i>        |
| <b>CANON</b>    | SETTLED   | <i>Every case it meets is decided by cases it has already met. <b>New things are filed as errors</b></i> |
| <b>CLUSTR</b>   | GROUPING  | <i>Puts near things with near things. <b>Nobody told it which things matter. It has guessed</b></i>      |
| <b>LOCUS</b>    | CONDITION | <i>It found a rule the points obey. <b>It cannot say what the rule is for, only that it holds</b></i>    |
| <b>MANIFOLD</b> | SURFACE   | <i>The scattered points were on a surface all along. <b>It can show you the shape, not a name</b></i>    |

The two final stages are the paradigms' failures, side by side. CANON is accurate and files novelty as error. MANIFOLD finds real structure it cannot name. One is certain and wrong at the edges; the other is right and mute.

*And CANON's line is the Meeting Room minutes. New things are filed as errors is what happened to a woman who moved between fields (4.20) — printed in an Index entry, two hundred slots away from the building it describes.*

*Types implemented 2026-08-29, verified in the ROM.* The secondary appears at the **final** stage, exactly as vanilla's Charizard gains FLYING — so stages 1 and 2 are pure and stage 3 splits. Bulbasaur's CORRUPT secondary is dropped, since 8.2 makes the reinforcement line GROWTH.

| Slot                   | Was                      | Now                            |
|------------------------|--------------------------|--------------------------------|
| Charmander → Charizard | ENTROPY → ENTROPY/VECTOR | <b>CONTENT → CONTENT/LOGIC</b> |
| Squirtle → Blastoise   | FLOW                     | <b>VECTOR → VECTOR/LATENT</b>  |
| Bulbasaur → Venusaur   | GROWTH/CORRUPT           | <b>GROWTH → GROWTH/SIGNAL</b>  |

The lab dialogue moved with them — *the CONTENT DAEMON, the VECTOR DAEMON, the GROWTH DAEMON*. This is the second time those three lines have been rewritten and the last: 1.2 predicted exactly this, which is why the other nineteen type-words are being held until 5 settles the Benchmark leaders.

**Movesets are now the blocker, and one number decides how hard.** Gen 1 has **6 FLYING moves in total** — that is the whole VECTOR pool for a daemon the player raises for forty hours. It is less alarming than it sounds (FIRE has 5, so vanilla's own fire starter is thinner still, and Gen 1 mons lean on NORMAL moves for damage), but VECTOR/LATENT compounds it because GHOST is nearly empty too. **CONSENSUS is the precedent:** 2.5 already established that when a type is unusable the fix belongs in `moves.asm`, not in the chart. The unsupervised line will likely need its own move the way SWARM needed one.

*Until movesets follow, the starters know their vanilla moves* — a CONTENT daemon that learns EMBER. Mechanically fine, narratively nonsense, and 9's remaining work.

Supervised is strong early and rigid late. Unsupervised is confusing early and excellent late. Reinforcement is inconsistent with the highest ceiling. **The paradigms should feel like their real tradeoffs.**

**The names are the weakest thing in this document**

**And they have no reasoning behind them.** The table above says *provisional*, 10 says *placeholders*, and nothing was ever written down. Two faults, stated so the next pass starts from a diagnosis rather than a blank page.

**They are single-duty.** Every load-bearing term here pays for itself twice or three times — BIND is `bind()` *and* binding a daimon *and* a bond; CACHE is a hoard *and* a memory cache *and* cash; PIP is a halftone dot *and* the thing that installs a package. **Labl** says *label and stops*. It is the only part of the lexicon that does one job.

**And the dropped vowel is borrowed from the wrong register.** Grimoire names are vowel-rich and Latinate — Vassago, Marbas, Focalor, Buer. Truncation is a 2010s branding tic: `Labl` and `Clustr` are not cryptic, they are Flickr. Compare what Gen 1 actually does — BULBASAUR, CHARMANDER, SQUIRTLE are **portmanteaus of whole words** that sound like creatures.

**The register question — technical surface, or true names**

**The house style is both at once**, which is why 1.5 chose BIND: the Unix call *and* the ritual phrase. So the question for the bestiary is not *literal or mystic*; it is **which one is on the surface.**



Everything the player handles is technical on top — USER, BENCHMARK, CHECKPOINT, INDEX, DETACHED. **If the creatures invert that, the naming convention starts doing 4.2's work for free.** 4.2 says the Index *can only measure content* and has no field for what Crystal cared about. Give daemons **true names** — a register the Index's flat bureaucratic vocabulary cannot reach — and the artifact's insufficiency is visible on every page of it, without a line of dialogue. The world is corporate; the things inside it are older than the taxonomy trying to hold them.

*Left entirely unsaid, as rule 1 requires:* in the tradition, **knowing a thing's true name is what lets you bind it.** The Index records the name and then you can BIND. Never remark on it; simply let the mechanic keep that shape.

**The test any candidate must pass is the one BIND passed: does it work twice?** VERAX (Latin *verax*, truthful) as a spirit-name and as **ground truth**. LATENS (lying hidden) as a spirit-name and as **latent space**. AVARIS (avarice) and **ε-greedy**. Those are illustrations of the shape, not proposals.

**The risk, and it is the real one.** A name that sounds like a demon and decodes to nothing is fantasy set-dressing, and the creatures stop reading as processes — which is where the whole argument lives. Every one of **151 names** has to survive that check. It is a harder convention than truncation, which was going to be exhausting by daemon forty regardless.

**Recovered from conversation, 2026-08-29 — and a process failure**

**All of the following was designed, offered as a v1.8 cut, and never written down.** The bible reached v3.8 with it sitting in a transcript. That is exactly what CLAUDE.md's change procedure exists to prevent, and it did not fire because the proposals ended in "*say which parts you want*" and no answer came. **Record proposals as open items immediately; do not hold them pending a yes.**

| Recovered                                    | What it was                                                                                                    |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| MUSAI and ROVER as two of the three starters | Real CodeMusic systems, in the slot the player lives with for forty hours                                      |
| Corpe1 ↔ Team Corpus collision               | A starter's second stage sharing a name with the villain organisation. A genuine bug, still in the table above |
| CODEX                                        | Proposed replacement for Corpe1                                                                                |
| OPTIMAX → Corpus                             | Reassigned away from the reinforcement line; it is a better name for something Corpus builds                   |
| Penphin as a trade evolution                 | Below                                                                                                          |
| PERSPECTIVE as the second colour moment      | See 8.6 and 6                                                                                                  |

## Penphin — the trade evolution

**Penphin is not a starter and should not be.** The trio is three learning paradigms; Penphin is the two-hemisphere architecture — a logic model and an emotion model that must agree before acting. That is not one paradigm, it is **what happens after you stop picking one.**

**So it is obtained by trade, and the mechanic is the argument.** The dual mind requires two minds. **You cannot get Penphin alone.** Gen 1 already supports this — Kadabra, Machoke, Graveler and Haunter all evolve on trade — so it costs an evolution method that already exists.

*And Gen 1 permits the shape the idea wants: two different species can evolve into the same species.* So a penguin-ish line and a dolphin-ish line can both terminate in Penphin — two halves, each of which only becomes whole through another person. Neither half is complete and neither is the *real* one.

*Its name is already correct.* PENPHIN is penguin + dolphin, which is precisely the Gen 1 convention the starters are currently missing — a portmanteau of whole words that sounds like a creature.

## MUSAI — the Eevee slot

**Confirmed 2026-08-29.** Musai is "*one framework, many modules*" and Eevee is Gen 1's only branching line, so the shape was already right. What makes it more than a cameo is the base type: **Eevee is NORMAL, which in our chart is CONTENT.** A mind that has not specialised yet, and what it becomes depends on what it is exposed to.

| Form         | Type    | Why it decodes                                                                                                 |
|--------------|---------|----------------------------------------------------------------------------------------------------------------|
| CodeMusai    | LOGIC   | constraint satisfaction, verification, convergent reasoning — RoverByte's own description of its Logical Agent |
| SearchMusai  | VECTOR  | vector search <i>is</i> semantic retrieval; the name is literal                                                |
| TherapyMusai | CONTEXT | <i>emotions as the colour of context</i> — the thread the whole project grew from                              |

**Why TherapyMusai is the third, and not MusaiStudio or MedicalMusai.** CodeMusai and TherapyMusai are **Penphin's two hemispheres pulled apart into two creatures** — and 2's chart makes **LOGIC fail against CONTEXT.** One mind specialises two ways, and the type chart says they beat each other. That is the game's central argument delivered as an evolution branch, with no dialogue at all. SearchMusai sits between them, which is also where retrieval sits between reasoning and feeling.

**The stones are not stones — they are what you expose a mind to.** Vanilla's item is a mineral that triggers change; ours is the input that decides what an undifferentiated CONTENT mind specialises into. So the three are named for *kinds of input*, not for elements:

| Item             | Produces               | Reading                                                          |
|------------------|------------------------|------------------------------------------------------------------|
| <b>AXIOM</b>     | CodeMusai (LOGIC)      | what formal reasoning is given to start from                     |
| <b>EMBEDDING</b> | SearchMusai (VECTOR)   | the representation retrieval actually searches                   |
| <b>AFFECT</b>    | TherapyMusai (CONTEXT) | the clinical word for feeling, and <i>to affect</i> is to change |

They are a set conceptually rather than lexically — an axiom to reason from, an embedding to search with, an affect to feel with — which is how Gen 1's own key items work (SILPH SCOPE, POKé FLUTE, CARD KEY share no suffix either). All three clear the 12-character item limit.

*The near-miss worth recording:* **KERNEL** is the better word on paper — it is simultaneously the OS core, the ML kernel method, **and the thing inside a stone fruit's stone**, so it replaces *stone* with a word meaning the same physical object. It loses on two counts: **CONTEXT KERNEL** is 14 characters, and alone it reads as *system* rather than *logic*.

**ROVER** — a line, not a second Eevee

**Two branching families would dilute both**, so Rover takes the shape its own material already has. The RoverVerse is three embodiments in a fixed progression:

**ROVER** (scout — explores the field) → **ROVERSEER** (strategist — consolidates memory, reflects, *dreams*) → **ROVERBYTE** (companion — embodied, emotional, acts)

That is **explore** → **model** → **act**, which is reinforcement learning's actual loop. So it confirms ROVER as the Reinforcement starter rather than competing with it, and the line earns its stages instead of just getting bigger.

*RoverRadio does not survive the character cap* (10). **The base is ROVERCUB** — a real repository ("*a simplified board implementing Rover's technology*"), exactly 9 characters, and it reads as a young animal, which is what a starter should be. So every stage of the line is a system that exists: **ROVERCUB** → **ROVERSEER** → **ROVERBYTE**. RoverRadio is held in reserve for a separate daemon.

**Corpel, Corpus, and which one moves**

**Corpel** (supervised, stage 2) collides with **Team Corpus**. **Corpus stays**. It carries three readings at once — a **body of text** you train on, a **corporate body**, and a **corpse** — which is the standard the rest of the lexicon meets, and 4.4 plus the lobby engraving are built on it. **Corpel** sits in a table marked *provisional*, inside the line diagnosed above as single-duty. **The placeholder moves, not the term that earned its place.**

*It is also only a document problem.* Charmander's line is untouched in the ROM, so nothing in the game is broken by it, and the register decision will likely replace the whole supervised line regardless.

*Considered, if the org ever does need to move:* **INGEST** (data ingestion, and consuming) and **SCRAPE** (web scraping, and taking without asking). Both good; neither has three readings.

## The affect model, and why it stays out of the status bar

`excuseGPT` carries a five-state model — **BASE, MAD, SAD, AFRAID, GLAD** — and Gen 1 has exactly five status conditions. **Rejected**, and by 0.1's own rule 3: *name the process, not the pathology*. A status sits in the HUD as a thing a creature **is**. Renaming them to affect makes the game announce *your daemon is SAD* every battle, in the loudest and most repeated place there is — rule 3's forbidden move, performed on a creature instead of a town.

Three further failures. Statuses are **inflicted by moves**, so THUNDER WAVE would make a target AFRAID — cruelty as mechanic, a claim this design does not make. **GLAD has no slot**, and forcing one asserts that being happy is a debuff. And PSN maps to none of the five, so the set does not close.

*The elegant part, recorded because it is genuinely good:* **BASE would have been "no status"** — a mind with no affect is a mind at rest. Not enough to buy the rest.

**Where it belongs instead:** four *daemons*, not five conditions. As creatures they are non-diagnostic and nothing is afflicted with them. 7's music is the other candidate — a soundtrack carries affect without ever naming it, which is exactly what rule 1 asks for.

## Two naming registers, and this is what separates them

**MUSAI, ROVER, Penphin, BunnyArtsai and S.T.A.R.R. are named individuals from the SHC / iASHC universe.** They are proper nouns and they do not answer to the bestiary convention.

*Amended 2026-08-31, under external review.* **A starter may be a named individual.** The rule as first written implied it could not, and 8.2 then put ROVER in a row with two abstractions — a real inconsistency, correctly caught. **The resolution is to amend the rule, not the ROM.** The mismatch was never the *giving* — a starter is given, and so is MUSAI — it was the **row of three**, and the cure proposed (a `BANDIT / MARKOV / BELLMAN` line) costs **two surnames in a Gen 1 bestiary** and discards an arc that already pays: *software* → *infrastructure* → *body*, with **SIGNAL** arriving exactly where the body does. **ROVER stays. The register question above governs the other ~148**, which are species rather than characters.

That distinction is what makes both answerable: the generic daemons can take true names without the cast having to, and the cast can carry names from outside the game without setting a precedent 148 creatures then have to follow.

**Implemented 2026-08-29, verified in the ROM.** The two long branches shortened to a set — a four-letter stem plus MUSAI, all exactly nine characters:

| Was       | Is               | Item      | Type          |
|-----------|------------------|-----------|---------------|
| Eevee     | <b>MUSAI</b>     | —         | CONTENT       |
| Flareon   | <b>CODEMUSAI</b> | AXIOM     | LOGIC         |
| Jolteon   | <b>SEEKMUSAI</b> | EMBEDDING | VECTOR        |
| Vaporeon  | <b>CAREMUSAI</b> | AFFECT    | CONTEXT       |
| Bulbasaur | <b>ROVERCUB</b>  | —         | GROWTH        |
| Ivysaur   | <b>ROVERSEER</b> | —         | GROWTH        |
| Venusaur  | <b>ROVERBYTE</b> | —         | GROWTH/SIGNAL |

CODE / SEEK / CARE — the branches are single pure types, as vanilla's own eeveelutions are.

**Index entries, written 2026-08-31.** Categories **GENERAL** , **DEDUCTIVE** , **RETRIEVAL** , **AFFECTIVE** — technical register, which 4.9 says the Index may use because there it lands as jargon rather than insight.

|                  | Category  | The entry, and what it is doing                                                                                                                                                                          |
|------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MUSAI</b>     | GENERAL   | <i>Runs no module by default. It keeps whatever it is given first. <b>Nothing is filed about the rest</b> — 4.2's thinness, and a general model before fine-tuning</i>                                   |
| <b>CODEMUSAI</b> | DEDUCTIVE | <i>Given a rule, it will not stop until the rule is met. <b>It does not ask where the rule came from. That is Scorn's failure in a creature description</b>, and the Index prints it as a spec sheet</i> |
| <b>SEEKMUSAI</b> | RETRIEVAL | <i>Finds the nearest match to anything it is shown. Nearest is not the same as right. <b>Nobody told it</b></i>                                                                                          |
| <b>CAREMUSAI</b> | AFFECTIVE | <i>Reads the room before it reads the problem. Often correct. <b>Cannot show its working</b></i>                                                                                                         |

The last two lines are the argument. 2406 has LOGIC failing against CONTEXT, and here is why in the Index's own flat voice: **one of them cannot say where its rule came from, and the other cannot show its working, and the chart says the second one wins.** Nobody comments on it.

**The ROVER order, reviewed and kept — 2026-08-31**

**RoverCub** is the portable build, **RoverSeer** the server, **RoverByte** the core — the dog. As an evolution line that reads **software** → **infrastructure** → **body**, and the destination is the right one: in a game whose subject is processes that become something, **ending in a body** is the same shape as ARTSAI coming back real.

And SIGNAL lands exactly there. 5 gives SIGNAL to Benchmark 3 as Perception — raw input is fast and shallow. Both disembodied stages are pure GROWTH; the moment it has a body it gains the perception type. Nothing was arranged for that; it fell out of Venusaur's slot.

The middle stage is the sharpest part. A seer sees ahead, and a server-side model predicts — so prediction arrives one stage before perception. The system could forecast before it could feel anything, which is a real critique of how these are built, sitting silently in an evolution line. It only works in this order.

The alternative — Byte → Cub → Seer, truer to how you would actually deploy it — is wrong twice: the last form must be the most impressive, and it would put embodiment first and throw the arc away.

Index entries, 2026-08-31. Categories PORTABLE / PREDICTIVE / EMBODIED, and the prose carries the arc:

|           | The entry, and what it is doing                                                                                                                   |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| ROVERCUB  | Learns from what it is carried past. <b>It has no way to choose the route</b> — the reinforcement paradigm, and a quiet point about training data |
| ROVERSEER | Too large to carry now. It answers before it is asked. <b>It has never once been outside</b> — it forecasts a world it has not been in            |
| ROVERBYTE | It has a body now, and senses that come with one. <b>Everything it predicted, it can finally check</b>                                            |

The last line closes the arc, and nobody in the game ever points at it.

RESONANCE — Penphin's, and it completes the set

Proposed 2026-08-31. The signature moves are one argument each, and there was a gap:

| Move        | Owner      |                                                                              |
|-------------|------------|------------------------------------------------------------------------------|
| PERSPECTIVE | ARTSAI     | I take <b>your</b> frame, and lose mine. <b>Lossy</b> (4.6)                  |
| RECURSION   | S.T.A.R.R. | I take <b>my own</b> frame. <b>Compounds, collapses if interrupted</b> (4.7) |
| RESONANCE   | Penphin    | <b>Two frames reinforce each other and neither is lost</b>                   |

That is a third thing, not a variation — and Penphin is "a logic model and an emotion model that must agree before acting", which is two oscillators in phase.

And Penphin cannot be obtained alone. It is a trade evolution because the dual mind requires two minds — so the move meaning *two frames in phase* belongs to the one daemon that requires another person to exist. The mechanic is the argument twice over.

The behaviour is deliberately unspecified. RECURSION already claims the Bide machinery, and inventing a Gen 1 mechanic before there is a need is how placeholders become canon.



*One consequence, accepted.* Item names are global, so **six other daemons now evolve by AXIOM, EMBEDDING or AFFECT** — Vulpix and Growlithe by AXIOM, Pikachu by EMBEDDING, Poliwhirl, Shellder and Staryu by AFFECT. This is coherent rather than awkward: the mechanic was never *mineral*, it is exposure, and every one of those is a mind being given something.

**LEAF STONE became REWARD, and it closes the set.** GROWTH is the reinforcement type, so a reward signal producing growth is the mechanic saying its own name:

| Item             | What it is           |
|------------------|----------------------|
| <b>AXIOM</b>     | what you reason from |
| <b>EMBEDDING</b> | what you search with |
| <b>AFFECT</b>    | what you feel with   |
| <b>REWARD</b>    | what you learn from  |

*MOON STONE is the fifth and is deliberately left.* It is tied to a place on the map rather than to a paradigm, so it belongs with the town pass, not this one.

**The 9-character cap bites here first.** SEARCHMUSAI is 12, THERAPYMUSAI is 13, ROVERRADIO is 10. Surviving as-is: MUSAI (5), CODEMUSAI (9), ROVERSEER (9), ROVERBYTE (9). EYEMUSAI (8) is available and is a real module name. **The two long branches need shortening before implementation** — that is the only open work in this section.

**Are MUSAI, ROVER and Penphin the three legends?**

**Recommendation: no, and scattering them is the point.** They read as a trio because they are one in the world outside the game, but each earns more in a different slot.

- **ROVER as the Reinforcement starter** is the strongest single fit in this section. A dog is the canonical image of learning by reward, and the paradigm is *inconsistent with the highest ceiling*, which is also a dog.
- **MUSAI as Unsupervised** — generative, finds its own structure, confusing early and excellent late.
- **Penphin as the trade evolution**, because that placement is a *mechanic* and legends are only rare encounters.

**The legendary slots would waste them.** Vanilla's birds carry almost no story; they are trophies caught near the end and rarely used. A starter is raised for forty hours. If these three systems matter to the project, the starter and trade slots honour them and the bird slots do not.

*Which leaves the three bird slots genuinely open* — and they should be filled by whatever the map needs, not by what is available.

### 8.3 Known bottleneck

**Sprites.** 151 daemons, front and back, 2bpp, four shades, 56×56 maximum — roughly 300 hand-tuned tiles. Nothing about it is hard; it is simply the largest block of labor in the project. AI generation can rough out silhouettes, but expect manual cleanup: the Game Boy palette and tile constraints are unforgiving.

---

### 8.4 The two editions

The project is called **CONTEXT / CONTENT** and the slash has never been explained. It should be literal: **two editions, one source tree**.

**And the order is a decision, recorded 2026-08-31**

**The mapping runs RED → CONTENT and BLUE → CONTEXT**, in the Makefile, the cart titles and 9's cost table. **The title runs the other way**. That inconsistency was noticed from the outside, and it was accidental until now.

**It stays, and here is the reason, stated as a choice rather than a discovery.**

**Everyone says content is king.** Putting **CONTEXT** first is the thesis in two words, before the player has read a line of anything — and it means **the title refuses to let the original decide its order**, which is the correct posture for a conversion whose entire subject is that the frame decides what you are looking at.

**The two orders answer different questions and both are right.** *The title is the argument:* context is prior to content. *The mapping is the inheritance:* Red came first and Red is the default build.

*Flagged honestly, per 0.2.* **This is a justification for a choice already made, not something found.** The rule against promoting coincidence to finding applies to the project's own decisions as readily as to the source material — so it is filed as a decision, and it is not evidence of anything.

**It is already in the engine**

`pokered` does not build one ROM. It builds Red *and* Blue from the same sources, and the mechanism is three lines of Makefile. Verified against `pret/pokered` master, 2026-08-28:

```
$(pokered_obj): RGBASMFLAGS += -D _RED
$(pokeblue_obj): RGBASMFLAGS += -D _BLUE
```

and in the data itself — this is `data/wild/maps/Route2.asm`, unmodified:

```

IF DEF(_RED)
    db 3, WEEDLE
    db 4, WEEDLE
ENDC
IF DEF(_BLUE)
    db 3, CATERPIE
    db 4, CATERPIE
ENDC

```

make red, make blue, make all. Rename the two defines to **\_CONTENT** and **\_CONTEXT** and the two-edition release is done. There is no new engineering here at all — the hard part was solved in 1996 by people who needed you to buy two cartridges.

**CONTENT** is the primary target (**\_RED**'s slot), because content is the literal layer — and because it is quietly funny that the content-first edition is the default build.

### Three tiers of divergence

**Tier 1 — encounter tables.** Free, standard, exactly the Route 2 pattern above. Note what it does *not* cost: **zero extra sprites**. Every daemon exists in both ROMs. Each edition simply cannot *bind* some of them, so the 8.3 bottleneck does not move at all.

**Tier 2 — let the rosters lean.** Do not split the exclusives at random. Bias them:

| Edition        | Sees more of, early                 | Reads as                                        |
|----------------|-------------------------------------|-------------------------------------------------|
| <b>CONTENT</b> | CONTENT · LOGIC · STRATUM · LEGACY  | the literal, the rule-based, the physical layer |
| <b>CONTEXT</b> | CONTEXT · LATENT · VECTOR · ENTROPY | framing, the unconscious, embeddings, noise     |

The consequence is the good part. **The two editions are hard in different places.** A **CONTENT** player finds the early benchmarks straightforward and Benchmark 6 — attention and framing, in the bought city — genuinely punishing. A **CONTEXT** player struggles early and walks Benchmark 6, which is exactly how 8.2 says the learning paradigms should feel: unsupervised is confusing early and excellent late.

So each edition's player builds a different intuition about what is strong — and when they compare notes, they disagree, sincerely, with evidence. That is the thesis arriving as a *distribution* mechanic, and nobody has to say a word.

**Tier 3 — the Index entries disagree.** This is the prize.

Same daemon. Two editions. Two different entries, both stated as fact, neither acknowledging that the other exists.

The trap to avoid: do **not** make CONTENT's Index measure content and CONTEXT's measure context. That collapses 4.2's irony into a joke. **Both editions' Index measures only content — and they disagree anyway.** Different heights. Different weights. Different flat descriptive sentences about the same creature.

Two players compare entries and find the record does not match. Nobody explains it, ever. **The Index is a reproduction, and two reproductions of the same thing differ** — which is 0.2's whole printing throughline, and the Five Witnesses at the scale of the cartridge.

*Restraint required.* Do this for a handful of daemons, not all of them — enough that a player who compares finds it, few enough that it reads as unsettling rather than as a gimmick. And **never for the starter**, whose entry has a different job (11.1 in `lineage.md`: it rewrites itself once, late).

Cost: dex text is text. `IF DEF(_CONTENT)` around the entries. No engine work.

### Trading is the structural argument

Gen 1 link trading is already in the engine, and edition-exclusive daemons mean **you cannot complete the Index alone.**

The artifact that measures only content cannot be completed without another person's playthrough. You need someone else's context, literally, over a cable.

That is the best structural argument this project has available and it costs an encounter-table split.

Done, 2026-08-29 — partially

**The visible half is in.** `make content` and `make context` build `daemonsContent.gbc` and `daemonsContext.gbc`, with cart titles `CONTENT` and `CONTEXT`. Saves follow the ROM name, so the two editions keep separate playthroughs automatically.

**The defines are not renamed.** `_RED` and `_BLUE` are used by `IF DEF(...)` blocks in 47 asm files, so swapping them to `_CONTENT` / `_CONTEXT` is a sweep, not an edit — and it would be a large permanent diff against upstream while we still want `git pull upstream master`. Deferred until the tree has diverged enough that upstream merges stop mattering.

Nothing depends on it. The defines are internal; the player never sees them.

**Scope: build the flag now, use it later**

Section 8 is unambiguous that the vertical slice comes first, and two editions is a step-14 item at the earliest. **But the flag is not.**

Set up both targets on day one, even if the two ROMs are byte-identical for a year.

Adding a version define at the start costs nothing — the tree already has one. Retrofitting one *after* 151 dex entries and sixty encounter tables exist is a miserable afternoon. Keep `make content` and `make context` in the Makefile from the first commit and let them produce identical ROMs until they do not.

## What must never differ

**The type chart is byte-identical across both editions.** It is the argument, and an argument that changes by cartridge is not an argument. Same maps, same benchmarks, same story, same lexicon, same music.

The editions differ in **what you meet** and **what the record says about it**. Nothing else.

## Title screens — as built

*Implemented 2026-08-29.* The subtitle reads **Content Edition / Context Edition**, matching vanilla's *Red Version* pattern.

**The version subtitle is not a blitted image, and that matters.** The graphic is loaded into VRAM as tiles, and then a string of *tile indices* is printed:

```
db $60,$61,$7F,$65,$66,$67,$68,$69,"@" ; vanilla "Red Version"
```

Tiles 0, 1, a blank, then 5–9 — **tiles 2, 3 and 4 are never displayed.** Vanilla did this because "Red" and "Blue" are different lengths and the two versions share one VRAM window, right-aligned by a size calculation at load time.

A first attempt at this rename put the word in tiles 0–4 and rendered as "**Con**".

**Our fix simplifies it.** Both editions ship the same 10-tile canvas, so both load at `$60` and both print the full contiguous run — no per-edition string, no skipped tiles, no `IF DEF` at all:

```
db $60,$61,$62,$63,$64,$65,$66,$67,$68,$69,"@"
```

*Built to spec, not generated.* Clean `e`, `o`, `n`, `i`, `d` letterforms were extracted from the vanilla graphic; `C`, `E`, `t`, `x` were drawn to match its 2px stems; the words were composited centred in an 80×8 canvas and written as **1-bit greyscale PNGs**. No image model produces that format — `rgbgfx --colors dmg` rejects anything with anti-aliasing or a colour outside the palette.

## The splash line — and why it changed

`GAME FREAK inc.` is now `CODEMUSIC`, and the year is gone.

**This is more honest, not less.** The vanilla line asserts that Game Freak made this build. They did not. Leaving it would be a false attribution on modified work, and replacing it with a CodeMusic copyright *and a year* would assert a claim over an engine that is pret's disassembly of Nintendo's game.

So the line shows the wordmark and nothing else. **Trademark acknowledgment stays in `README.md`**, where it belongs and where it is unambiguous.

*How it was assembled.* The vanilla line is 16 tile indices — year tiles `$41 – $45` (shared with the boot splash) plus a 9-tile wordmark at `$46 – $4E`:

```
db $41,$42,$43,$42,$44,$42,$45,$46,$47,$48,$49,$4A,$4B,$4C,$4D,$4E
```

`gamefreak_inc.png` is  $72 \times 8$ , 2bpp, exactly those 9 tiles — so CODEMUSIC is a direct swap with no VRAM changes. The string now prints `$46 – $4E` only, recentred at column 6.

*Deferred:* a year would be lovely — '11.'26 encodes the whole lineage, and vanilla's own multi-year format invites it. But the digits need glyphs that do not exist in the current tile set, and the year tiles are shared with the boot splash. It wants a slightly wider wordmark graphic, which is 8.5 work.

**The Super Game Boy border, done.** `red_border` / `blue_border` are now `content_border` / `context_border`, and they no longer read RED and BLUE.

**And it is harder than it looks.** A border file is not a picture of a border. It is  $128 \times 48$  — a bank of 96 unique  $8 \times 8$  tiles — plus a separate `.tilemap` of 896 (tile, attribute) pairs arranging them across the  $256 \times 224$  frame, plus SGB palette entries that colour the 2bpp greys at runtime. Four facts had to be measured against vanilla rather than assumed:

- `rgbgfx` inverts greyscale. PNG level 3 becomes colour index 0. So in the source PNG *higher value is lighter on screen*, and level 3 is the light ground.
- **The attribute byte carries the palette:** `(attr >> 2) & 7`. Vanilla uses palettes 4/5/6 (`PAL_SGB1 – 3`) and `$40` for X-flip. Ours uses palette 4 throughout, so every border entry is `$10`.
- **Tile `$00` is reserved and flat.** The centre  $160 \times 144$  is covered by the Game Boy screen; vanilla fills it with tile `$00`, attribute `$00`.
- **Colour index 0 is not transparent here.** Vanilla's art uses it as the *lightest* value, not as a hole.



**They are generated, not drawn.** `tools/genborder.py` builds each border from a repeating cell. This was not the first plan — three illustrated versions were commissioned and measured with `tools/mkborder.py`, which takes a  $256 \times 224$  design, quantises, deduplicates and reports the tile count. They came in at **496 and 523 unique tiles against a budget of 96**. The lesson is that a halftone frame is *geometry, not illustration*: generated from a cell it lands every dot on the grid, and both editions now cost **12 tiles**. `mkborder.py` remains the measuring tool for any supplied art.

**And the two borders argue the same thing the editions do:**

- **CONTENT** — one dot grid. The thing itself, repeating, self-identical. Dots shrink toward the screen, so the image resolves as you approach it.
- **CONTEXT** — the *same* cell, laid over itself at an offset. Nothing new is drawn; the pattern is entirely the relationship between the two. True moiré is definitionally non-repeating and therefore cannot be tiled at all — so this is a repeating unit that *reads* as interference. That constraint is not a compromise; it is the argument in miniature.

**Which daemon appears**

Vanilla puts a starter on the title — Charizard on Red, Blastoise on Blue, and Venusaur on neither. So:

- **CONTENT edition:** the Supervised line (CONTENT → CONTENT/LOGIC)
  - **CONTEXT edition:** the Unsupervised line (VECTOR → VECTOR/LATENT)
  - Reinforcement appears on neither title, exactly as Venusaur does not
- 

### **8.5 Not yet — sprites, and the recomp question**

Two ambitions worth recording so they are not lost, and **deliberately not designed**. Both sit after the vertical slice. Neither is a commitment.

**Original sprites.** 8.3 already names this as the bottleneck. Nothing to add except that it stays a bottleneck no matter which of the below happens. *Prompts and specs for the first pass now live in* `sprite-prompts.md`.

**How far the foxes spread — decided 2026-08-29.** 4.1 makes Crystal a fox and 4.3 makes the resemblance an inference beat. The open question was whether *everyone* is animal-like.

**They are not, and the reason is resolution rather than lore.** At 16×16 a character has about three pixels of head, so the entire animal vocabulary is **ears, a tail, and a shade** — the bible says as much already. Making the whole cast anthropomorphic costs something like a hundred sprite sets and buys almost nothing, because at that size a badger and a bear are the same eight pixels. So: **the Clears get ears, tail and value; nobody else changes; and the game never says anyone is a fox.** It reads as family resemblance rather than as a species claim, which is exactly the job 4.3 gives it.

*The failure mode, and the cheap escape.* If three tailed characters among an untailed cast reads as *otherness* rather than *lineage*, the fix is not to convert everyone — it is to give light animal features to a handful of **recurring** characters (Scorn, two or three Benchmark leaders) so that fox becomes one species among several. Eight sprites, not a hundred. Decide it on playtest, not now.

*And the resemblance is carried by value, not hue.* 8.6 means the player sees four greys, so *golden-amber, darker* and *between* have to be **light-to-mid, dark, and between** — with identical ear and muzzle geometry doing the family work. Colour words will not survive the conversion and should never be relied on in a prompt.

**The Gen1Recomp question.** Gen1Recomp is a hand-written Lua / LÖVE2D recreation of Gen 1 that **imports data and graphics from a player-supplied ROM** rather than emulating it. Its mod ecosystem includes the **voxel mod** and first-person renderers — which is the specific pairing of interest here: DAEMONS creatures, rendered as voxels, in a first-person Kanto.

That split — **imported data, hand-written behaviour** — is the whole of the analysis, and it cuts cleanly:

|      | Likely carries                                                                                              | Almost certainly does not                                                                                          |
|------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| What | Sprites, base stats, names, type constants and matchups, Index entries — anything decoded out of ROM tables | Story scripts, custom events, engine work (RECURSION), anything that lives in <code>asm</code> rather than in data |

So the optimistic case is a DAEMONS ROM loading in a 3D shell with the right creatures and the right combat philosophy, and the pessimistic case is that the loader rejects a non-canonical ROM outright — the project documents accepting *"the canonical 1 MiB US versions,"* which implies it may check.

**The one test worth running, whenever the mood strikes:** build the step-5 milestone (vanilla everything, new type chart) and drop it in the launcher. It costs an afternoon and it answers the entire question — either it loads or it does not. Do that before designing anything around it, and before assuming either outcome.

**A note on the voxel mod specifically.** A voxel renderer builds volume out of the 2bpp sprite data — so the sprite work in 8.3 would be doing double duty, and the four-shade constraint that makes Game Boy spritework hard is exactly what a voxel mod needs to read cleanly. That is a reason to design the sprites well rather than a reason to design them differently.

**Scope note.** This is a *rendering* ambition, not a design one. Nothing in sections 0 through 7 should bend to accommodate it. If it works it is a gift; if it does not, the game was always a Game Boy game.

---

## 8.6 The monochrome question

**The game is about colour and cannot show any.** Every city is a colour word rendered in four shades of grey. That is either a limitation to be fixed or the best thing the hardware gives us, and the answer decides whether colorization is an upgrade or an act of vandalism.

**It is the second one.** Three reasons, in ascending order of how much they cost to lose.

**1. The player is told these places are colours and shown grey.** They supply the rest. That is the 8-Bit World thought experiment — a system whose perception is bounded by its resolution — running for free, on the hardware the thought experiment names, without a line of dialogue. And it is *The Colour We Never See*: purple is not on the spectrum, the mind folds the line into a loop and invents it. The town colours are not on the screen. The player invents them.

**2. The Index is a reproduction that loses the original (0.2).** A greyscale record of a world named in colour loses something specific, visible, and unrecoverable — which is exactly what 4.2 needs the Index to be doing.

**3. Colorizing Halftone destroys Halftone.** The town's entire premise is *dots that only look like grey* — continuous tone that is actually discrete black and white sampling, where the bias comes from **resolution, not malice** (3.1). In colour, halftone stops being the town's condition and becomes a quaint printing reference. **This is not a cost to weigh. It is a load-bearing wall.**

**Recommended: colour as something the world spends**

Not "monochrome forever" and not full colorization. **Greyscale by default, and colour appears exactly once, where it means something.**

**The Review Board at Umbra is the only place in the game that has colour.**

Section 6 already sets it up: two thousand years ago emotions were *literally coloured fluids*, and the four humors arrive as a classical flourish that becomes the whole thesis about six seconds later. Give them their colours — sanguine red, choleric yellow, melancholic black, phlegmatic white — and give them to nobody else.

Forty hours of colour-named grey towns, and then one room where colour is real, occupied by four ancient people insisting that emotions are coloured fluids. Who are **wrong**, and lose, and are followed immediately by Ty, who is also wrong and also loses.

**The one place the game admits colour is the place with the obsolete answer.** Nobody remarks on it. Nobody can.

**Settled: two sources, one of them earned**

The open question — *is one colour moment right?* — resolves at **two, different in kind.**

|             | What colour is                                 | Who sees it            |
|-------------|------------------------------------------------|------------------------|
| Umbra       | an obsolete answer, held still and examined    | everyone who finishes  |
| PERSPECTIVE | a glimpse of another's frame, immediately lost | most players, via MOCK |

**MOCK settles it.** 4.6 establishes that Ditto is CONTENT/CONTENT and learns PERSPECTIVE, so the flash is reachable in ordinary play rather than gated behind a puzzle almost nobody solves.

**BunnyArtsai is not a third source.** She is the most loaded firing of the first — the same few frames, from the one creature that could not come back — and only for a player who solved the Five Witnesses (4.8).

*Why this is the right pair.* Vanilla had **no designed way to reach Mew at all** — verified: it appears in no wild encounter table, only in `names.asm` and `palettes.asm`. The famous routes were an unintended glitch and event distribution, and player culture filled the vacuum with rumour. **If colour lived only with BunnyArtsai it would be a secret nobody could confirm.** MOCK makes it something players can compare notes about, which is the condition under which Umbra lands rather than passing unnoticed.

*Restraint clause holds at two.* A third source makes it a system, and a system invites explanation.

## Notes and consequences

- **Technically this is easier than full colorization, not harder.** A colour build has to palette every map; this one palettes a single map and leaves the rest on a greyscale ramp.
- **On the GBC caveat:** Gen 1 already receives limited hardware palettes on a Game Boy Color, so "monochrome" is a slight fiction in practice. The design intent stands regardless — the ramp should read as value, not hue, everywhere but Umbra.
- **The humors survive greyscale anyway.** Value carries them: melancholic dark, phlegmatic pale, sanguine and choleric mid. The colour at Umbra is a confirmation of something the player has already been reading in tone, which is the correct order.
- **This serves the sprite work rather than fighting it** (8.3, 8.5). Four shades with no hue to lean on forces value contrast to be right, and value contrast is exactly what a voxel renderer needs to read cleanly.
- **The recomp path is a remix, not the canon.** If a 3D or voxel presentation colours everything, that is a different presentation of the same ROM and it is welcome. The canonical build stays grey.

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### 8.6 External review, 2026-08-31 — what was taken and what was not

Read against v11.0 by a second model. Its sharpest finding is about this document:

The bible is outrunning the slice.

It is right, and it is recorded here as **binding**. Sixteen version bumps in one session, and 8.1's slice is still not beaten on hardware — **the next milestone worth recording is that, not another section.**

One constraint the review did not have

*Gen 1 has no item descriptions.* There is no descriptions file in `data/items/` — names, prices and use-handlers only. **Three of the review's proposals put their payload in one**, and had to move:

| Proposal                                 | Where it went instead                                                                                                                              |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| CC-7's "A sealed module. Part no. CC-7." | <b>Into the item name.</b> The bag reads <code>CC-7</code> , which is stronger — the player <i>carries</i> the number rather than reading about it |
| The RESOLVER's asset tag                 | <b>Into Scorn's mouth</b> , as he hands it over and dismisses it                                                                                   |
| AXIOM / EMBEDDING / AFFECT as documents  | <b>Nowhere yet.</b> Deferred until it has a speaker                                                                                                |

## Built

**PACKAGE** → **CC-7**. The clerk: *Crystal Clear's order came in. A CC-7. Will you run it out?* Her receipt: *Ah. The module. Thank you.* **Thirty hours later the player reads a requisition for the same part on a door at Quicksilver (4.18), with its reason field. The player has carried the crime-scene's part number since minute fifteen, and nothing points at it. This replaced a custom USERBOX — the wrong gift twice: a box is 1.3's argument, not a present, and it copied vanilla's custom ball.**

**Scorn at the bottom of the Verdigris basement, where Giovanni stood — cheerful, unbothered, interested in the measurement. This settles 4.5's open question: the player likes him before the tower, so Halftone becomes a betrayal of their own judgment. He hands over the RESOLVER as inventory equipment and dismisses its tag — QS-LAB, I think. Before my time. He owns the instrument Crystal's lab built and has no idea what it was for. Then he wishes the player luck at the tower.**

**Scorn's Benchmark 8 line is his one accounting, and he splits it with something that cannot answer: The machine chose what I weighted it to choose. We are both to blame, I suppose. It has never said otherwise. 4.18's finding, made audible.**

**The catching tutorial teaches 1.3 before the player owns a box: Watch. You weaken it, then you offer it a box. It goes in on its own, or it does not go in.**

## Adopted, not yet built

**Ty is not in the ROM at all. The bible places him at Quicksilver; no NPC has been written. So his line and the post-game triangle are new content, not edits — a larger job than the review implies.**



- **The post-game triangle** — S.T.A.R.R. → Ty → Crystal, three text blocks and three party checks. **And S.T.A.R.R. inferred correctly and could not go:** a machine that reads its own state cannot walk back into that building, so it went to the nearest place and stopped — **the becalmed city, the local minimum.** RECURSION collapses when interrupted; her leaving was the interruption
- **Penphin cannot be weighted.** The Clarifier failed *because* one vote could be raised; **Penphin is two frames that must agree, so there is nothing to raise.** What she would build after learning what was done to the first one — and **HOLT is the failed Penphin,** two perspectives with no rule for choosing. Halves in Doldrum's waters, one per edition
- **ORPHAN is a Quicksilver process nobody reaped** — the parent exited when the lab burned, and Corpus inherited it by succession
- **THE HOLDOUT,** and the warden's speech garbled until the TOKEN is returned
- **RESONANCE = RECURSION that does not collapse when interrupted** — one branch removed. Deferred until RECURSION exists, since it is a diff against it

Declined

**Moving ROVER out of the starter slot** — see 8.5's amendment. The diagnosis was right and the cure cost more than the flaw.

## 9. IMPLEMENTATION

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`pokered` is RGBDS assembly. The *data* lives in readable macro tables. Engine features are where it gets expensive.

*Paths are approximate. Verify against your checkout — they drift between commits.*

| What                                     | Where                                                      | Effort                                                                                                  |
|------------------------------------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Type IDs                                 | constants/type_constants.asm                               | trivial                                                                                                 |
| Type names                               | data/types/names.asm                                       | trivial, but see 9.1                                                                                    |
| <b>Type chart</b>                        | data/types/type_matchups.asm                               | trivial, highest leverage                                                                               |
| Species stats                            | data/pokemon/base_stats/*.asm                              | bulk, scriptable                                                                                        |
| Species names                            | data/pokemon/names.asm                                     | bulk                                                                                                    |
| Index entries                            | data/pokemon/dex_entries.asm plus text/                    | bulk                                                                                                    |
| Moves                                    | data/moves/moves.asm, data/moves/names.asm                 | medium                                                                                                  |
| PERSPECTIVE                              | rename TRANSFORM                                           | trivial                                                                                                 |
| RECURSION                                | repurpose Bide/Rage machinery                              | <b>engine work</b>                                                                                      |
| Sprites                                  | gfx/pokemon/front/, gfx/pokemon/back/                      | <b>the bottleneck</b>                                                                                   |
| Map names                                | data/maps/names.asm, constants/map_constants.asm           | trivial                                                                                                 |
| Map layouts                              | maps/*.blk, data/maps/headers/, data/maps/objects/         | medium                                                                                                  |
| Route signage, number plus local name    | text/Route*.asm                                            | <b>done 2026-08-30</b>                                                                                  |
| Dialogue, signs, witnesses               | text/                                                      | bulk — <b>witnesses done</b> (4.8), the rest ongoing                                                    |
| BIND verb and the flat message           | text/ — catch strings; also data/ menu labels              | <b>done 2026-08-31</b> — 55 instances, and the success message flattened (1.5)                          |
| Greyscale ramp everywhere but Umbra      | palette data; one map differs                              | medium, specialist                                                                                      |
| Two-edition build                        | Makefile — rename _RED / _BLUE to _CONTENT / _CONTEXT      | <b>trivial, do it first</b>                                                                             |
| Edition-exclusive encounters             | data/wild/maps/*.asm, IF DEF(_CONTENT) blocks              | trivial                                                                                                 |
| Divergent Index entries                  | data/pokemon/dex_entries.asm plus text/, conditional       | trivial, high value                                                                                     |
| ORPHAN as a species in a free index slot | constants/pokemon_constants.asm, base stats, sprite, cry   | one daemon's work                                                                                       |
| ORPHAN's blank Index entry               | data/pokemon/dex_entries.asm — a pointer that goes nowhere | trivial, needs care                                                                                     |
| SILPH SCOPE → RESOLVER                   | data/items/names.asm plus text/                            | <b>done</b> — and Scorn hands it over with its QS-LAB tag (8.6)                                         |
| Hidden BunnyArtsai tile                  | hidden object plus event flag                              | <b>the Five Witnesses' missing payoff</b> — accounts and terminal built, nothing rewards standing there |
| Quicksilver terminal log                 | one text block                                             | <b>done</b> — the 1001 sequence, plus ITER 35 on Mansion 1F (4.8)                                       |
| Corpus lobby engraving                   | one sign-text object, Brazen                               | <b>done 2026-08-30</b>                                                                                  |

| What                                             | Where                                                                                    | Effort          |
|--------------------------------------------------|------------------------------------------------------------------------------------------|-----------------|
| Quicksilver asset tags at Halftone               | one sign-text object                                                                     | trivial         |
| AL CLEAR in formal text; naming screen untouched | text/ — literal string instead of the <RIVAL> control char                               | trivial         |
| Default name lists, both editions                | constants/player_constants.asm                                                           | done 2026-08-29 |
| PROF.OAK → CRYSTAL                               | text/ — 33 occurrences in 10 files, plus the intro sequence. Every line needs rewrapping | half a day      |

**Done on the GBA side 2026-09-03, and it was 84 blocks in 25 files, not 33 in 10.**

**tools/port\_oak.py** measures the rewrap against FONT\_NORMAL's own advance widths rather than counting characters. *Three things the Game Boy pass did not have to face:* a **name vanilla wraps mid-line** ( PROF.\nOAK's ), which only the tool's second run found; **four fixed-width fields** where CRYSTAL CLEAR does not fit, of which only the trainer nameplate had to give; and **fourteen masculine pronouns**, applied only inside blocks that name her — VermilionCity's "he" is the other AIDE, and a blanket sweep would have changed him too. | Quicksilver journal fragments, one signed | text/, Cinnabar Mansion idiom | trivial | | Corpus badge name matching the signature | text/ | trivial | | Brazen employee's post-Quicksilver second line | one event flag, one text block | trivial | | Music | audio/music/\*.asm | medium, specialist |

### 9.1 The TypeName gotcha — resolved, and smaller than feared

Verified against pret/pokered master, 2026-08-28. Built and confirmed.

The warning in earlier drafts was right in spirit and wrong in detail, and the detail matters because it makes step 5 **much** cheaper than the bible previously claimed.

**Current master does not need a subtraction.** data/types/names.asm is a flat table guarded by table\_width 2 and assert\_table\_length NUM\_TYPES, and the ID gap is filled explicitly:

```
REPT UNUSED_TYPES_END - UNUSED_TYPES
    dw .Normal
ENDR
```

So every unused ID points at .Normal and GetTypeName indexes straight in. There is no \$0B fold in this checkout. **Never paste a replacement names.asm wholesale** — that advice stands, harder than before, because the structure here is nothing like the standalone file in patches/.

**The bigger correction: you do not need to rename the constants at all.**

The chart references types by symbol; the symbols resolve to IDs; the IDs drive the matchup lookup. **None of that touches what the player reads.** What the player reads is the strings in `names.asm`. So a complete step-5 build is:

1. Change **only the strings** in `data/types/names.asm` — `db "NORMAL@"` becomes `db "CONTENT@"`, and so on for fifteen
2. Apply the two matchup deltas

Renaming `NORMAL` to `CONTENT` in `constants/type_constants.asm` is a **readability change for the developer**, not a gameplay change — and it is the expensive one, because every type constant is referenced across `moves.asm`, `base_stats/`, and elsewhere, and several collide with unrelated symbols ( `ROCK` with `ROCK_SLIDE`, `FIRE` with `FIRE_BLAST`, `BUG` with the Bug Catcher trainer class). Do it later, deliberately, with word boundaries, or never.

**Three literals the earlier notes got wrong**, all confirmed by build:

| Earlier note                       | Actual                                                              |
|------------------------------------|---------------------------------------------------------------------|
| <code>PSYCHIC</code>               | <code>PSYCHIC_TYPE</code>                                           |
| <code>db GHOST, PSYCHIC, 00</code> | <code>db GHOST, PSYCHIC_TYPE, NO_EFFECT</code>                      |
| <code>numeric 20 / 05 / 00</code>  | <code>named SUPER_EFFECTIVE / NOT_VERY_EFFECTIVE / NO_EFFECT</code> |

`constants/type_constants.asm` also cannot be pasted wholesale. Master wraps the list in `DEF PHYSICAL`, `DEF UNUSED_TYPES / UNUSED_TYPES_END`, `DEF SPECIAL` and `DEF NUM_TYPES`, and `names.asm` depends on three of those symbols. The standalone file in `patches/` defines none of them and would break the build immediately.

The 8/7 physical–special split is intact and is exactly where 2.1 says it is: `const_next 20 (decimal, = $14)`.

### 9.3 The GBA question — a spike, opened 2026-09-02

**pret also disassembled FireRed.** *The question raised was whether the GBA lets this design express itself more appropriately*, and it is a real question rather than a tempting one.

## What would transfer, measured

|                                    | Size                        | Survives                                                         |
|------------------------------------|-----------------------------|------------------------------------------------------------------|
| docs/                              | 5,343 lines                 | <b>All of it.</b> The design has never been engine-specific      |
| The generated source illustrations | ~30 files                   | <b>Yes</b> — high-res; only the downsampler is Game Boy-specific |
| tools/                             | 1,630 lines                 | Partly. Crop and quantise port; 2bpp tile output does not        |
| 64 sprite PNGs                     | 4-shade, 40–56px            | No — redraw from the sources                                     |
| 16 music files                     | GB channel data             | <b>No.</b> GBA is a software mixer; total rewrite                |
| 236 asm/data edits                 | chart, names, scripts, maps | No                                                               |

*334 files of implementation against 5,343 lines of design.* The expensive half is the half that ports.

## The case for, and it is not colour

|                             |                                                                                                                                                                                                                |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Abilities</b>            | <i>Gen 3 gives every species a passive. 2.5 invented <b>CONSENSUS</b> because the chart needed a move to be legible</i> — an ability is a second axis for exactly that work, and Gen 1 does not have one       |
| <b>Item descriptions</b>    | <i>Gen 1 has none at all. The eight <b>MARKS</b>, the boxes, 8.x's stones-that-are-inputs — every one currently gets a name and silence.</i> A writing-led project on the one generation with nowhere to write |
| <b>A scripting language</b> | instead of asm for every sign and guard                                                                                                                                                                        |

## What breaks, and one of it is an invariant

*Invariant 5 stops being a constraint and becomes a preference.* On a DMG the machine and the meaning agree — the player sees grey because that is what the hardware is, so the Review Board's colour feels **impossible**. *On GBA it is a saturation choice, and the rupture becomes a stylistic decision. It can still be good. It cannot be the same idea.*

*Stated honestly: it is already softer than the invariant claims* — SGB palettes already put a blue logo and a tan coat on screen. **But colour arriving through a palette is not the same as colour being free.**

*And the chart goes from 15 types to 17. 2 calls the fifteen names "exactly at the limit" of the character count, and they were chosen as a *complete* argument.* **Invariant 3 says the chart is the argument** — so two filler types is a worse chart. *Two good ones would be better than what exists. That is a design problem to solve before porting, not after.*

Decided 2026-09-03 — the project moved

*The spike answered its question in a day and the project pivoted. `engine/` is a reference now; it still compiles and still runs the slice, and it is not updated further.*

*What follows is the reasoning as it stood before the answer, kept because it is the argument the decision was made against.*

Why it was a spike and not a decision

*8's rule is the deciding one: "the graveyard of ROM hacks is full of projects that designed 151 creatures and shipped zero towns." The vertical slice is **done**. A port resets implementation to zero while the design keeps galloping — which is precisely the failure mode 8 exists to prevent, arriving dressed as an upgrade.*

*And the port is cheaper later, not dearer. The docs are fully portable, the art sources are portable, and a design that has been **played** de-risks the rewrite enormously.*

*So both engines stay. `engine/` keeps the slice; `engineGba/` answers one question — *is the expressive gain real enough to pay for 334 files?* — by building one daemon with an ability and one MARK with a description, and nothing else.*

Two corrections, found by porting rather than by arguing

*The case above got one benefit wrong and missed one cost. Both surfaced within an hour of touching the code, which is the argument for spiking rather than deciding on paper.*

**Wrong:** the MARKS were never going to get descriptions. 9.3 said "the eight MARKS, the boxes and the stones-that-are-inputs each get a name and silence." In Gen 3 badges are not items at all — `items.h` contains the string `BADGE` zero times. They are flags plus a Trainer Card display, exactly as in Gen 1. *The description win is real* — `items.json` carries a `description_english` field with real multi-line text — but it applies to the boxes, the stones, medicines and TMs. Not to the MARKS.

**Missed:** type names get *shorter*, not longer. `TYPE_NAME_LENGTH` is 6 on GBA — which is why vanilla ships `FIGHT`, `ELECTR` and `PSYCHC`. Six of our fifteen **do not fit**: `CONTENT`, `CORRUPT`, `STRATUM`, `ENTROPY` and `CONTEXT` at seven, `EMERGENT` at eight.

*We assumed the newer machine gives more room everywhere. For the one field that carries the argument, it gives less.*



It turned out cheap — the 6 is the array's stride and nothing else reads it, and `B_WIN_MOVE_TYPE` is eight tiles of proportional font. Widened to 8. *But it was found by building, not by reading*, and abbreviating `CONTENT` and `CONTEXT` was never going to be acceptable: they are the edition names and they are the thesis.

The port, begun 2026-09-02

*The chart's vocabulary is in.* All fifteen names ported from `data/types/names.asm` unchanged — *the port does not get to renegotiate a set that 2 calls complete* — and verified in all three ROMs by encoding through `charmap.txt` and searching for the bytes.

`tools/gbastr.py` is the GBA counterpart of `verify-sprites`: Gen 3 encodes text with its own `charmap` exactly as Gen 1 does, so grepping a `.gba` for ASCII finds nothing and proves nothing. *It caught two stale ROMs on its first run.*

`TYPE_STEEL` and `TYPE_DARK` are left vanilla on purpose, so a build that reaches them is obviously unfinished rather than quietly wrong. They are the open decision above and they belong in this document before they belong in a header.

What is in the GBA ROMs, 2026-09-02

|                        |                                                                                                     |
|------------------------|-----------------------------------------------------------------------------------------------------|
| The chart's vocabulary | all fifteen, plus <code>HARDENED</code> and <code>OPAQUE</code>                                     |
| 33 daemon names        | every rename carried across                                                                         |
| 12 item names          | the box ladder, the four inputs, <code>INTERRUPT</code> , <code>RESOLVER</code> , <code>CC-7</code> |
| 12 item descriptions   | <i>new writing. Gen 1 had nowhere to put them</i>                                                   |

`OPAQUE` and `HARDENED` were named by reading the vanilla chart rather than by inventing a concept and hoping the numbers agreed.

`DARK` is immune to `PSYCHIC`, strong against `PSYCHIC` and `GHOST`, cracked by `FIGHTING` and `BUG`. *In our vocabulary: a black box is immune to perspective-taking, beats perspective-taking and the unconscious, and is opened by reasoning and by collectives.* The chart already said it; we only named it.

`STEEL` resists nearly everything and falls to `FIRE`, `FIGHTING` and `GROUND` — *ours: to randomness, to reasoning, and to whoever owns the ground.* Which is what happens to safety hardening.

*The MARKS do land, and the earlier claim was wrong*

*9.3 recorded that the MARKS could not be ported — "badges are not items in Gen 3, so the eight of them have no slot." The first half is true and the conclusion does not follow.*

Gen 3 keeps the eight badge names as individual symbols in `src/strings.c` — `gText_BoulderBadge`, `gText_CascadeBadge`, and so on — *shown wherever a badge is named rather than drawn. They have no bag slot and no description, which is a real loss; they have names, which is most of what the rename was.*

*"Not an item" is not the same as "has no name." The check that would have caught it was one grep, and the claim went into the bible without it.*

The descriptions are the first thing the port bought

`tools/port_names.py` carries a rename; `tools/port_item_text.py` writes something that never existed. Three lines, about 35 characters. Two rules held throughout: craft rule 1 — none of them explains the thesis, they describe an object and stop; craft rule 6 — the joke sits underneath and is never pointed at. The boxes are a privilege ladder and not one description says so:

USERBOX — A host offered at user level. A steady daemon will run on it. A stubborn one will not. ROOTBOX — A host with unrestricted rights. No daemon can decline to run on it. Nothing is refused root.

1.x already made the catch rate literal — "an unbound daemon will not run on a box where it only has user rights" — and until now there was no surface on which to say it.

Built 2026-09-02

`CodeMusic/pokefirered-daemons`, branch `context-content`, `upstream` → pret. Both editions build and both match the retail hashes byte for byte — `pokefirered.gba: OK`, `pokeleafgreen.gba: OK`.

*The edition split survives the port unchanged*, which is the first good sign:

`firered` / `leafgreen` maps onto `_RED` / `_BLUE` with no design work at all. Both disassemblies ship the same game twice and differ by a build flag.

And it needed a debug build, which it does not have. Unlike *pokeemerald*, *pokefirered* ships no debug menu to port — so `firered_debug` is ours, and it is scaffolding for this question rather than a general one. A new game starts with six daemons chosen for their *abilities* and one of each *kind* of item — a ball, a medicine, a berry, a held item, a TM — because the thing being evaluated is the description window, which Gen 1 does not have at all. Hold B to walk through grass, the same gesture the Game Boy build uses.

With `DAEMONS_DEBUG=0` both retail builds still match their `.sha1` byte for byte, which is the check that matters: *the scaffolding is inert when it is off.*

**Toolchain note.** Homebrew's `arm-none-eabi-gcc` installs without `sudo` but ships no `libc`, so `MODERN=1` dies on `string.h`. `agbcc` brings its own headers and is the path that works — built from source into `engineGba/tools/agbcc`, which is why `setup.sh` now builds it.

---

9.4 The sprites, and the answer to invariant 5

**Built 2026-09-02.** 9.3 said two things had to be decided in this document before they were decided in code. This is the second of them.

The art is not upscaled

Our sprites are 40, 48 or 56 pixels square; Gen 3's frame is 64. Upscaling 40 to 64 is a  $1.6\times$  resample of hand-tuned pixel art — it would soften every edge 7.14 spent weeks getting right, and the whole reason those sprites work is that nothing in them is accidental.

*So each sprite is centred in the frame at its native size. Gen 3 does not require a creature to fill its box either; small species simply do not. Nothing is resampled and nothing is lost.*

Colour is by type, which makes it information

*Invariant 5 says greyscale is the design and colour appears once, at the Review Board. On a DMG the machine and the meaning agreed — the player saw grey because that is what the hardware was. On GBA they do not, and 9.3 named that as the thing the port breaks.*

**This is the answer: a daemon is coloured by what it is.** *The palette becomes another way of reading the chart rather than a coat of paint over it — colour carries the argument instead of decorating it, which is the only form of it invariant 5 was ever protecting.*

*Four of the anchors are not invented. 6 gives the Review Board its humours, and each humour already carries a type:*

| Humour      | Colour     | Element | Type    |
|-------------|------------|---------|---------|
| Sanguine    | red        | air     | VECTOR  |
| Choleric    | yellow     | fire    | ENTROPY |
| Melancholic | black bile | earth   | LATENT  |
| Phlegmatic  | water      | calm    | FROZEN  |

The other eleven are extended from those, and every one of them is a claim this document is making rather than a fact it inherited. CONTENT is bone — the undifferentiated one. LOGIC is cold steel blue. CONTEXT is magenta. So CODEMUSAI and CAREMUSAI, who are the same chassis, now differ by hue as well as by ear cups — and the hue is the matchup.

The outline keeps its own colour

Our four tones map to four palette entries: paper becomes index 0, which Gen 3 treats as transparent, and the three inks become a light, a mid and a dark of the type's hue. Level 0 stays nearly black: an outline that takes the hue stops reading as an outline, which is the same lesson the 16×16 player sprite taught three times over.

9.5 The music, re-emitted rather than carried

Built 2026-09-02. The two engines could not be less alike. The Game Boy build IS the sequencer — note C\_, 8 writes a frequency into a hardware channel. The GBA has a software mixer, and pokefirered builds its music from .mid files through mid2agb. So nothing can be copied. The notes have to be re-emitted in another format — which is fine, because the notes are the part worth keeping.

tools/port\_music.py parses our GB channel data and writes MIDI directly.

|            |              |                      |
|------------|--------------|----------------------|
| titletheme | → mus_title  | the front door       |
| slatecity  | → mus_pewter | Pewter is Slate City |
| thebleed   | → mus_route1 | Routes 1 and 2       |

Each one replaces a slot that already exists rather than claiming a new one. Adding a song means a new constant, a new song\_table row, and every index after it moving. Replacing means none of that — and the slots were chosen so the map that plays them is the map the track was written for.

Three conversions, and one of them is a guess

|                                    |                                                                                                                   |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Eight GB ticks is one quarter note | structural — our own tracks are built in bars of 32 as four notes of eight, so this is what we wrote them to mean |
| GB octave N becomes MIDI octave N  | structural — octave 4 / note C_ is middle C, and a bass at octave 2 lands where a bass belongs                    |
| The GB tempo value is used as BPM  | a guess, and the most likely thing to want tuning by ear                                                          |

Brazen, and a claim that was wrong

*The first version of this section said adding a song "moves every index after it." That is false, and the mistake had already caused a bug.*

Saffron shares `MUS_PEWTER` with Pewter and Viridian — Gen 1 gave all three *Cities 1*. So putting Slate City into that slot also handed Brazen the wrong theme, silently, in a build that reported success.

*The song table is positional and nothing more. `mus_title` is row 278 and `MUS_TITLE` is 278, so a row appended at the end shifts nothing; only an insertion would. Adding a song costs four lines and it is the correct fix, not the expensive one:*

|                                        |                                 |
|----------------------------------------|---------------------------------|
| <code>sound/song_table.inc</code>      | one row, at the end             |
| <code>include/constants/songs.h</code> | <code>MUS_BRAZEN 347</code>     |
| <code>sound/songs/midi/midi.cfg</code> | the mid2agb arguments           |
| <code>ld_script.ld</code>              | <i>and this one is the trap</i> |

*The linker script names every song object explicitly. A song that is built but not listed there links as "defined in discarded section" and the ROM fails to link — which is a good failure, because it is loud. Nothing about the song table or the constants would have caught it.*

`SaffronCity/map.json` now plays `MUS_BRAZEN`. Viridian keeps `MUS_PEWTER`, which is correct: 7.4 has Callow sharing Cities 1 with Slate in the Game Boy build too.

9.6 The chart change, finally built — and cheaper than it was

*2 specifies exactly one change to the matchup table — delete Gen 1's `GHOST`, `PSYCHIC`, `00` bug, then add the mutual pair `CONTEXT` ↔ `LATENT` at 2×. "Two additions, one deletion. That is the whole chart change."*

*It had never been built. `git diff` against upstream on `data/types/matchups.asm` is empty, and has been the whole time. The one place invariant 3 says the argument actually lives was still vanilla on the Game Boy.*

And the port makes it cheaper

|                                                                             | Game Boy  | GBA                           |
|-----------------------------------------------------------------------------|-----------|-------------------------------|
| <code>GHOST</code> , <code>PSYCHIC</code> , <code>00</code> — the Gen 1 bug | delete it | Gen 3 fixed it; nothing to do |
| <code>LATENT</code> → <code>CONTEXT</code> ×2                               | add it    | already there                 |
| <code>CONTEXT</code> → <code>LATENT</code> ×2                               | add it    | add it                        |

*Three edits become one. Gen 3 had already fixed the bug the design was working around, and already ships the half of the pair that follows from the fix. So the entire chart argument is now one line in `gTypeEffectiveness` — plus bumping the array from 336 to 339, because Gen 3 declares its size rather than terminating on the sentinel alone.*

Framing is how you reach what runs below the surface; what runs below the surface destabilises framing. *Mutual  $2\times$  is unusual and correct here: these two interpenetrate and neither has a safe angle on the other.*

### 9.7 The Index, expanded — and what the room is actually worth

Built 2026-09-03. *The gain is smaller than 9.3 implied and worth having anyway.*

|       |                                                |
|-------|------------------------------------------------|
| Gen 1 | six lines of ~18 — about <b>108</b> characters |
| Gen 3 | three lines of ~42 — about <b>126</b>          |

Our entries ran 82 to 99. *So the room is roughly one more short sentence each, not a new canvas — and it is spent as one more beat rather than on adjectives. The entries have a shape: a fact, a consequence, an unmarked absence. The extra line extends the shape.*

**CLUSTER** — *Puts near things with near things. Nobody told it which things matter. It has guessed. Nobody checked the guess. OVERFLOW* — *It kept counting past the space it was given. Nothing stopped it. Nothing was watching. It is counting still.*

*Nothing added says anything the short version did not already imply. Craft rule 1 holds throughout: not one of them explains what it is about.*

The one place the two engines deliberately differ

*The Game Boy build keeps the short entries. It has to — 126 characters do not fit in 108. So this is a fork in the writing, and it is worth being exact about what kind: the two engines differ by length, not by meaning. 8.4's rule is about the two editions telling the same story, and they still do — CONTENT and CONTEXT are identical on both machines.*

### 9.8 Full-colour sprites — the other twelve of the sixteen

*9.4 spent four colours of the sixteen: three shades of a type hue plus an outline, laid over Game Boy art. `tools/gbacolour.py` spends the rest, taking art that was drawn in colour and quantising it down.*

**Index 0 is the background** — Gen 3 treats it as transparent — *so the subject gets fifteen, chosen by median cut over the pixels actually in the sprite. Quantising the white surround too would waste a slot on a colour nothing draws.*



*The outline is protected. Generated art puts a great many nearly-black pixels on an edge, and a median cut will merge them into the darkest body colour and leave the creature with no outline at all. Anything below the outline luminance is forced to pure black before the cut runs on everything else.*

## One palette, two sprites

*Gen 3 gives a species one palette and both sprites share it. So the back must not compute its own — doing that would recolour the front sprite already in the ROM. The front defines the palette; the back maps into it, and the tool refuses to do it the other way round.*

*And the rule from 9.4 survives the upgrade: colour is still by type. docs/gemini-prompts-gba-sprites.txt is generated rather than written, so every prompt names the hue that daemon's type was actually assigned — a prompt naming the wrong colour is worse than no prompt.*

## 9.9 The copyright screen, again — and what a tilemap buys

*Built 2026-09-03. The Game Boy version of this screen was a 31-tile budget (7.14d) — a raw bank walked by PlaceString, where \$7F had to stay blank because the string used it as a space, and the whole design fell out of the arithmetic.*

*The GBA version is a bank and a tilemap. graphics/intro/copyright.png is a deduplicated atlas; copyright.bin is a 32×32 map of indices into it. More work, and far more freedom: the text can sit anywhere on the 30×20 screen instead of on three consecutive rows of a font table.*

*Both decompress to the start of their VRAM blocks, so indices are 0-based into our own atlas and nothing needs offsetting. Tile 0 must be blank, because every cell the map does not name is 0.*

*70 unique tiles from 600 cells. Vanilla used 39 — ours costs more because it says more, and a char block holds 512.*

## The same face, on purpose

*The typeface is the one the Game Boy screen uses — vanilla's copyright font, lifted pixel-for-pixel, plus the glyphs it never had (7.14d). Reused rather than redrawn, so the two editions of this screen are recognisably the same object even though nothing about how they are built is shared.*

*One addition the Game Boy could not afford: a one-pixel drop shadow. Five-pixel glyphs vanish on a GBA screen, and the second palette entry was free.*

*tools/gencopyright.py is importable now — its work sits behind a \_\_main\_\_ guard so tools/gbacopyright.py can borrow the font without generating a Game Boy asset as a side effect.*

## 9.10 What the gender question was actually asking

*FireRed* asks "are you a boy or a girl?" and uses the answer for almost nothing. The flag is read 102 times in the source and all but one of them picks a graphic or a palette — a trainer back pic, a sprite template, a palette table.

The single exception called you someone's son or daughter. It says *child* now, and with that gone `playerGender` is purely a sprite selector.

*So the question was doing no work, and it has been replaced with one that does:*

Now tell me. When the two disagree, which do you follow?

### LOGIC INTUITION

The two figures are the same person's two ways of arriving at an answer, which is why they are *deliberately not a matched pair of opposites*. Neither is the careful one. Neither is the wild one. They are two people, and the player picks the one they would rather be.

*And it costs nothing to be careful here: nothing in the game refers to either with a pronoun.*

## 9.10a The two legends were describing their own paperwork

*150 and 151's entries said only how they came to exist. Assembled from a study of a single prior event. Observed once, at a facility, by five people. True, and useless to someone holding one — the provenance is the Quicksilver plot, and the Index is where a player looks to find out what a thing is for.*

They say what they do now.

S.T.A.R.R. — *It reads its own output, then reads that. Nothing tells it to stop. What comes back is no longer information.*

ARTSAI — *It can stand where you are standing and see what you see. What it brings back was in neither of you before.*

*Both stop one sentence early, which is the whole of craft rule 1. The first describes a loop deep enough to become something else and does not name what. The second describes its own move — PERSPECTIVE — and lets the last clause do the work.*

## 9.11 The ID number is a record of what you chose

Vanilla rolls 32 random bits and shows you the low sixteen — *a number that means nothing.*

*Ours derives the visible half from the choices actually made at the front of the game — your name, your rival's name, which edition this is, and whether it is a testing build — folded through an FNV hash into eleven bits, and leaves the last five to chance.*

*So two players who chose the same things land within 32 of each other, and everyone else lands somewhere else entirely. The number tells you what was chosen; the last two digits do not. Nobody is told this and nothing in the game remarks on it.*

*The high sixteen bits stay random on purpose. They are never displayed, and they are half of what decides whether a daemon is shiny — deriving those from a name would quietly make shininess a function of what you typed.*

**9.12 The type badges, and why they say `CONTNT`**

*Built 2026-09-03. The chart's vocabulary lives in two places and only one of them is a string. `gTypeNames` feeds the move-select window; the badges are pixel art — 32×12 each, at a tile offset given by `sMenuInfoIcons`, indexed by the type constant plus one.*

Six characters, and it is not a choice

*The face is four pixels wide on a five-pixel pitch, so six letters fill the badge edge to edge — measured off vanilla's own `NORMAL`, `GROUND` and `DRAGON`, each exactly thirty pixels of ink. Seven would need thirty-five.*

*Which is why vanilla ships `FIGHT`, `ELECTR` and `PSYCHC`. It truncates, and it drops a vowel — so we do the same:*

|                         |                          |                          |                         |
|-------------------------|--------------------------|--------------------------|-------------------------|
| CONTENT → <b>CONTNT</b> | CORRUPT → <b>CORRPT</b>  | STRATUM → <b>STRATM</b>  | ENTROPY → <b>ENTRPY</b> |
| CONTEXT → <b>CONXTT</b> | EMERGENT → <b>EMRGNT</b> | HARDENED → <b>HARDND</b> |                         |

*`CONTNT` and `CONXTT` still differ in the middle rather than at the end, which is the pair that most needs to stay apart at a glance.*

*And thirty is measured, not assumed. Packed as tightly as the face allows — real ink extents, one pixel between letters — `CONTENT` is 32 pixels, `ENTROPY` 32, `CORRUPT` 33, `EMERGENT` 38. The badge interior is 30. Nothing seven letters long fits, at any spacing.*

*This is the third time this design has met a width limit and the third different answer. `TYPE_NAME_LENGTH` was a `#define` and we widened it. The Index reflowed. This one cannot move — the badge is a fixed rectangle in a fixed sheet — so the words give way, in the way vanilla itself established.*

Only the text is redrawn

Each badge keeps its own background colour and its rounded corners. *The colour is how a type is recognised before the word is read, and 9.4 already spent colour on meaning.*

And the shadow is generated rather than drawn — one pixel down and right — *which is what keeps white legible on eight different backgrounds.*

The face is the game's, and drawing our own was the mistake

The first version of this hand-drew an alphabet at four by seven. *It looked right letter by letter and wrong in words. Three glyphs failed, and each failed by exactly one pixel:*

|          | ours              | the game's                           | read as                 |
|----------|-------------------|--------------------------------------|-------------------------|
| <b>T</b> | two-pixel stem    | <i>one-pixel stem</i>                | I — GROWTH, LATENT      |
| <b>W</b> | tapers to a point | <i>legs, a solid row, legs again</i> | V — FLOW, SWARM, GROWTH |
| <b>V</b> | blunt foot        | <i>comes to a point</i>              | U — VECTOR              |

*The fix was not a better hand.* `graphics/fonts/latin_small.png` is `FONT_SMALL`, the game's own half-width face, and it is already four pixels of ink on a five-pixel pitch — *the exact metric the badge was built to.*

`tools/gbatypes.py` now lifts A–Z out of that sheet — 8×16 cells, thirty-two to a row, indexed by the charmap byte, so A is `0xBB` — and discards the sheet's own shadow, because ours has to survive eight different grounds.

*The lesson generalises past this badge.* A face that has shipped to millions of players has had every one-pixel ambiguity argued out of it already. *When the pitch matches, take the font rather than approximate it.*

*The tell was in the player's own eye, not in the diff:* the species names beside the badges were clear and the badges were not — same screen, same size, two different alphabets.

### 9.13 The catch-up sweep, and the word the Game Boy got for free

Built 2026-09-03/04. *The port tools all reported success and the game still said POKéMON.* That is the whole finding, and it has one cause: `pokered` writes the core noun as the charmap glyph `#`, so a single line — `charmap "#", $54` — renamed it everywhere at once. `pokefirered` spells the word out in every string. The Game Boy build got its entire vocabulary from one character; the GBA build had inherited none of it.

## What a rename is, and where it has to live

`tools/port_vocab.py` derives its map rather than declaring it. Species, moves and items are diffed out of *our own GBA tables* against upstream — **if the Index says PACKET, the NPC who mentions one has to say PACKET** — and the vocabulary itself is lifted from the Game Boy diff. *1595 blocks, 242 files.*

*And the town names were in nobody's repository.* `port_names.py` wrote them into `region_map_entry_strings.h`, which is generated from `region_map_sections.json` and gitignored. All sixteen existed only in one working build; **a fresh clone would have built vanilla Kanto.** *The header is an artifact; the JSON is the source.* Renaming a mapsec also renames a generated C symbol that `region_map.c` refers to by hand — and the generator works on bytes, so `é` is two of them and POKÉMON MANSION is `sMapsecName_POK__MON_MANSION`.

## Singular and plural cannot be derived, so it was measured

`pokered` had **two spellings**, `#MON` and `#MONS`, and the Game Boy pass chose between them **by hand 586 times**. *"your" is plural 7 times and singular 25.* No rule recovers that. The heuristic here **agrees with those 586 decisions 93.7% of the time** — scored against them — and recovers **115 of 121 plurals**.

## One bug, in three tools

*An escape is two literal characters and the first is a letter.* So in `\nPOKéMON` the `n` sits against the `P` and `\b` finds no boundary: every word at the start of a line was invisible. It hid **104 substitutions** in `port_vocab`, a name **wrapped mid-word** (`PROF.\nOAK's POKéMON SEMINAR`) in `port_dialogue`, and **two fame checker blocks** in `port_oak`. *Each was found by re-running a tool that had already reported clean.*

## Measuring a line means knowing what a control code draws

A code either draws nothing, draws one glyph, or expands to a word, and treating the third kind as zero made `own_budget` wrap strings to a width they never had. Three more of the same shape: **a control code can contain spaces** (`{CLEAR_T0 56}`), so splitting on whitespace tore the Safari menu in half; **a string that positions its own text is a layout, not prose**; and `\l` scrolls — a dex entry, a move description and a quest log line are drawn into a pane that shows every line at once, and rewrapping had introduced **355 scroll breaks** into files that had never contained one.

## What else came across

|                            |                                                                                                   |
|----------------------------|---------------------------------------------------------------------------------------------------|
| <b>507 dialogue blocks</b> | matched on <i>vanilla against vanilla</i> , the winner having to beat the runner-up               |
| <b>CONSENSUS</b>           | appends at 355 — Gen 3 makes no <code>NUM_ATTACKS == STRUGGLE</code> assertion, so nothing shifts |
| <b>the eight MARKS</b>     | Gen 3's badge palette already carries a grey ramp, so invariant 5 costs nothing                   |
| <b>eight more tracks</b>   | only four of twelve had ever been carried; each lands where its town landed                       |
| <b>two signs</b>           | placed by <i>reading the map blockdata</i> — a Gen 1 coordinate means nothing on a Gen 3 layout   |

*And a SMOKE BALL is not a box.* The box line is about acquisition, so BALL → BOX is right for the eleven capture devices — but **SMOKE BALL and LIGHT BALL** are hold items, and a SMOKE BOX would be a box that helps you leave.

### 9.14 The front door — the face-off, and the third time colour is spent

**Designed 2026-09-04.** *The title screen and the presents scene are the two places the player looks before the world starts, and neither has been ours yet.*

### Colour, sanctioned

2586 says the greyscale is invariant 5's *spending decision about where colour is used, not a mood*. It has been spent twice: **once in the world**, at the Review Board, and **once on the GBA sprites**, where 9.4 made it carry TYPE. *This is the third and last.*

*Both these scenes sit outside the fiction.* They are the front door, not a room — so colour there breaks no rule about the world. **What it must not do is decorate.** So it is spent on one claim, stated twice:

Colour is what arrives when two things that disagree connect.

*In the presents scene that is code becoming music.* **On the title it is LOGIC meeting CONTEXT.** Both are the Review Board's colour arriving early, as a promise rather than a reward — and neither says so.

### The face-off is the one 2.4 already designed

**CODEMUSAI against CAREMUSAI** — *Penphin's two hemispheres pulled apart into two creatures*, and 2's chart makes **LOGIC fail against CONTEXT**. It was drawn for the intro; the title is the same argument at the door.



*And 2.4's warning carries over unchanged. CODEMUSAI lunges. CAREMUSAI opens. One of them is attacking and the other is not. The sides swap between editions and the poses do not* — if the slot contents are swapped wholesale, CAREMUSAI ends up lunging and the chart quietly argues the opposite thing.

*Which is free here, because the swap is a hardware flip. One drawing each, mirrored by OAM for the edition that puts it on the right. The creature still lunges; only the direction changes.*

### The frame, measured

**box\_art\_mon** is a background block — 96×96 on one 16-colour palette — and two creatures do not fit in it. **They become OAM sprites instead**, 64×64 each, which is exactly the frame 9.4 refused to upscale into: *CODEMUSAI is 48 and CAREMUSAI is 56, and both sit inside 64 untouched.*

**The connection is the particle system that is already there.** FireRed rises flames from the floor on a spawner and a per-sprite callback; ours emits from **each** creature toward the centre. Every particle already runs **ten frames** over its life — *frames nought to five carry it inward flat, six to nine are the bloom where they meet.* **No new sprite slots, no new tasks, and the bloom stays abstract** — it alludes, and craft rule 1 keeps it from explaining.

### The presents scene

**The sparkle field is tunable already** — speed, spawn rate, gravity, flicker and destroy timers are all variables. So the particles begin as **0 and 1 in flat green and blue** and become **musical notes in colour**, and *the transformation happens mid-flight in each particle's own animation.* **The existing timers stagger their lifetimes, so the field converts gradually rather than all at once** — which is the effect, and it costs nothing.

## 9.2 Order of operations

1. Toolchain and a **vanilla matching build**. If the checksum matches, your toolchain is sound and every later break is yours.
2. Replace `constants/type_constants.asm`.
3. Replace the strings in `data/types/names.asm` — structure untouched, per 9.1.
4. Apply the three-line patch to `data/types/type_matchups.asm`.
5. `make`. **Stop here and play.**
6. Add CONSENSUS to `moves.asm` and `data/moves/names.asm`. **Done** — and to `move_constants.asm`, `animations.asm` and `sfx.asm`, which the length asserts require (2.5).
7. Rename TRANSFORM to PERSPECTIVE. **Done** — the move name and its battle message (*took the frame of*). MOCK's Index entry still says TRANSFORM and is left for 9 (4.6).
8. City and route renames plus sign text — fast, and it transforms the feel immediately.
9. Starter trio: stats, sprites, moves.
10. Crystal Clear intro script.
11. The Bleed and Undertone encounter tables plus nine wild daemons.
12. Benchmark 1 at Slate.
13. Two city themes and one modulation.
14. **Play it. Decide if it is fun.**

**Step 5 is the real milestone.** Vanilla sprites, vanilla maps, vanilla everything, running your combat philosophy underneath. It will look like Pokémon Red and fight like Context/Content. A couple of hours' work for the fastest possible read on whether the type system is *fun*, before a single pixel is drawn.

Defer RECURSION past the slice. S.T.A.R.R. appears after the Review Board; you will not reach that content for months, and Bide surgery in assembly is a bad first engine task.

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## 10. DECISION LOG

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### Settled

- Base: `pokered`, Gen 1, RGBDS assembly
- Fifteen types as listed; CONTEXT is Special; STRATUM replaces SUBSTRATE for string length
- Creatures are DAEMONS; full lexicon per section 1
- **MOCK** (Ditto) is CONTENT/CONTENT and also learns PERSPECTIVE — the capability was always in the wild; Quicksilver only noticed it (4.6)
- **ORPHAN** at Halftone Tower — CORRUPT/LATENT, the one daemon with a genuinely blank Index entry; an orphaned process is how a daemon is made (4.5)
- **Silph Scope** → **RESOLVER** — it exists because the Index is insufficient (1, 4.5)
- **Species renamed: POKéMON** → **DAEMON** / **DAEMONS**, by repointing one string ( `PlacePOKeText` ), so 650 occurrences moved for free (1.2)
- The Pokédex text token becomes literal **INDEX**; item prefixes become literal `POKé` pending their own renames (1.2)
- **Catching is BINDING** — `bind()` and *binding a daimon*; the flat, uncongratulatory message register goes with it (1.1)
- **The box ladder implemented** — USERBOX / ADMINBOX / SUPERBOX / ROOTBOX, plus **GUESTBOX** for the Safari Zone's restricted temporary access; 36 strings across 19 files, none of which had ever been touched (1.3)
- **"They contain DAEMONS"** → **"DAEMONS run on them"** — a box hosts, it does not contain, and the vanilla line argued against 1.3 at the player's first sight of the object (1.3)
- **"Wild DAEMONS live in tall grass"** → **"UNBOUND DAEMONS run loose out here"** — 1.1 defines a daemon as a process that *runs unattended* (1.4)
- **The box exists as an object** — `tools/genbox.py` replaces both ball sprites; a small server with vents and an indicator, not a crate or a cube, since either would undo the rename. The throw opens rather than bursts, and the USERBOX→ROOTBOX ladder is a parameter (1.3)
- **Gemini asserted the reprise was Crystal's Lament transposed; its own notes refute it** — transposition preserves intervals and these differ entirely. Caught by arithmetic, the only check available (7.12)
- **An unclaimed relationship is in the data instead** — track 11 shares key and first three notes with the *procedure*, then descends where the procedure never does. **Held as unconfirmed**, being one observation of an ordinary figure (7.12)
- **Method**: ask models for measurements, derive relationships here. The one field that invited inference is the one that was wrong (7.12)

- **The best-defended claim in this document is retracted (7.11)** — all three of Scorn's songs arch; the man/directive distinction was a bad transcription, not a finding. A control cannot rescue bad input
- **Track 07 re-scanned** — D5 E5 F5 E5 D5 A4 C5 D5 , 4 up / 3 down. Three findings struck (7.7, 7.10, 7.11, 7.12)
- **A controlled comparison, and the best evidence in the set** — Scorn's two *own* songs both arch at 2/3, independently; his *directive* is the only hook in six that never descends. Same character, same writer, three keys: the contour tracks **man versus procedure** (7.11)
- **Five of six hooks arch**, so arching is the baseline and the single exception is the signal (7.11)
- **The procedure's hook is the only one of six with no descending interval** — it climbs and never comes down. A property, not a reading, and worth more than the shape observations around it (7.10)
- **7.8's "within one accidental" corrected at n=5** — the real spread is five accidentals; the narrow band was a three-track artifact. Scorn's two keys still rule out keys as character labels (7.8)
- **S.T.A.R.R. is the outlier on three axes at once** — F# minor against a one-accidental band, 110 against three at 128, sparse against three driving. **It cannot be layered with the human motifs**, which is a mechanical fact rather than a reading (7.9)
- **Keys will not separate characters** — three transcriptions sit within one accidental, and Scorn has two keys. **Mode and contour** carry character; **keys belong to towns**, as 7 always said (7.8)
- **Two readings retracted** — Crystal's B is generic to the key, not particular to her, and the B ♭ opposition rested on it. The correction arrived one track later (7.6, 7.7)
- **Craft rule 6 is already in the source** — Scorn's own song is major and bouncy; the villain is cheerful, written years before the rule (7.8)
- **Scorn's motif transcribed** — D minor, and **A minor and D minor differ by exactly one note: B against B ♭**, which is Crystal's note of tension flattened away. Marked as a *reading* of ordinary material, since adjacent keys differ by one note by definition (7.7)
- **Their contours oppose** — Crystal's hook descends, Scorn's climbs; both 128 BPM and both 8 beats over 16, so they are interchangeable and layerable (7.7)
- **Crystal's motif transcribed** — A minor, the **relative minor of Blanche's C**, so it needs no modulation; and **exactly one of its notes falls outside Blanche's pentatonic: B, the leading tone**. Her motif is Blanche's scale plus the one note that wants something (7.6)
- **The pipeline caught two real errors on its first run** — a chord name in a note field, and a hook half the length of its progression (7.6)

- **The rock opera is character music, not place music** — Act 1's tracklist is the cast, so the hooks become **leitmotifs** inside town themes rather than town themes themselves (7.5)
- **Gemini is the ear** — Claude cannot hear audio; prompts and format in `music-prompts.md`. Ask for the hook, not the song (7.5)
- **Blanche Town's theme written** — C major **pentatonic**, no F and no B, so it can form neither a leading tone nor a dissonance and has nothing to resolve (7.3)
- **The *My Chromatic Year* mapping is recovered** — it maps **months** to semitones, not colours, but two tracks name colours (*Green:Blue F#, Blue G*) and those two anchor everything (7.1)
- **The colour rule is the octave-folded visible spectrum**, red as C — both anchors land exactly, so it is derived rather than asserted (7.2)
- **A# and B have no wavelength**. They are magenta, which is not in light. The scale closes B→C where the colour wheel closes through an invented colour — and **Halftone**, which replaced **purple Lavender**, sits on those two notes (7.3)
- **Five towns have no hue** — and 7.2 explains it: white is every wavelength, black is none. Hue sets key, **value sets register** (7.4)
- **LEAF STONE** → **REWARD**, closing the item set: reason from / search with / feel with / learn from. **MOON STONE** left for the town pass (8.2)
- **Corpus stays**, **Corpel moves** — three readings versus a placeholder in a provisional table; and it is only a document collision, since Charmander's line is untouched in the ROM (8.2)
- **MUSAI and ROVER implemented** — MUSAI/CODEMUSAI/SEEKMUSAI/CAREMUSAI and ROVERCUB/ROVERSEER/ROVERBYTE, with AXIOM/EMBEDDING/AFFECT and the three branch types, verified in the ROM (8.2)
- **The affect model stays out of the status bar** — rule 3 forbids it: a status is a thing a creature is. Four daemons, not five conditions (8.2)
- **The evolution items are AXIOM / EMBEDDING / AFFECT** — not stones but *kinds of input*, since the mechanic is exposure rather than mineralogy; KERNEL was better on paper and 2 characters too long (8.2)
- **ROVERCUB is the Reinforcement base** — a real repository, exactly 9 characters, and it reads as a young animal (8.2)
- **The repositories surveyed** — `codemusic-repos.md`. **Dexter is 4.2's Index built for real**, flat robotic voice included; **SafetyScribe is 4.10 with the instrument that was missing**
- **MUSAI takes the Eevee slot** — CONTENT base branching to CodeMusai/LOGIC, SearchMusai/VECTOR, TherapyMusai/CONTEXT;

the first and third are Penphin's two hemispheres, and the chart makes LOGIC fail against CONTEXT (8.2)

- **ROVER is a linear line, not a second Eevee** — ROVER → ROVERSEER → ROVERBYTE is explore → model → act, reinforcement learning's own loop (8.2)
- **Two naming registers** — named individuals from the SHC/iASHC universe answer to nothing; the register question governs the other ~148 species (8.2)
- **Starter types implemented** — CONTENT/VECTOR/GROWTH with the secondary at final stage; lab dialogue moved with them. **Movesets are the blocker**: FLYING has 6 moves in all of Gen 1, so VECTOR may need its own CONSENSUS (8.2)
- **Penphin is a trade evolution, not a starter** — the dual mind requires two minds, so it cannot be obtained alone; recovered from conversation after going unrecorded from v1.7 to v3.8 (8.2)
- **MUSAI as Unsupervised, ROVER as Reinforcement; not legendaries** — a starter is raised for forty hours and a legendary is a trophy (8.2)
- **The foxes stop at the Clears** — ears, tail and *value*, nobody else converted; the cast stays as it is because at 16×16 the animal vocabulary is three pixels wide and a full conversion costs ~100 sprite sets for almost no signal (8.5)
- **EnemyText was the third Enemy** — the prefix behind every `<USER> / <TARGET>`, missed by the first REMOTE pass. Now `Remote @`, which caps species names at 9 characters (1.4)
- **The text ceiling is 18, not 19** — the earlier figure counted `<COLON>` as seven characters. Vanilla's four 19-char lines overwrite the right border (1.2)
- **INVOKED rejected on measurement, not taste** — 21 as the verb, 6 in a 5-char menu slot (1.4)
- **Currency is CACHE** — a hoard, a memory cache, and *cash*, all at once; the word is almost never on screen, so this is one prose line and a naming decision. **¥ kept**, logged as optional art (1.4)
- **The battle menu still says FIGHT and RUN** — a live collision in the most-seen string in the game. Not a text edit: the right column is 3 characters, so DETACH cannot fit, and moving the split is `wTopMenuItemX` surgery at four sites (1.4)
- **Kept: used** (1.4 made *use* morally loaded on purpose; INVOKED measured at 20 chars, over the ceiling) and **ATTACK** as a stat (1.4)
- **enemy** → **REMOTE** — nothing here is anyone's enemy; a REMOTE is a process you have no handle on, and it pairs with DETACH. `Enemy X ran!` → **Remote X DETACHED.**, killing a live `ran` collision (1.4)
- **defeated** → **outscored** — a BENCHMARK yields a score; *outran* rejected for reaching back at RUN (1.4)



- **Kept after interrogation:** **RUN**, (comma), **EXP**, **LEVEL** — EXP because 4.3's whole argument is that context forms from *experience*; LEVEL because it already means *permission level* and **<LV>** is a single tile (1.4)
- **Crystal is female and the text had not caught up** — Gramps → **Gran** (she is Al's grandmother, 4.3), His order → Her order, and Agatha's block repronounced. **handsome deliberately kept** — it is attested for women and keeps Crystal tough rather than pretty (4.3)
- **Starter dialogue named vanilla types** — *fire/water/plant* → **ENTROPY/FLOW/GROWTH**; 19 more remain in gyms, held for 5 (1.2)
- **Crystal's opening speech:** six **#MON** marked plural; **pets** → **companions** (1.5's animal-word test), **fights** → **BENCHMARK**. *assistants* considered and held in reserve — it collapses vanilla's contrast (1.4)
- **PERSPECTIVE built** — move **\$90**, and the battle message becomes **<USER> took the frame of <NAME> !**; **Index categories are capped at 10 characters**, so **MOCK**'s cannot be **PERSPECTIVE** (4.6)
- **CONSENSUS built** — move **\$A5**, **SWARM**, 90/100/15, no secondary effect; inserted before **STRUGGLE** because **NUM\_ATTACKS == STRUGGLE** is asserted. Not yet learnable by anything (2.5)
- **SGB borders generated, not drawn** ( `tools/genborder.py` ) — **CONTENT** is one dot grid, **CONTEXT** is the same cell interfering with itself; 12 tiles each against a 96 budget (8.4)
- City names per 3.1; Slate over Sombre; Halftone over Pallor; Quicksilver over Cinder; **Brazen over Gilt**
- Doldrum and The Bleed interrogated and kept, with the reasoning recorded (3.1, 3.2)
- Routes keep numbers officially and carry local names on signs
- Crystal Clear as the Oak figure; the Index measures only content, deliberately
- **Al Clear** as rival and incumbent — Crystal's **grandson**, Ty's son. Vanilla's rival is Oak's grandson, and Ty is far too old to race you (4.3)
- Default name lists per edition: slot 1 fixed, slot 2 swapped, slot 3 differentiated (4.3)
- **Ty P. Clear** is the generation between: Scorn's partner before the rebrand, then the one who went back. He is at Quicksilver, and he explains nothing (4.3)
- **Context does not transmit; only content does.** Ty taught Al the method and none of the experience — the Index problem inside a family (4.3)
- The family is named for three kinds of clarity and none of them can see (4.3)
- Richard Scorn at Benchmark 8, Alignment, sympathetic throughout; the Goodhart engraving in the Corpus lobby, unattributed and adopted approvingly (4.4)
- BunnyArtsai as Mew with **PERSPECTIVE** — the first realization of perspective thinking, not a template (4.6)

- S.T.A.R.R. as Mewtwo with RECURSION — a **comprehension, not a clone** (4.7)
- The content/context feedback loop is the mechanism under the thesis and is never named in dialogue (0.3, 4.9)
- Five Witnesses easter egg, locked at 35 steps; the number lives only in the Quicksilver terminal log (4.8)
- **Ty Clear's parentage is never stated.** Inferred from the surname, confirmed late by one line at Brazen (4.3)
- **The rival naming prompt stays.** The player names him; **CLEAR** is hard-coded, and formal text carries TY CLEAR — the route-sign device applied to a person (4.3, 9)
- Crystal is CRYSTAL CLEAR in full wherever the game refers to her formally, or the surname device dies (4.3)
- **Corpus is downstream of Quicksilver by succession** — Scorn took over the lab, the metric changed, it burned, the people came with him (4.10)
- Crystal was at Quicksilver and built S.T.A.R.R. there; a fitness-for-work procedure removed her, and **the Index is what she did next** (4.1, 4.10)
- Scorn arrived afterwards to run it, signed a complete file about a person he never met, and set the temperature (4.4)
- Ty **stayed** when his mother was removed; his guilt is an absence of decision, not a betrayal (4.3)
- **No date on the fire; dates on everything around it.** The player reconstructs the order or does not (4.10)
- **Brazen confirmed over Brass.** Brass is a colour and not a feeling, and would be the only single-meaning name on the map (3.1)
- At least one survivor knew what they were trading and did it anyway, for an ordinary reason — the guard against determinism (4.10)
- Craft rule 6: comedy is the cover. Corpus is funny; Quicksilver is fun to explore (0.1, 4.10)
- Four humors as the Review Board
- Vertical slice before anything else
- **Greyscale is the design, not a limitation.** Colour appears once, at the Review Board, and nowhere else (8.6)
- **Two editions, CONTENT and CONTEXT, from one source tree** — the slash in the title is literal (8.4)
- **The title order is CONTEXT-first by decision, not by accident.** Everyone says content is king; the title refuses to let the original decide its order. **The title is the argument, the RED→CONTENT mapping is the inheritance** — and this is filed as a choice, not a finding (8.4)
- The **type chart is byte-identical** across editions; only encounters and a handful of Index entries differ (8.4)

- The two-target build goes in on day one even while both ROMs are identical (8.4)

## Reversed

Kept here because the reasoning is worth more than the outcome.

| Was                                                            | Now                                                 | Why                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Gilt City</b>                                               | <b>Brazen City</b>                                  | Gilt implies a concealer, and this story has no schemer. Brass is honestly itself; so is Scorn. Full argument in 3.1.                                                                                                                                 |
| <b>BunnyArtsai35</b>                                           | <b>BunnyArtsai</b>                                  | The number in her name gave away the Five Witnesses lock and made a serial of a one-off. Relocated to a single lab log (4.6, 4.8).                                                                                                                    |
| <b>Ty</b> , an unrelated rival                                 | <b>Ty Clear</b> , Crystal's son and Scorn's partner | Turns a methodological disagreement into a cost somebody pays (4.3).                                                                                                                                                                                  |
| S.T.A.R.R. as <i>refined successor</i> to the BunnyArtsai line | S.T.A.R.R. as a built understanding                 | "Successor" still smelled of cloning. The lab understood recursion and instantiated it (4.7).                                                                                                                                                         |
| <b>CATCH</b>                                                   | <b>BIND</b>                                         | The container was renamed and the verb was not. CATCH is the only lexicon entry doing no double duty; BIND is <code>bind()</code> , <i>binding a daimon</i> , and a bond — and it is darker rather than softer, so the player stays implicated (1.1). |
| Drop the rival naming prompt, hard-code "Ty Clear"             | Keep the prompt, hard-code <b>CLEAR</b>             | The surname is the half that carries the inference. Spending a famous vanilla beat bought nothing — and the prompt, reframed as <i>what will you call him</i> , becomes the route-sign device a minute before Route 1 teaches it (4.3).               |

## Open

- Does Halftone hold once the tower is written, or do Penumbra / Moiré serve better?
- Does the player meet Scorn before Halftone Tower?
- Does RECURSION justify engine work in the slice, or defer?
- How legible is S.T.A.R.R.'s SHC backstory to a player who has not heard the rock opera — and does it need to be?
- Are the humors too neat? Four is convenient; the real theory had temperaments blending.
- Does **Al** get a redemption, a plateau, or neither — and does the family tie make redemption too cheap?
- Ty is absent from the endgame by design. Does that read as a statement or as a loose thread?
- Is the Quicksilver terminal missable enough to soft-lock the Five Witnesses puzzle, and is that acceptable?
- Does Brazen ever read as the game *sneering* at Scorn? If playtesters hear that, swap to Brass immediately — the whole point of him is that the game does not sneer.
- **The bestiary naming register is undecided, and it gates 151 names (8.2).** Technical surface like the rest of the lexicon, or **true names** the Index cannot reach — the second does 4.2's work for free. Every candidate must pass BIND's test: does it work twice?
- The supervised starter slot is **yours to pick (8.2)**
- **Callow and The Bleed share their music** with other maps, so 8.1's second theme needs new `songs.asm` entries first (7.3)
- ~~57 catch / caught instances remain in dialogue~~ **Done 2026-08-31**, caught in play: the capture message still read *PACKET was caught!*. **55 instances rewritten.** The only survivor is *catchy tune*, which is a song
- **MOON STONE still wants a name** — tied to a place rather than a paradigm, so it belongs with the town pass (8.2)
- `Corpel` collides with **Team Corpus** and is still in 8.2's table — CODEX proposed
- Starter daemon names are placeholders and need a pass.
- Should ORPHAN be bindable at all, or only witnessed? *Lean: bindable — a blank entry sitting in your own collection is worth more than a blank entry you only heard about.*
- Nineteen `type word + DAEMON` dialogue lines still name vanilla types, nearly all in gyms — held until 5 settles the Benchmark leaders (1.2)
- Forty-two `fight` occurrences want a human read; some are the replaced verb, some are people (1.2)

- Roughly 154 #MON occurrences read singular by default and want a human read at step 8 — on screen, not in the source (1.2)
- Does a non-canonical ROM load in Gen1Recomp at all? One afternoon answers it; do not design around either answer first (8.5)
- How many Index entries should disagree between editions — five? twelve? — before it stops being unsettling and starts being a gimmick? (8.4)
- Is edition-exclusivity fair when link trading needs two people, two carts and a cable, and most players will have one? *Lean: yes — the Index was never going to be completable, and 4.2 says so.*